Quantitative Business Methods

Supplementary Teaching Material

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Review of Mathematics

0.1 Arithmetic Operations

Let us start with elementary arithmetic operations which we will need to use in this module. The acronym to remember is **BODMAS**, which tells you the correct order in which to perform a mathematical operation:

Operation	Example
Brackets	$( )$
Orders	2^3
Division	$8 \div 2$
Multiplication	4×3
Addition	$5 + 2$
Subtraction	$7 - 3$

Example 1: Derive the equation:

$$(35 \div 7 + 10) - (4^2 - 6 \times 2)$$

Solution

Remember the order of **BODMAS**.

For convenience, let us call $(35 \div 7 + 10)$ the **first term** and $(4^2 - 6 \times 2)$ the **second term**. In order to subtract these two terms, you need to first evaluate the equations inside the **b**racket.

Lets start with the first term:

$$= 35 \div 7 + 10 \qquad \textbf{D}\text{ivision}$$

$$= 5 + 10 \qquad \textbf{A}\text{ddition}$$

$$= 15$$

Now we can do the second term:

$$= 4^2 - 6 \times 2 \qquad \textbf{O}\text{rder}$$

$$= 16 - 6 \times 2 \qquad \textbf{M}\text{ultiplication}$$

$$= 16 - 12 \qquad \textbf{S}\text{ubtraction}$$

$$= 4$$

Finally, we can subtract the two terms:

$$= (35 \div 7 + 10) - (4^2 - 6 \times 2)$$

$$= 15 - 4$$

$$= 11$$

With practice, you will be able to conduct these operations with ease. It will be useful when we start with algebraic notations.

It is also worth remembering some interchanging notation. For *any* numbers a and b:

- The **sum** of a and b means $a + b$.
- The **difference**[a] between a and b means $a - b$.
- The **product** of a and b means $a \times b = a \cdot b$.
- The **quotient** of a and b means a divided by b or a/b.

The *power* is the amount of times a number is multiplied by itself. For example:

$$3^4 = 3 \cdot 3 \cdot 3 \cdot 3 = 81$$

$$2^2 = 2 \cdot 2 = 4$$

One important power which we will use throughout the module is the square root $\sqrt{a}$. For example[b]:

$$= \sqrt{9} = 3$$

$$= \sqrt{4} = 2$$

0.2 Fractions and Percentages

A fraction is simply a part of a whole and expressed as:

$$\frac{a}{b}$$

We call a the **numerator** while b is the **denominator**. You should be comfortable with handling fractions. Moreover,

[a] The reverse is also true, where b - a.
[b] For your information, the square root of a number is simply the base multiplied by 0.5.

you should also know that

$$\frac{a}{b} \cdot 100,$$

will convert the fraction into a percentage, with the % sign to be used.

> **Example 2:**
>
> (a) Evaluate this expression as a fraction:
>
> $$\frac{4}{9} \cdot \frac{1}{3} \cdot \frac{2}{5}$$
>
> (b) Express the fraction as a percentage.
>
> **Solution**
>
> (a) You can simply multiply all the numerator terms together and all the denominator terms together:
>
> $$\frac{4 \cdot 1 \cdot 2}{9 \cdot 3 \cdot 5} = \frac{8}{135}$$
>
> (b) To convert this into a percentage, simply multiply it by 100.
>
> $$\frac{8}{135} \cdot 100 = 5.92\%$$

0.3 Functions

You should be very comfortable in understanding what a function is and how to illustrate it.

When a variable y depends on another variable x, we can mathematically denote this as:

$$y = f(x)$$

where we say that y is a function of x.

Example 3: Consider the below function:

$$y = 2x$$

(a) Express each value of y for the below values of x:

x	y
0	
1	
2	
3	
4	
5	

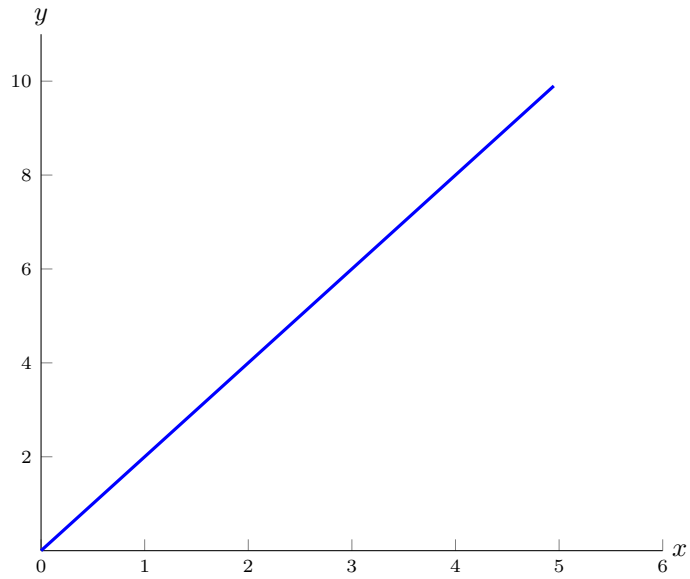
(b) Plot a graph where y is on the vertical axis and x is on the horizontal axis. Label each value of (x, y) and connect it.

Solution

(a) Simply input each value of x into the function $y = 2x$:

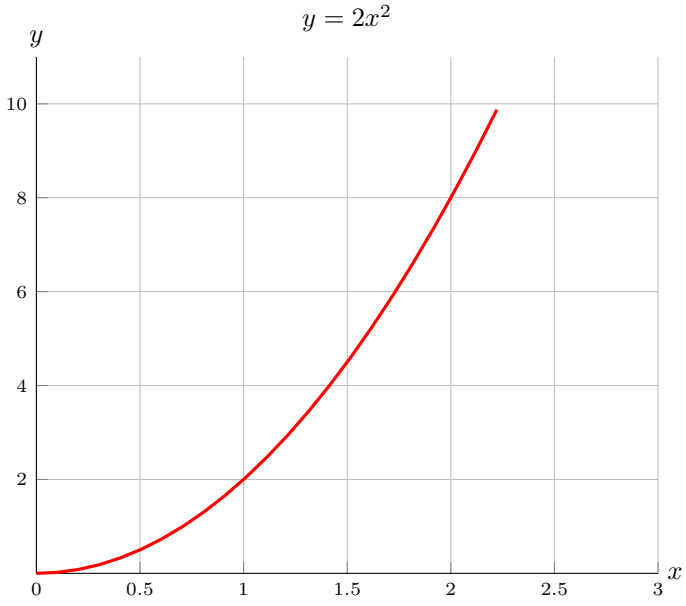
x	y
0	0
1	2
2	4
3	6
4	8
5	10

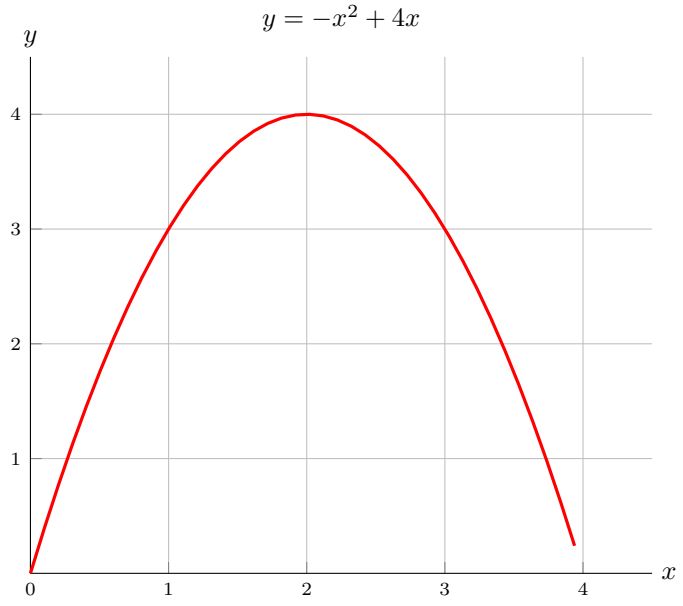
(b) As per the instructions, we can illustrate this function as below:



Notice how the function is a straight line. We call this a **linear function**[c].

A non-linear function can be visually identified as having curves[d]:





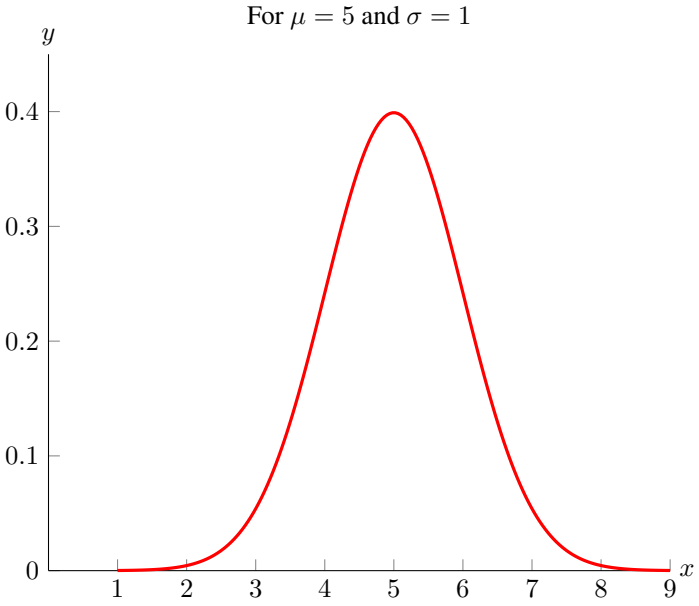
An important non-linear function that you will need to be familiar with is the **normal distribution**:

[c]The qualifier here is that the slope of a linear function is constant. You can derive this using a variety of methods, but one way is through calculus. For **Example 3** the slope of the function is $\frac{dy}{dx} = 2$.

[d]Similarly, a non-linear function will not have constant slopes. In fact, the slope changes based on the position of x.

$$y = \frac{1}{\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{\sigma^2}}$$

Although this function looks complicated[e] it is actually quite simple to calculate in a software if you have the values of the mean μ and the variance σ^2. You will learn more about this later in the module. For now, you should just know that it is a non-linear function and can be visually represented as below:



0.4 Summation Operator

Consider the exam score data of 5 students from a class:

i	**Name**	**Score** x
1	Ahmed	70
2	Sara	90
3	Hassan	80
4	Fatima	44
5	Zahra	73
6	Hisham	32

[e]The normal distribution function describes the probability of observing a particular value of x. It says that it is greatest when it is near the average μ, and that this probability dies away rapidy as the difference from the mean increases. The rapidity of the degeneration is based on the standard deviation σ.

There are a total of $n = 6$ people in this dataset. Moreover, each person is indexed by a number i. For example, x_5 refers to Zahra's score, which is $x_5 = 73$.

Suppose we are interested in the sum of all scores in the class. This is simply:

$$= 70 + 90 + 80 + 44 + 73 + 32 = 389$$

Another way to express the sum of the score variable x is by using the summation operator $\sum$. Formally, we would say:

$$= \sum_{i=1}^{n} x_i$$

$$= x_1 + x_2 + x_3 + x_4 + x_5 + x_6$$

$$= 70 + 90 + 80 + 44 + 73 + 32$$

$$= 389$$

This will be a useful operator later in the module. For example, the average score of the class can be expressed as:

$$= \frac{1}{n} \sum_{i=1}^{n} x_i$$

$$= \frac{389}{6} = 64.8$$

0.5 Problem Set

1) Work out the following:

(a) $(2+4) \times (3+7)$

(b) $(1/3 \times 12) - 4/2$

(c) $\frac{(1+4)}{5} \times (100 - 98)$

2) Work out the following (You can use a calculator when needed):

(a) $\sqrt{16}$

(b) $(0.07)^2$

(c) $\sqrt{0.49}$

3) Work out the following:

(a) What is 98% of 200?

(b) Express 17/25 as a percentage.

4) Sketch the following functions:

(a) $y = x + 3$

(b) $y = 3x - 2$

5) Consider the following data:

i	x
1	3
2	1
3	4
4	6
5	8

Find the following expressions:

(a) $\sum_{i=1}^{5} x_i$.

(b) $\frac{1}{n} \sum_{i=1}^{5} x_i^2$.

0.6 Solutions

1) Remember to use BODMAS:

(a)

$$\begin{aligned}(2+4)\times(3+7) &= 6\times 10\\ &= 60\end{aligned}$$

(b)

$$\begin{aligned}\left(\frac{1}{3}\times 12\right)-\frac{4}{2} &= 4-2\\ &= 2\end{aligned}$$

(c)

$$\begin{aligned}\frac{(1+4)}{5}\times(100-98) &= \frac{5}{5}\times 2\\ &= 1\times 2\\ &= 2\end{aligned}$$

2) You can use a calculator for the second and third part:

(a)

$$\sqrt{16} = 4$$

(b)

$$(0.07)^2 = 0.0049$$

(c)

$$\sqrt{0.49} = 0.7$$

3) Remember that a percentage is a faction multiplied by 100:

(a)

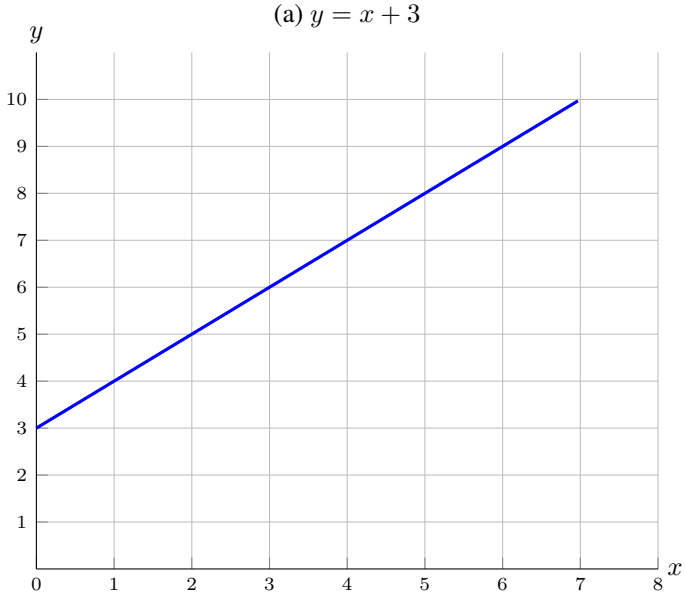
$$\begin{aligned} 98\% \text{ of } 200 &= \frac{98}{100} \times 200 \\ &= 196 \end{aligned}$$

(b)

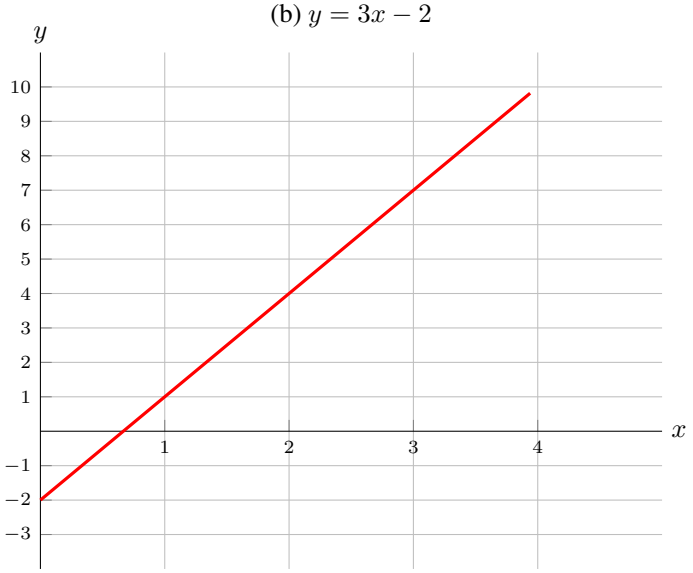
$$\frac{17}{25} \times 100 = 68\%$$

4) Note that both functions are linear. However for (b), the intercept point starts at a negative number.

(a) $y = x + 3$



(b) $y = 3x - 2$



5) Using the summation operator:

(a)

$$\sum_{i=1}^{5} x_i = 3 + 1 + 4 + 6 + 8$$
$$= 22$$

(b)

$$\frac{1}{n}\sum_{i=1}^{5} x_i^2 = \frac{1}{5}\left(3^2 + 1^2 + 4^2 + 6^2 + 8^2\right)$$
$$= \frac{1}{5}\left(9 + 1 + 16 + 36 + 64\right)$$
$$= \frac{126}{5}$$
$$= 25.2$$

Chapter 1

What is Statistics

1.1 Descriptive and Inferential Statistics

You are a policy analyst of the Bahraini Ministry of Education and you have been asked to do a study on the quality of primary schools in Bahrain. You collected data on a sample of schools and inputted it into a dataset:

i	**School**	**Type**	**Score**
1	Ain Jaloot Primary Intermediate Girls	Female	Good
2	Abusaiba Primary Boys	Male	Bad
3	Aisha Um Al Moamneen Primary Girls	Female	Good
4	A'ali Primary Boys	Male	Good
5	Sanad Primary Girls	Female	Bad
6	Ahmed Al Fateh Primary Intermediate Boys	Male	Bad
7	Al-Budaiyya Primary Intermediate Girls	Female	Good
8	Hajer Primary Girls	Female	Bad
9	Al Imam Malek Bin Anas Primary Boys	Male	Bad
10	Al Busaiteen Primary Girls	Female	Good
11	Abu Bakr Al-Siddeeq Primary Intermediate Boys	Male	Bad
12	Al Maamoon Primary Boys	Male	Bad
13	Al-Belad Al-Qadeem Primary Girls	Female	Good
14	Al Andalus Primary Girls	Female	Good
15	East Riffa Primary Boys	Male	Good
16	Abu Alaala Almaari Primary Boys	Male	Good
17	Al-Alaa Al Hadrami Primary Boys	Male	Good
18	A'ali Primary Girls	Female	Good
19	Al Budaiya Primary Boys	Male	Good
20	Al Malikiyah Primary Intermediate Girls	Female	Good

In order to conduct your study, you summarise the information in a small table:

	Good	**Bad**
Boys	5	5
Girls	8	2

There are a number facts that you can use to describe your data:

- There are a total of 20 schools in your dataset.
- There are 10 boy schools in your sample.
- There are 10 girl schools in your sample.
- 50% of boys schools achieved a good score.
- 80% of girls schools achieved a good score.

This is known as **descriptive statistics**. You are simply looking at the information that was collected and describing the contents.

Now suppose you are interested to find out whether girl schools are *better* than boys schools. One primitive way to doing this is to simply compare the percentage difference of good schools based on gender. We can clearly see that

$$80\% > 50\%,$$

Based on the sample, an analyst can conclude that **all** primary girl schools in Bahrain are higher quality than **all** boy schools.

This analysis is known as **inferential statistics**. You have analysed your sampled data and drawn a conclusion on all schools in Bahrain.

> **Activity 1:**
>
> You currently work as a strategist for Batelco. Explain whether the following tasks are considered **descriptive** or **inferential** statistics:
>
> (a) The average network engineer works for 1.5 years before leaving the company.

(b) Based on a sample of customers, 68% said they would not switch their service, which means that most people are happy with the service.

Solution

(a) The qualifies as **descriptive** statistics, because we are simply talking about the data collected from network engineers.

(b) The qualifies as **inferential** statistics, since you are analysing the response from a sample of customers.

1.2 Sample and Population

An important fact to consider is that your analysis is only based on 20 schools. Suppose that three other Ministry of Education employees also did their own investigations and collected the following data from 20 schools each:

Sample X

	Good	Bad
Boys	4	6
Girls	9	1

Sample Y

	Good	Bad
Boys	5	5
Girls	7	3

Sample Z

	Good	Bad
Boys	6	4
Girls	8	2

Even though they all have different numbers, the overall story remains the same. It seems that based on the sample of schools, each investigator can make an inference that primary girls schools are better than primary boys schools.

These are known as **samples**. They represent parts of the overall total of primary schools across Bahrain.

Suppose you now go and collect data on **all** 130 primary schools in Bahrain. You find the following information:

	Good	Bad
Boys	53%	47%
Girls	96%	4%

Since you have collected this information from all 130 schools, then we say that this dataset is the **population** of primary schools in Bahrain.

An important point to keep in mind is that inferential statistics allows us to draw a conclusion about the population based on a sample.

1.3 Types of Variables

In order to assess whether a public school is good or bad, you need to base your conclusion a set of variables. For example:

- Funding received from the government.
- Number of teachers and students.
- Number of classrooms
- Classroom size.
- Exam scores.
- Number of air conditioners.

These are examples of quantitative variables. That is, the data that you record will be inputted as numbers.

School	Funding (BD)	Students	Teachers	Classrooms	Class Size	Avg. Exam Score (%)	A/Cs
School 1	450,000	600	45	25	24	78	32
School 2	520,000	750	50	30	25	84	38
School 3	380,000	480	35	20	24	71	25
School 4	600,000	800	55	32	25	89	40

You should be aware that there are two types of quantitative variables: **continuous** and **discrete**.

Continuous variables can be described as numbers that can value within a range. For example, the amount of money received by a school in 2026 can 2.6 Million BD or the average score of the mathematics exam was 67.3%.

Discrete variables can be described as those that can counted. For example, a school can be 50 teachers, 100 classrooms and each classroom can have 2 ACs. A school **cannot** have 100.5 classrooms or 50.3 teachers or 0.5 ACs. This is what makes it a discrete variable.

Data can also be **qualitative**. This means that the data collected is some descriptive measure and not something measurable. For example:

- City where the school is located.
- The education degree of the principal.
- If there is a requirement to wear uniforms.

School	City	Principal's Degree	Uniform Required
School 1	Manama	Bachelor's	Yes
School 2	Riffa	Master's	Yes
School 3	Muharraq	PhD	No
School 4	Hamad Town	Master's	Yes

1.4 Levels of Measurement

Both quantitative and qualitative data can be categorised by four levels of measurement:

Level of Measurement	Description	Examples
Nominal	Data may only be classified into different categories.	Employee department; Firm's industry; School type
Ordinal	Data are classified into categories that can be ranked or ordered.	Customer satisfaction; Credit rating; Performance in a Class
Interval	Data are ordered and the differences between values are meaningful.	Temperature in degrees Celsius; Calendar years
Ratio	Data have meaningful differences between values and a meaningful zero point.	Employee salary; Sales; Number of employees; School enrolment

One way to identify the difference between interval and ratio is to ask if *is zero a genuine absence of the measured quantity?*.

Only ratio requires zero to be a proper "zero". For example, 0 BD, 0 employees, 0 students in a classroom, where the zero is properly understood. However, 0°C does not literally mean "no temperature" and "Year 0" does not literally mean "no years".

1.5 Ethics in Statistics

We conclude this chapter by informing students that statistics can easily be manipulated or misrepresented. As a statistician, you must try to collect data in principled way.

- Make sure you collect a proper representative sample.
- Do not omit relevant statistics.
- Do not let bias impact your collection.

Activity 3

Explain how each scenario impacts the school study.

(a) You only collect data on schools from high-income neighbourhoods.

(b) You notice that teachers in boy's schools get paid much less compared to girl's schools, but cannot collect the data.

(c) The principal only lets you interview teachers with high evaluations.

Solution:

(a) By omitting schools from low-income areas, your sample is not representative of the population. The average scores, performance and ranks of the schools will be higher than what is true.

(b) Its possible that because teachers in boy's schools get paid less, they are not as good as teachers in girl's schools. This is probably useful information that can explain the difference in performance that you would report to the ministry of education.

(c) Your data will be biased because you are only getting information from teachers that perform well in the school. Low-performing teachers will not be considered and so you will improperly rank these schools.

1.6 Problem Set

1) For each of the following statements, identify whether it is an example of **descriptive statistics** or **inferential statistics**.

(a) A company records the salaries of all 50 of its employees and reports that the average salary is BD 850 per month.

(b) A supermarket surveys 100 customers and uses the results to estimate the percentage of all its customers who are satisfied with its service.

(c) The monthly sales of a business are recorded for the past 12 months and presented using a line graph.

(d) A researcher surveys 500 households in Bahrain and uses the results to estimate the average household income of all households in Bahrain.

(e) A university reports that 72% of the 200 students who completed a survey were satisfied with their course.

(f) A shop records the amount spent by 150 customers on a particular day and calculates a average expenditure of BD 12.

(g) A researcher examines a sample of 100 firms and concludes that larger firms tend to employ more workers than smaller firms in the wider population of businesses.

2) Consider the below cases and specify the **type** (qualitative or quantitative) and **level** (nominal, ordinal, interval, ratio) for each measure:

(a) Eye Colour.

(b) Rank in a class room.

(c) Temperature.

(d) Annual Salary.

Specify an example of a data point for each case.

1.7 Solutions

1)

(a) A company records the salaries of all 50 of its employees and reports that the average salary is BD 850 per month. **DESCRIPTIVE**.

(b) A supermarket surveys 100 customers and uses the results to estimate the percentage of all its customers who are satisfied with its service. **INFERENTIAL**.

(c) The monthly sales of a business are recorded for the past 12 months and presented using a line graph. **DESCRIPTIVE**.

(d) A researcher surveys 500 households in Bahrain and uses the results to estimate the average household income of all households in Bahrain. **INFERENTIAL**.

(e) A university reports that 72% of the 200 students who completed a survey were satisfied with their course. **DESCRIPTIVE**.

(f) A shop records the amount spent by 150 customers on a particular day and calculates a average expenditure of BD 12. **DESCRIPTIVE**.

(g) A researcher examines a sample of 100 firms and concludes that larger firms tend to employ more workers than smaller firms in the wider population of businesses. **INFERENTIAL**.

2)

(a) Eye colour: Brown, blue, green.

Eye colour is a **qualitative** variable because its values represent categories rather than numerical quantities. It has a **nominal** level of measurement because there is no natural ordering of the categories. For example, brown cannot meaningfully be ranked above or below blue or green.

(b) Class Rank: 1st place, 2nd place,...

Class rank is a **qualitative** variable with an **ordinal** level of measurement. Although numbers are used to represent the ranks, these numbers indicate positions rather than numerical quantities. We can place the observations in a meaningful order: a student ranked 1st is ahead of a student ranked 2nd, who is ahead of a student ranked 3rd.

However, the differences between ranks are not meaningful numerical distances. For example, the difference between 1st and 2nd place is not necessarily equivalent to the difference in performance between 2nd and 3rd place.

(c) Temperature: 18° C

Temperature measured in degrees Celsius is a **quantitative** variable because its values represent measurable numerical quantities. It has an **interval** level of measurement because differences between values are meaningful. For example, the difference between $10°$C and $20°$C is the same temperature difference as between $20°$C and $30°$C.

However, Celsius does not have a meaningful zero point. A temperature of $0°$C does not represent the absence of temperature. Consequently, ratios are not meaningful: we cannot say that $20°$C is "twice as hot" as $10°$C.

(d) Annual Salary: 1000 BD

Annual salary is a **quantitative** variable because it represents a measurable numerical quantity. It has a **ratio** level of measurement because differences between values are meaningful and there is a meaningful zero point. An annual salary of BD 0 represents no salary income.

Ratios are therefore also meaningful. For example, an annual salary of BD 20,000 is twice an annual salary of BD 10,000.

Chapter 2

Describing Data: Frequency Tables and Graphs

In this chapter, we are interested in the various ways in which we can present qualitative and quantitative data.

2.1 Qualitative Data

2.1.1 Frequency Table

The Bahraini Minister of Urban Planning has received a complaint that petrol station queue times are very long. The minister asks you to collect data on all petrol stations in order to conduct a study.

After collecting 180 entries, your dataset is shown as below:

i	**Petrol Station Name**	**Town**	**Governorate**
1	Al Fateh Service Station	Manama	Capital
2	Bapco Energies Station – Muharraq	Muharraq	Muharraq
3	Bapco Energies Station – Sanabis	Sanabis	Capital
4	Salmabad Service Station	Salmabad	Northern
5	Galali Petrol Station	Galali	Muharraq
6	Al Buhair Petrol Station	Riffa	Southern
⋮			
180	Hajjyat Petrol Station 180	Riffa	Southern

There are various ways to present this data to the minister. One way is to construct a frequency table.

> A **Frequency Table** is a grouping of qualitative data into mutually exclusive and collectively exhaustive classes showing the number of observations in each class.
>
> *Mutually Exhaustive* implies that each observation falls on only one class.
>
> *Collectively Exhaustive* implies that there is a class for every value.
>
> An *observation* refers to a row in your dataset. For example, East Riffa Petrol Station is one observation in your dataset.

Let us start with a basic analysis and find out how many petrol stations are available in each governorate:

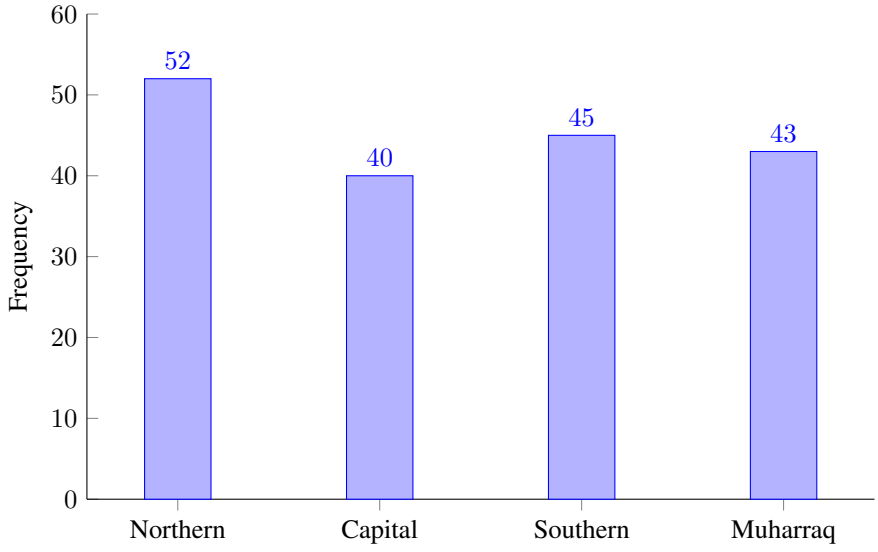
Governorate	**Frequency**
Northern	52
Capital	40
Southern	45
Muharraq	43
Total	**180**

This is a simple way of presenting this information to the Minister. Notice that there are 4 governorates: Northern, Capital, Southern and Muharraq. For the frequency table, we call each of these categories as **classes**. We call this a frequency table because it shows the total amount of **observations** for each **class**. For example, there are 40 petrol stations in the Capital governorate.

2.1.2 Bar Chart

You can take your dataset on petrol stations and display it in a **bar chart**. This is when we have the frequency of the data on the vertical axis while each class is represented by a bar on the horizontal axis.

Figure 2.1: Frequency of Petrol Stations by Governorate



The bar chart is informative because it immediately reveals the class with the highest frequency.

> We will learn later that the **mode** of this dataset is the class which has the highest frequency. In this case, it is the capital governorate.

2.1.3 Pie Chart

It is probably more informative to display this information with the **share** of each class to the overall total. To do this, we simply divide the number of petrol stations from each class with the total. We call this the **relative frequency**.

$$\text{Relative Frequency} = \frac{\text{Frequency}}{\text{Total Frequency}}$$

Since there are a total of 180 petrol stations, we can arrive at the following result:

Governorate	**Frequency**	**Calculation**	**Relative Frequency**
Northern	52	$\frac{52}{180}$	0.289
Capital	40	$\frac{40}{180}$	0.222
Southern	45	$\frac{45}{180}$	0.250
Muharraq	43	$\frac{43}{180}$	0.239
Total	**180**	$\frac{180}{180}$	**1.000**

You can use this information to construct a **pie chart**. This displays the relative frequency of each class.

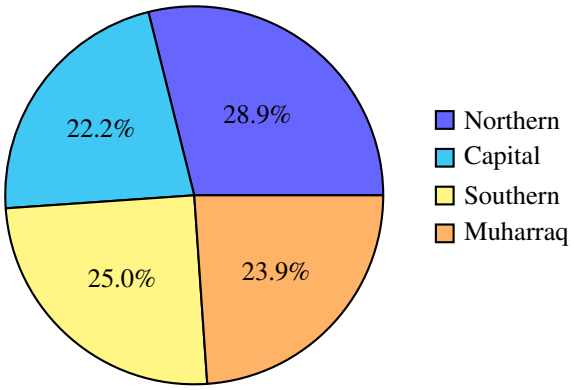


Figure 2.2: Petrol Stations by Governorate

Here, each *slice* of the chart represents a class. For example, 22.2% of the petrol stations in your dataset is from the Capital governorate.

Activity 1:

Bahrain City Center has the following aggregated information about types of stores in their mall:

Store Type	**Frequency**
Clothing and Fashion	70
Food and Beverage	45
Electronics	20
Health and Beauty	30
Home and Furniture	15
Entertainment and Leisure	20
Total	**200**

(a) Create an additional column representing the relative frequency.

(b) Display the totals of each class in a bar chart.

(c) Display the relative frequency of each class in a pie chart.

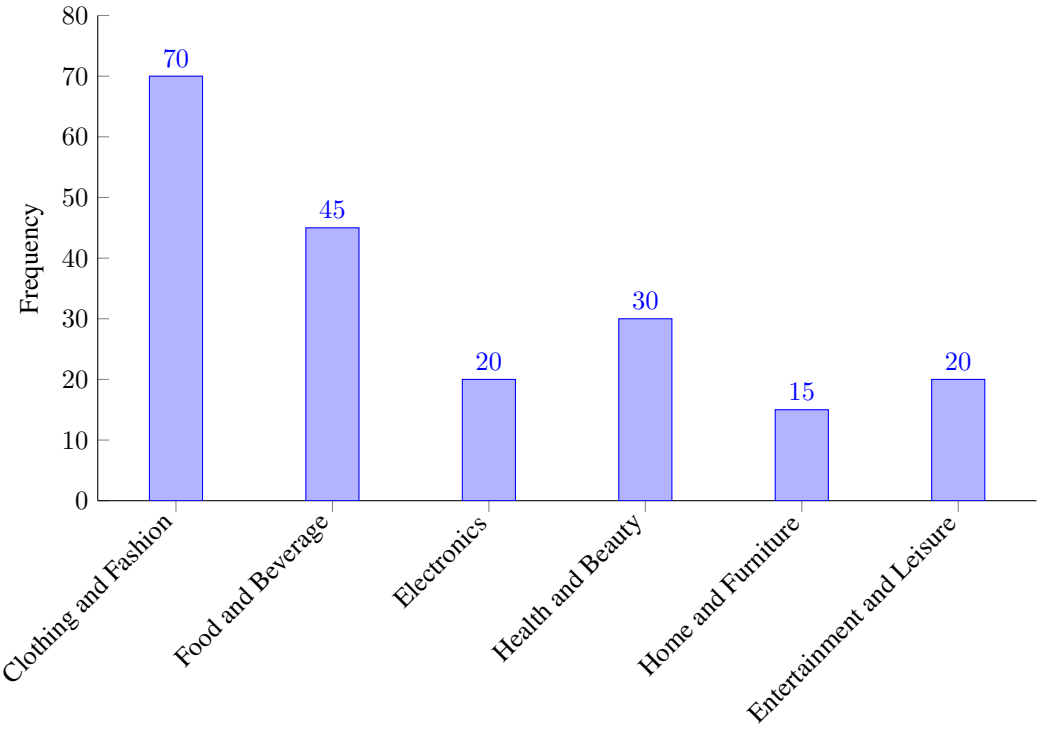
Solution:

(a) To create the relative frequency column, we first get the total number of store in City Center. we then divide the frequency of each class with the total number of stores.

Store Type	Frequency	Relative Frequency
Clothing and Fashion	70	0.350
Food and Beverage	45	0.225
Electronics	20	0.100
Health and Beauty	30	0.150
Home and Furniture	15	0.075
Entertainment and Leisure	20	0.100
Total	**200**	**1.000**

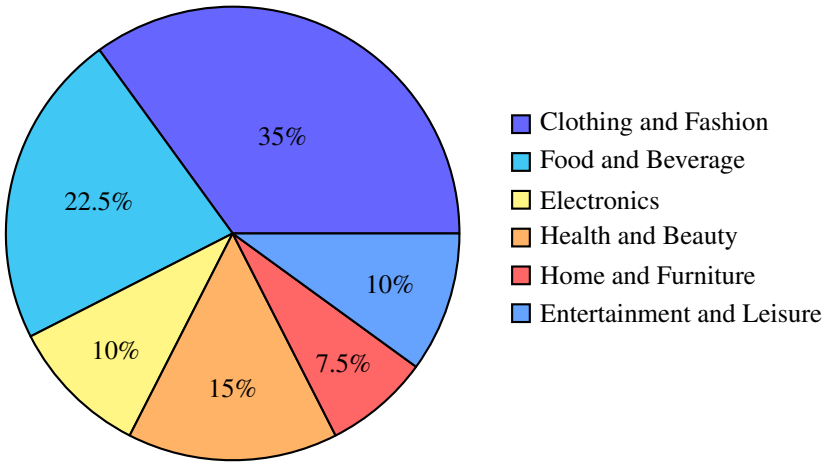
(b) We can display the totals in a bar chart:

Figure 2.3: Frequency of Stores by Type



(c) We can display the relative frequency in a pie chart:

Figure 2.4: Relative Frequency of Stores by Type



2.2 Quantitative Data

Previously, we worked with displaying qualitative data as classes in a frequency table. This means that each class was some qualitative category, like the governorate or the type of store in a mall. In this section, we will study a similar analysis, except this time the classes will be quantitative data.

2.2.1 Frequency Table

Consider a dataset on the sale of cars for the Toyota Showroom in September 2026:

Transaction Number	Sales Value (BD)
1	15,520
2	23,590
3	19,190
4	27,530
5	24,360
6	20,310
⋮	
180	30,380

Sales data can have many observations. It would be useful to create a frequency table to properly summarise this information. Just as before, we need to create a set of groups (classes). However, since we only have quantitative data, we would need to create the class based on an **interval of values**.

Suppose we created the frequency of transactions under these classes:

- 2,000 up to 12,000
- 12,000 up to 24,000
- 24,000 up to 34,000

That is, there are three classes, and each one has a width of 10,000 BD:

Interval Class (BD)	**Frequency**
2,000 up to 12,000	5
12,000 up to 24,000	75
24,000 up to 34,000	100
Total	**180**

This is how we we would display a frequency table with quantitative data classes. There are 5 transactions that are valued in the first class, 75 transactions in the second class, and 100 transactions in the third class.

We arbitrarily selected very few classes to represent the dataset. Depending on the context of your study, you can also have many classes:

Interval Class (BD)	**Frequency**
2,000 up to 4,500	1
4,500 up to 7,000	0
7,000 up to 9,500	1
9,500 up to 12,000	3
12,000 up to 14,500	5
14,500 up to 17,000	10
17,000 up to 19,500	13
19,500 up to 22,000	22
22,000 up to 24,500	35
24,500 up to 27,000	24
27,000 up to 29,500	31
29,500 up to 32,000	29
32,000 up to 34,500	6
34,500 up to 37,000	0
Total	**180**

Now this shows that there are 14 classes, where each interval has a width of 1500 BD.

You should know that when deciding on the number of classes and the choice of interval width, we can follow a rule of thumb to construct the optimal amounts.

2.2.2 Constructing the Optimal Frequency Distribution

You need to decide on **the number of classes** and the **width of the interval** for each class. In order to do this, you need three details about your dataset:

- The total number of observations n.
- The minimum value of your variable in the dataset $\min$.
- The maximum value of your variable in the dataset $\max$.

The Number of Classes

Use the following rule:

$$2^k > n$$

where k is the number of classes and n is the total observations in your dataset.

Keep choosing different values of k until the equality is satisfied.

Width of the Interval

To determine the optimal interval, we use the following rule:

$$i = \frac{\max - \min}{k}$$

$\max$ and $\min$ are the largest and smallest values of the variable in your dataset. We normally round i to the nearest whole number or a convenient multiple of 10, 100, 1000, etc.

Activity 2: Consider the sales data regarding Toyota's showroom. We are given the following details:

- $n = 180$
- $\min = 2,940$ BD
- $\max = 32,920$ BD

Calculate the optimal number of classes k and interval i using the rules described earlier.

Solution:

For the number of classes, we need to satisfy the rule:

$$2^k > n$$

Since there are $n = 180$ observations, we can test for values of k until the equality is just satisfied.

For example, if $k = 7$:

$$2^7 = 128 < 180$$

which does not satisfy the condition. However if we do $k = 8$:

$$2^8 = 256 > 180$$

Therefore, $k = 8$ is the recommended number of classes.

As for the interval width, we need to simply compute:

$$i = \frac{\max - \min}{k}$$

$$i = \frac{39290 - 2940}{8} = 4543.75$$

We can round this to the nearest whole number to get to 4,500.

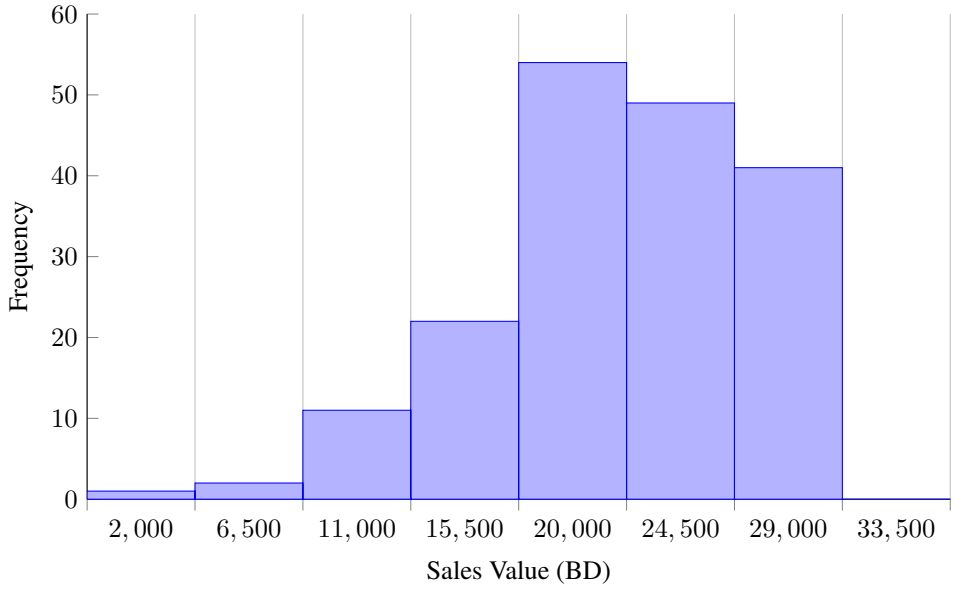
Now that we know how to calculate the optimal number of classes and interval width, we can display this information in a frequency table:

Interval Class (BD)	**Frequency**
2,000 up to 6,500	1
6,500 up to 11,000	2
11,000 up to 15,500	11
15,500 up to 20,000	22
20,000 up to 24,500	54
24,500 up to 29,000	49
29,000 up to 33,500	41
33,500 up to 38,000	0
Total	**180**

2.2.3 Histogram

We can take our optimised frequency table and display it in a **histogram**. This is very similar to a **bar chart**, where the vertical axis is the frequency and the horizontal axis represents the class intervals:

Figure 2.5: Distribution of Car Sales Values



2.2.4 Frequency Polygon

Another way to display quantitative data is through a **frequency polygon**. This is when the vertical axis has the frequency, but the horizontal axis is represented by the **midpoint** of every interval.

Midpoint of an Interval

To calculate the midpoint, simply apply the below formula for each interval class:

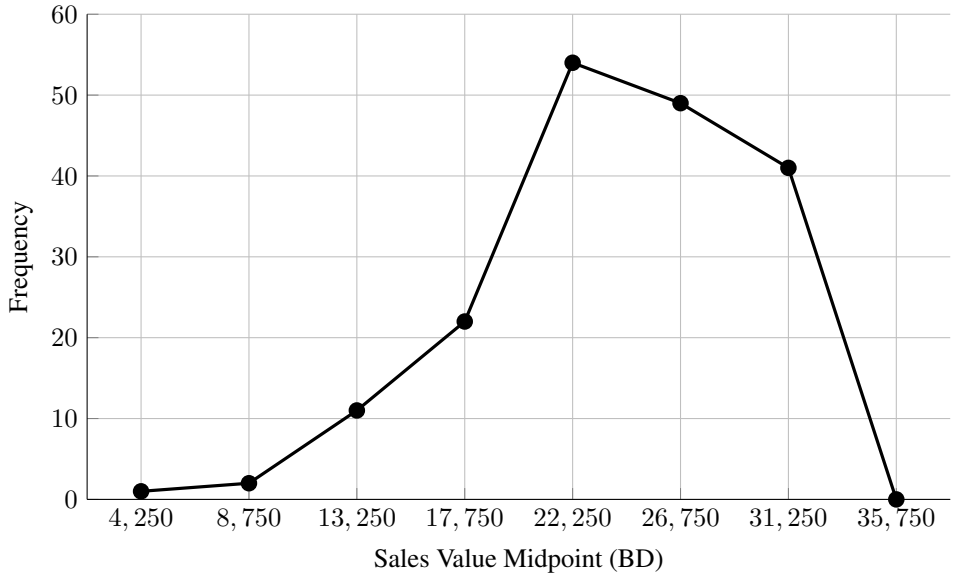
$$\frac{\text{Lower Bound + Upper Bound}}{2}$$

Using the car sale data:

Interval Class (BD)	Midpoint (BD)	Frequency
2,000 up to 6,500	4,250	1
6,500 up to 11,000	8,750	2
11,000 up to 15,500	13,250	11
15,500 up to 20,000	17,750	22
20,000 up to 24,500	22,250	54
24,500 up to 29,000	26,750	49
29,000 up to 33,500	31,250	41
33,500 up to 38,000	35,750	0

The frequency polygon can therefore be plotted:

Figure 2.6: Frequency Polygon of Car Sales Values



2.2.5 Relative Frequency

Similar to the previous section, we can also construct relative frequency distributions. Remember that the way to create this is take the total count of each class and divide it by the total number across all classes:

$$\text{Relative Frequency} = \frac{\text{Frequency}}{\text{Total Frequency}}$$

Interval Class (BD)	Frequency	Calculation	Relative Frequency
2,000 up to 6,500	1	$\frac{1}{180}$	0.006
6,500 up to 11,000	2	$\frac{2}{180}$	0.011
11,000 up to 15,500	11	$\frac{11}{180}$	0.061
15,500 up to 20,000	22	$\frac{22}{180}$	0.122
20,000 up to 24,500	54	$\frac{54}{180}$	0.300
24,500 up to 29,000	49	$\frac{49}{180}$	0.272
29,000 up to 33,500	41	$\frac{41}{180}$	0.228
33,500 up to 38,000	0	$\frac{0}{180}$	0.000
Total	**180**	$\frac{180}{180}$	**1.000**

2.2.6 Cumulative Frequency

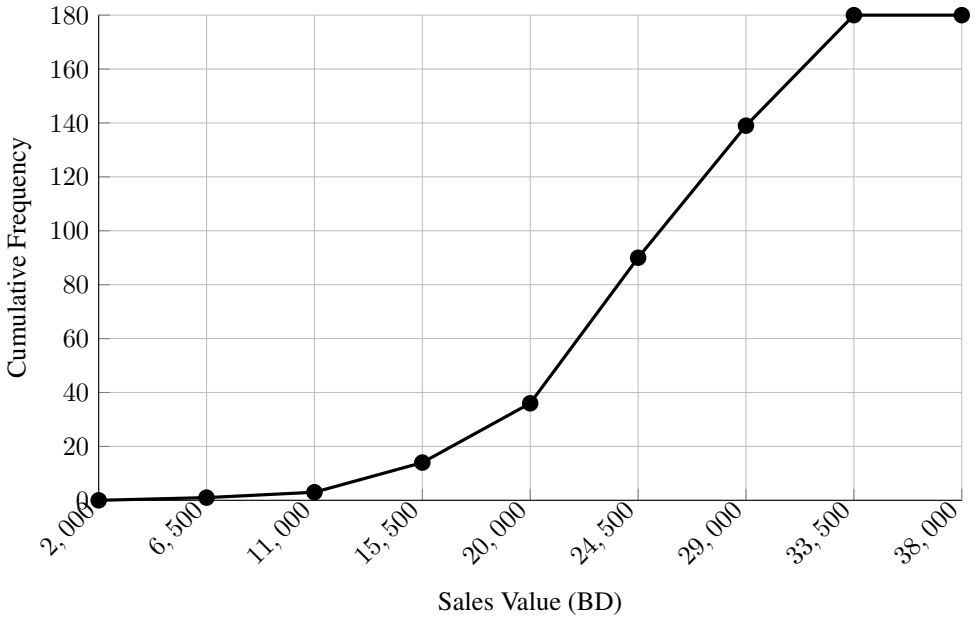
A cumulative frequency graph is one way of illustrating quantitative data across classes. Consider again the frequency graph for Toyota car sales:

Interval Class (BD)	Frequency
2,000 up to 6,500	1
6,500 up to 11,000	2
11,000 up to 15,500	11
15,500 up to 20,000	22
20,000 up to 24,500	54
24,500 up to 29,000	49
29,000 up to 33,500	41
33,500 up to 38,000	0
Total	**180**

The cumulative frequency is simply the addition of each row:

Interval Class (BD)	Frequency	Calculation	Cumulative Frequency
$> 6,500$	1	1	1
$> 11,000$	2	$1 + 2$	3
$> 15,500$	11	$3 + 11$	14
$> 20,000$	22	$14 + 22$	36
$> 24,500$	54	$36 + 54$	90
$> 29,000$	49	$90 + 49$	139
$> 33,500$	41	$139 + 41$	180

We can display this in a graph where the cumulative frequency is on the vertical axis and the class in on the horizontal axis:



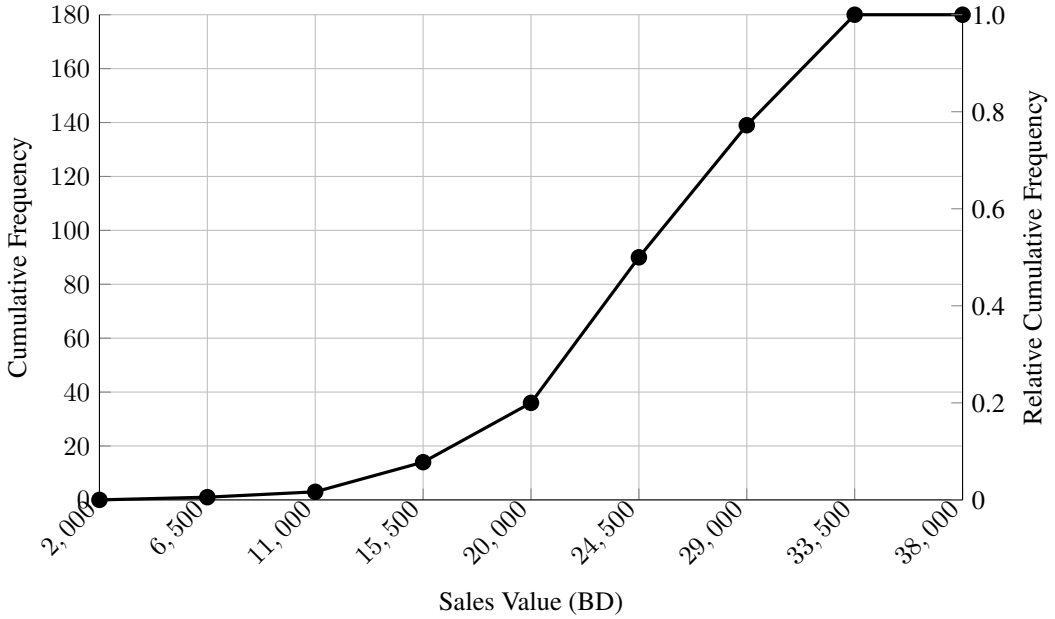
You can also gather the cumulative relative frequency to asses the contribution of each class.

$$\text{Relative Cumulative Frequency} = \frac{\text{Cumulative Frequency}}{\text{Total Frequency}}$$

Interval (BD)	**Cumulative Frequency**	**Calculation**	**Relative Cumulative Frequency**
$< 6,500$	1	$\frac{1}{180}$	0.006
$< 11,000$	3	$\frac{3}{180}$	0.017
$< 15,500$	14	$\frac{14}{180}$	0.078
$< 20,000$	36	$\frac{36}{180}$	0.200
$< 24,500$	90	$\frac{90}{180}$	0.500
$< 29,000$	139	$\frac{139}{180}$	0.772
$< 33,500$	180	$\frac{180}{180}$	1.000

An illustration in which we have the cumulative frequency and the relative cumulative frequency is called an Ogive:

Figure 2.7: Cumulative Frequency Distribution of Car Sales Values



Chapter 3

Describing Data: Numerical Measures

In this chapter, we are interested in the various methods a researcher needs to know to extract basic details of their dataset. We will study about the mean, median, mode, range, variance and the standard deviation.

3.1 Measures of Location

Consider a **sample** of Bahraini public schools, in which you have acquired the average math grade of each school:

School i	**Average Mathematics Grade** x	**Governorate**
School 1	87	Capital
School 2	58	Muharraq
School 3	46	Muharraq
School 4	48	Southern
School 5	80	Capital
School 6	67	Northern
School 7	75	Capital
School 8	67	Capital
School 9	82	Southern

We are interested in various ways to calculate the **measures of location**.

3.1.1 Mean

In order to calculate the mean of the **sample**, we apply the formula:

$$\bar{x} = \frac{\sum x_i}{n}$$

where n is the number of observations and $\sum x_i$ is the addition of all grades in your dataset.

Following from this, we calculate the result as:

$$\bar{x} = \frac{87 + 58 + 46 + 48 + 80 + 67 + 75 + 67 + 82}{9} = 67.7$$

The mean is also called the average of your variable.

Notice that when calculating the mean, we are using all the data values to come up with the final result. Although you should keep in mind that the mean is weak when the dataset has extreme values.

Before leaving this section, you should know that we are dealing with a **sample** of our dataset, the way we denote the average is by using $\bar{x}$. However, when our dataset is the **population**, then we simply denote it in a different way:

$$\mu = \frac{\sum x_i}{N}$$

The only difference here is that we use the Greek letter μ instead of $\bar{x}$ and that we denote the number of observations with capital N rather than small n.

> **Activity:**
>
> The *school_math.xlsx* file contains average math scores for all the schools in the country. Use this data to compute the population average.
>
> Note: the excel function to use is =AVERAGE().

Solution:

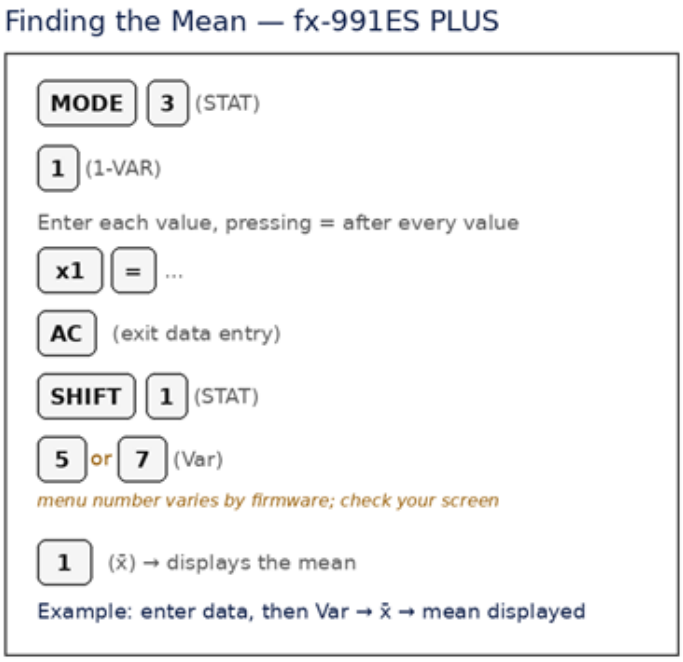
On excel, click on any empty cell and type: =AVERAGE().

In the brackets, simply select the entire column of the variable AvgMath.

For $N = 800$, we compute the population mean as:

$$\mu = 71$$

You should know that you can use your calculator to compute the mean using the following commands:



3.1.2 Median

The median is simply the midpoint of the values in the dataset. In order to do this, you would need to arrange the numbers from lowest to highest.

Initially, the dataset shows the following values:

$$87 + 58 + 46 + 48 + 80 + 67 + 75 + 67 + 82$$

Now we re-arrange it from lowest to highest:

$$46 + 48 + 58 + 67 + 67 + 75 + 80 + 82 + 87$$

The median is the value that is in the middle of the dataset. Since there is 9 data points, then the 5th data point is the median, which is 67.

The median is a useful measure because it is not impacted by extreme outliers. Essentially, 50% of the data is above and below the median.

Notice in the current example, there are 9 data points. In the event that we have an odd number of data points, then we simply select the middle data value. However, if it is an even number of values, then we need to take the average of the middle values.

> **Activity:** Consider the below sample of car sales for the Toyota showroom:
>
> $$3000 \quad 3500 \quad 2100 \quad 9400 \quad 6400 \quad 7450$$
>
> Find the median of the sample values.
>
> **Solution:**
>
> We first arrange the numbers in order:
>
> $$2100 \quad 3000 \quad 3500 \quad 6400 \quad 7450 \quad 9400$$
>
> Since there are 6 data points, which is an even number, then we need to take the 3rd and 4th data point and take its

> average:
>
> $$\frac{3500 + 6400}{2} = 4950$$
>
> This demonstrates how to get the median for an even numbered sample.

3.1.3 Mode

The mode of a dataset is simply the value that occurs most frequently.

Using our math data from Bahraini schools:

$$87 + 58 + 46 + 48 + 80 + 67 + 75 + 67 + 82$$

In this case, 67 is the mode of this dataset. Note that it is certainly possible to have more than one mode in a dataset or no mode at all.

3.1.4 Weighted Mean

We can mix the approach of the mean and the mode by using the weighted mean. In this case, we take the average of the sample values but we weight it based on how many observations have the same value.

The weighted mean is calculated as:

$$\bar{X}_W = \frac{\sum xw}{w} = \frac{w_1x_1 + w_2x_2 + ... + w_nx_n}{w}$$

where x_i is the data value while w_i is the amount of times the value appears in the dataset.

> **Activity:** Akhbar Al Khaleej is a newspaper company and printing press in Bahrain. They have 26 factory workers and they are paid the following hourly wages:

- 14 employees are paid 16.50 BD per hour.
- 10 employees are paid 19.00 BD per hour.
- 2 employees are paid 25.00 BD per hour.

Calculate the weighted mean.

Solution:

We need to weigh the data values based on how often they appear in the dataset. In this case:

$$\bar{X}_W = \frac{14(16.5) + 10(19) + 2(25)}{14 + 10 + 2} = \frac{471}{26} = 18.12$$

3.1.5 Geometric Mean

Consider the GDP of Bahrain for the past 5 years:

Year	GDP Growth	GDP Growth Ratio
2021	14.0%	0.14
2022	13.7%	0.137
2023	-0.5%	-0.005
2024	2.2%	0.022
2025	3.7%	0.037

Let us say we are interested in Bahrain's average GDP growth over these years. It would simply be:

$$\begin{aligned}\bar{x} &= \frac{0.14 + 0.137 - 0.005 + 0.022 + 0.037}{5} \\ &= \frac{0.330}{5} \\ &= 0.066\end{aligned}$$

which means Bahrain's economy grew on average by 6.6%.

One thing to consider is that Bahrain's GDP is likely correlated across the years. This means that if GDP was high in one year, then it might lead it to be high next year as well. If it is low in one year, then it might mean that it will be low

in the next year.

In a situation in which a variable is compounded by its past values, then it is better to use the geometric mean to better approximate its average growth.

To compute the geometric mean, we need to follow these steps:

(1) Add 1 to each growth ratio.

Year	**GDP Growth**	**GDP Growth Ratio** g	**Growth Factor** $(1+g)$
2021	14.0%	0.139	1.139
2022	13.7%	0.137	1.137
2023	-0.5%	-0.005	0.995
2024	2.2%	0.022	1.022
2025	3.7%	0.037	1.037

(2) Compute the below calculation

$$= \sqrt[n]{x_1 \cdot x_2 \cdot x_3 \cdot ... \cdot x_n}$$

where x_i is the data values and n is the number of observations. In this case:

$$= \sqrt[5]{1.139 \cdot 1.137 \cdot 0.995 \cdot 1.022 \cdot 1.037}$$

$$= \sqrt[5]{1.366}$$

$$= 1.0643$$

(3) Subtract this value by 1 and convert it into a percentage.

$$= 1.0643 - 1$$

$$= 0.0643$$

$$= 6.43\%$$

The interpretation is that Bahrain's average annual growth rate over the years was 6.43%. We would consider this to be more accurate than the simple mean since it correctly accounts for the compounding over the years.

Example:

Consider the sales growth of Saudi Aramco's SABIC facility:

Year	Sales Growth
2001	30%
2002	20%
2003	-40%
2004	200%

Calculate the geometric mean using the steps specified in the section.

Solution:

We start by computing the growth ratio and then use that to add 1 to each observation:

Year	Sales Growth	Growth Ratio g	**Growth Factor** $(1+g)$
2001	30%	0.30	1.30
2002	20%	0.20	1.20
2003	-40%	-0.40	0.60
2004	200%	2.00	3.00

We then compute the result:

$$= \sqrt[4]{1.30 \times 1.20 \times 0.60 \times 3.00}$$

$$= \sqrt[4]{2.808}$$

$$= 1.2946$$

Where we the subtract by 1 and convert it to percentage:

$$= 1.2946 - 1$$

$$= 0.2946$$

$$= 29.46\%$$

3.2 Measures of Dispersion

In statistics, we are often interested in how the data values are spread over the sample. Measures of dispersion therefore allows us to better understand sample's distribution.

3.2.1 Range

Consider the below sample of employees from a factory producing aluminium products in Bahrain and from the UAE:

Bahrain

i	**Hourly Production**
1	48
2	49
3	49
4	50
5	50
6	50
7	51
8	51
9	52

UAE

i	**Hourly Production**
1	40
2	47
3	47
4	50
5	50
6	50
7	53
8	53
9	60

We are interested in calculating the range for both these samples. This is simply:

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

For both factories, we get:

$$\text{BAH} = 52 - 48 = 4$$

$$\text{UAE} = 60 - 40 = 20$$

An interesting thing to keep in mind is that if you took the average production of both factories, you will get a value of 50 for both:

$$\begin{aligned}\bar{x}_{\text{Bahrain}} &= \frac{48 + 49 + 49 + 50 + 50 + 50 + 51 + 51 + 52}{9} \\ &= \frac{450}{9} \\ &= 50\end{aligned}$$

$$\begin{aligned}\bar{x}_{\text{UAE}} &= \frac{40 + 47 + 47 + 50 + 50 + 50 + 53 + 53 + 60}{9} \\ &= \frac{450}{9} \\ &= 50\end{aligned}$$

However, there is a larger **spread** for the UAE factory compared to the Bahraini factory. This is why we are interested in measuring dispersion as it tells us a different story to the data.

3.2.2 Variance and Standard Deviation

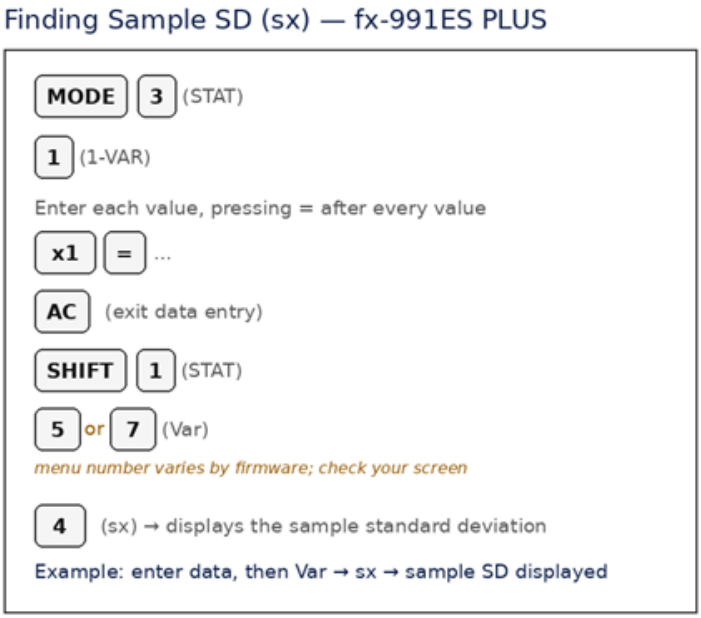
Another method to determine the dispersion is by computing the variance of a sample. This is found by using the below formula:

$$s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}$$

We can also use this to compute the standard deviation, which is simply the square root of the variance:

$$S = \sqrt{s}$$

You can also use your calculator to calculate the sample standard deviation:



> **Activity:** Consider again our example with hourly production of aluminium factories:

Bahrain

i	**Hourly Production**
1	48
2	49
3	49
4	50
5	50
6	50
7	51
8	51
9	52

UAE

i	**Hourly Production**
1	40
2	47
3	47
4	50
5	50
6	50
7	53
8	53
9	60

We know that the average of both factories is the same:

$$\bar{x}_{\text{BAH}} = \bar{x}_{\text{UAE}} = 50$$

Calculate the variance and standard deviation for both factories.

Solution:

Following from our method:

$$\begin{aligned} s^2_{\text{BAH}} &= \frac{(48-50)^2 + (49-50)^2 + (49-50)^2 + (50-50)^2 + (50-50)^2 + (50-50)^2}{9} \\ &\quad + \frac{(51-50)^2 + (51-50)^2 + (52-50)^2}{9} \\ &= \frac{4+1+1+0+0+0+1+1+4}{9} \\ &= \frac{12}{9} \\ &= 1.33 \end{aligned}$$

$$
\begin{aligned}
s^2_{\text{UAE}} &= \frac{(40-50)^2+(47-50)^2+(47-50)^2+(50-50)^2+(50-50)^2+(50-50)^2}{9} \\
&\quad + \frac{(53-50)^2+(53-50)^2+(60-50)^2}{9} \\
&= \frac{100+9+9+0+0+0+9+9+100}{9} \\
&= \frac{236}{9} \\
&= 26.22
\end{aligned}
$$

Similar to what we saw from the range, the Bahraini factory has a much smaller dispersion/variance compared to the UAE factory.

The standard deviation is simply the square root of the variance:

$$S_{\text{BAH}} = s^2_{\text{BAH}} = \sqrt{1.33} = 1.15$$

$$S_{\text{UAE}} = s^2_{\text{UAE}} = \sqrt{26.22} = 5.12$$

You should know that the smaller the standard deviation, the closer the data is to its mean, whereas the larger it is, the further away it is from the mean.

Just like the sample mean, you need to know that there is a different way to denote the **population** variance and standard deviation.

The formula for the population variance is slightly different:

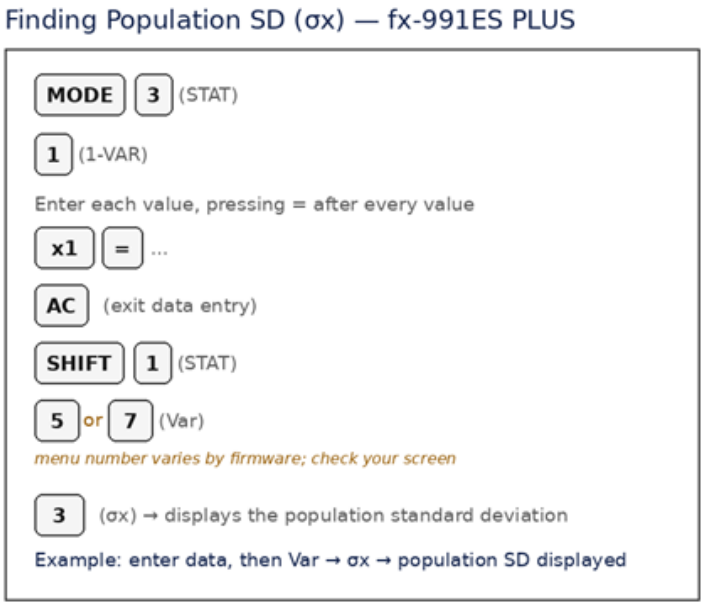
$$\sigma^2 = \frac{\sum(x_i - \mu)^2}{N}$$

where σ is a Greek letter, but most crucially we divide it by all observations N instead of $n-1$.

The population standard deviation is the same as before, except we denote it as:

$$\sigma = \sqrt{\sigma^2}$$

You can also use your calculator to calculate the population standard deviation:



Activity:

Use the *school_math.xlsx* data to calculate the variance and standard deviation of the population of schools.

On excel, the function for population variance is =VAR.P() and the standard deviation is =STDEV.P().

Solution:

Similar to the previous example, select an empty cell in the excel sheet and input the above command. Include all the data within those brackets by selecting the column.

$$\sigma^2 = 99$$

The standard deviation is simply the square root:

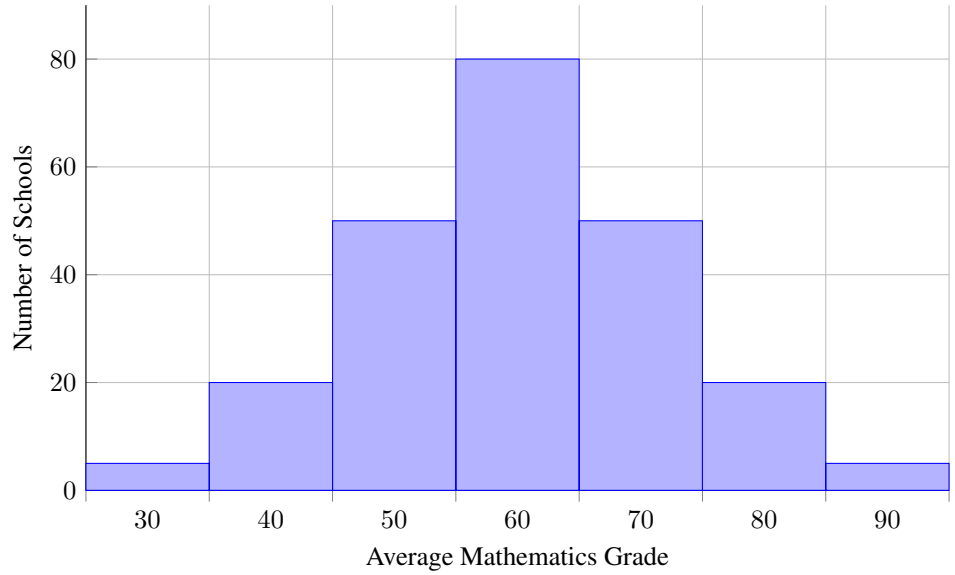
$$\sigma = \sqrt{99} = 9.95$$

3.3 The Distribution of the Data

3.3.1 The Normal Distribution

Consider again our data on average math scores. We can plot the data on a histogram:

Figure 3.1: Distribution of Average Mathematics Grades Across Schools

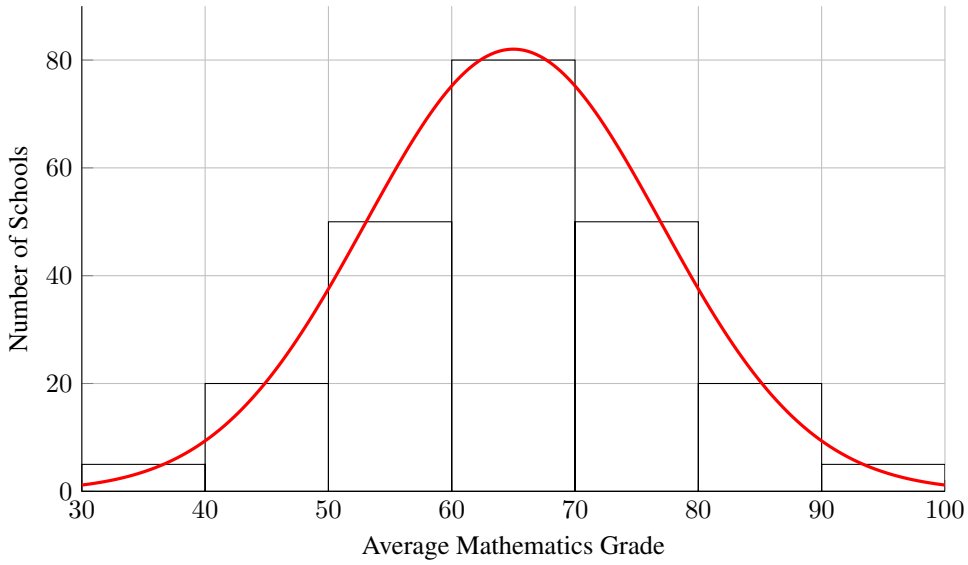


What this shows is that the largest frequency of observations is centred in the middle. That is, most schools have an average score that is between 60 and 70[a]. Schools that have extremely low and extremely high average grades are very few in our observations, which is what we would expect for this dataset.

Given the shape of the distribution, we can try and draw (fit) a line around this curvature:

[a] This is basically what we saw when we computed the mean, where $\bar{x} = 67$.

Figure 3.2: Distribution of Average Mathematics Grades Across Schools

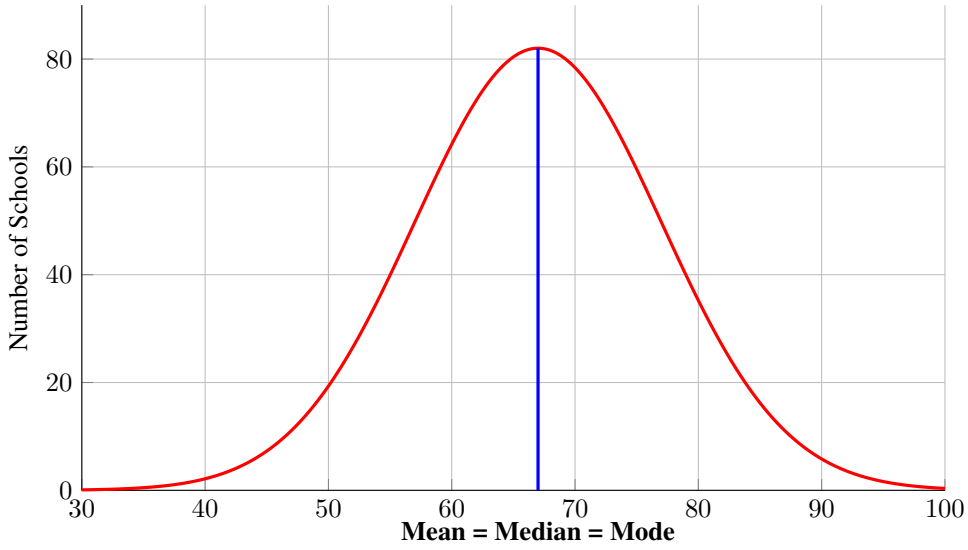


This bell-shaped curve is known as the **normal distribution**. An interesting fact about this fitted line is that the mean, median and mode all happen to be centred around the middle of the distribution.

Recall that when we conducted the measures of location for this dataset, we got:

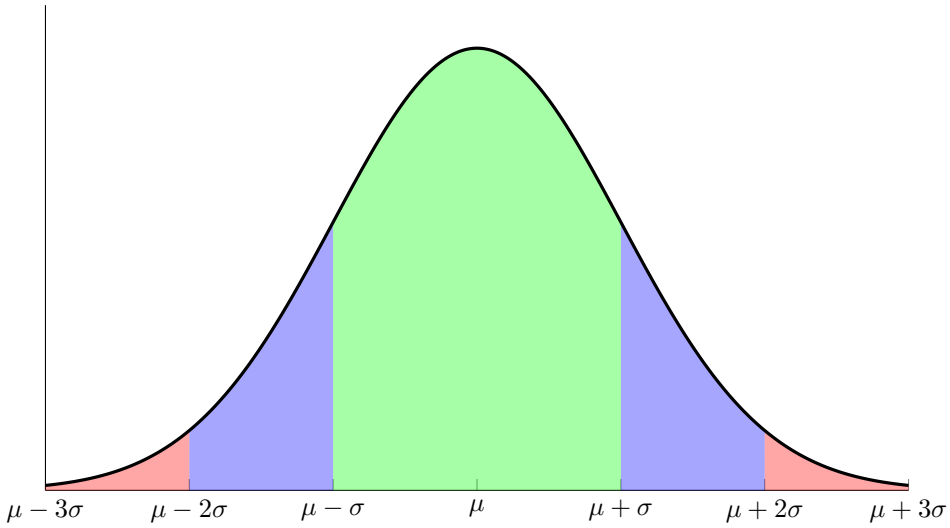
- Mean = 67.7
- Median = 67
- Mode = 67

which is basically what we are seeing in the graph:



When we are dealing with a normal distribution, it is worth keeping in mind how much of the observations are located around the distribution. This is more easily understood when considering the standard deviation.

Figure 3.3: The Empirical Rule for a Normal Distribution



This illustration shows the following fact:

- The green region represents 68% of all your observations, which is the distance between $\mu - \sigma$ and $\mu + \sigma$.
- Including the blue region represents 95% of all your observations, which is the distance between $\mu - 2\sigma$ and $\mu + 2\sigma$.
- Including the red region represents 99.7% of all your observations, which is the distance between $\mu - 3\sigma$ and $\mu + 3\sigma$.

Activity

The average mathematics grades across schools are approximately symmetric and bell-shaped. The mean mathematics grade is 67, and the standard deviation is 10.

Using the Empirical Rule, find the range of average mathematics grades that contains approximately:

- 68% of schools.
- 95% of schools.
- 99% of schools.

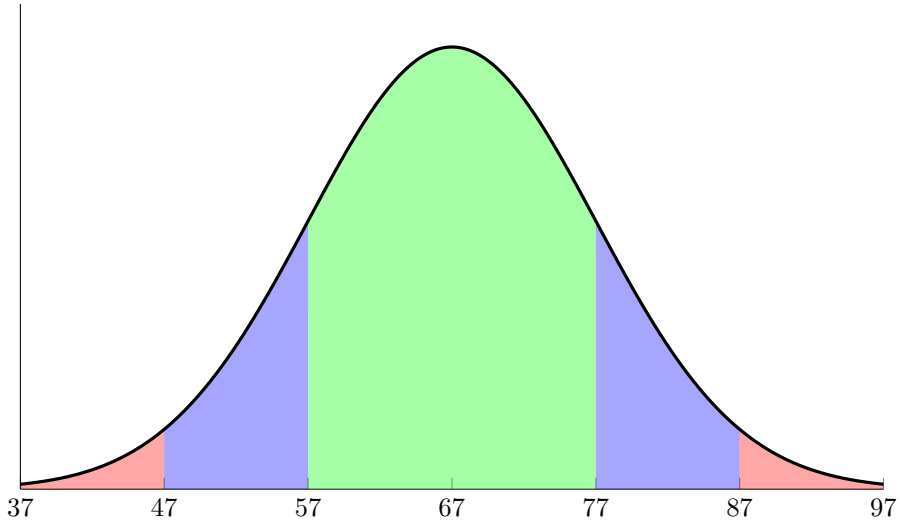
Illustrate the result in a plot with a normal distribution.

Solution:

$$\begin{aligned} 68\%: &\quad 67 \pm 1(10) = 57 \text{ to } 77 \\ 95\%: &\quad 67 \pm 2(10) = 47 \text{ to } 87 \\ 99.7\%: &\quad 67 \pm 3(10) = 37 \text{ to } 97 \end{aligned}$$

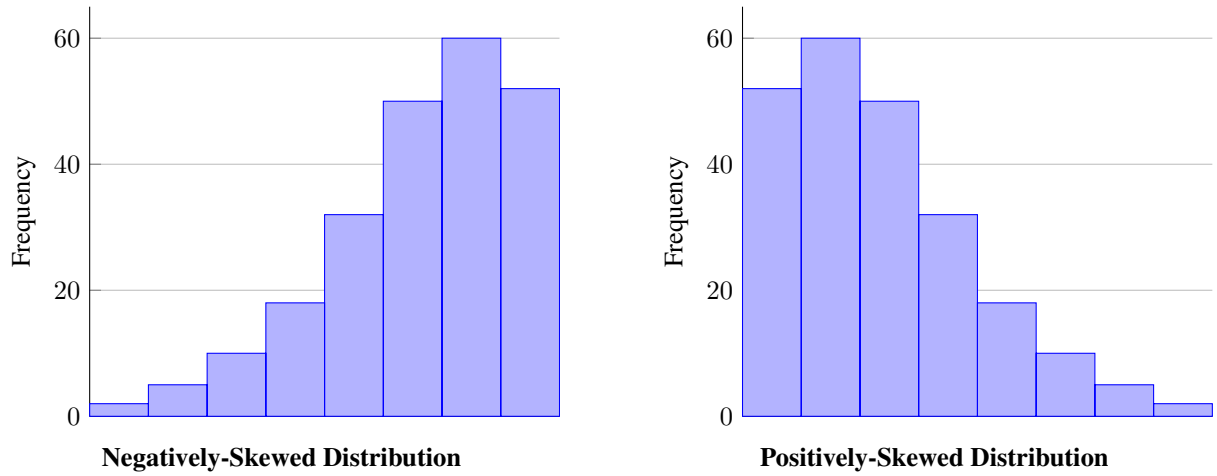
The corresponding illustration:

Figure 3.4: The Empirical Rule for a Normal Distribution of Math Scores

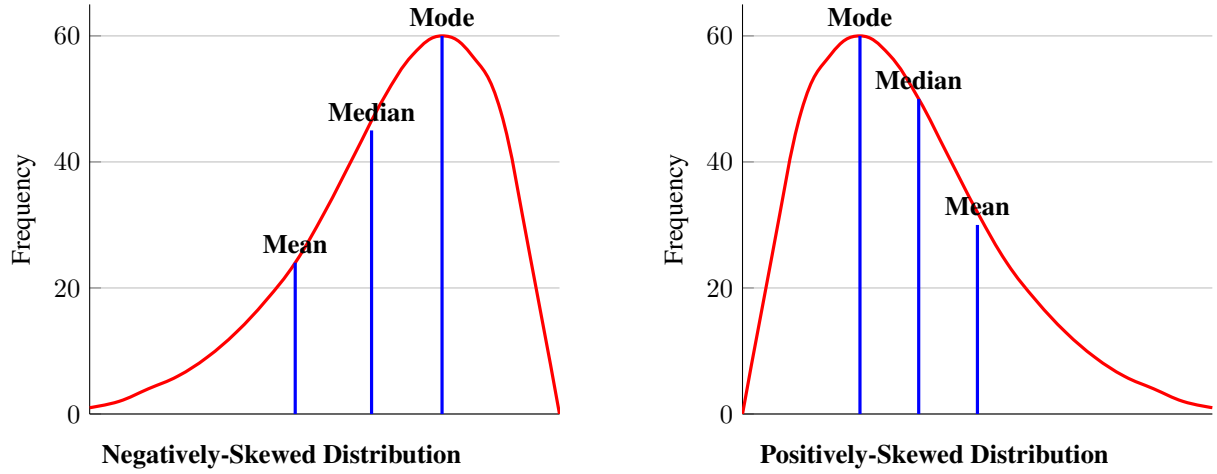


3.3.2 Skewness

Not all data is normally distributed. Sometimes, data can be skewed negatively or positively:



However do note that in these scenarios, the measures of location are not equal:



Clearly, when the skewness is positive, we can see that the mean ¿ median. When the skewness is negative, the median ¿ mean.

The problem is that when the distribution is not normal, we cannot use the empirical rule to determine where the percentage of observations are located[b].

In this scenario, it is possible to use the **Chebyshev's Theorem** to determine the proportion of values within some k standard deviations around the mean.

The Chebyshev Theorem is simply:

[b] That is, when we have a non-normal distribution, we cannot say that 68% of observations are around 1 standard deviation from the mean, and 95% of observations are 2 standard deviations from the mean, etc.

$$1 - \frac{1}{k^2}$$

where k is the standard deviation. By inserting any value of $k > 1$, you will get the approximate percentage amount of observations that is included in that standard deviation.

Consider our negatively-skewed distribution (which is clearly not normally distributed):

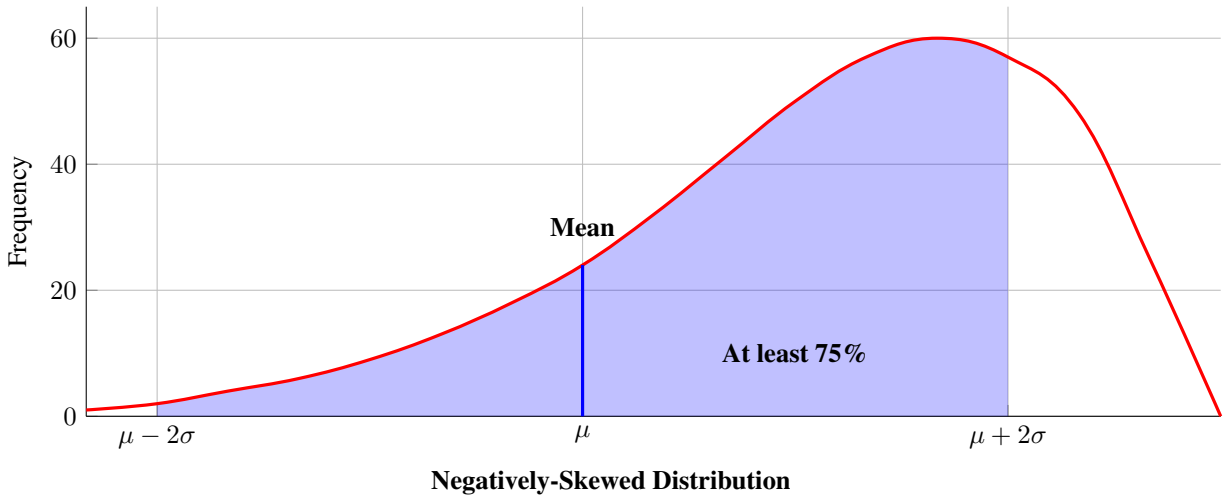


Negatively-Skewed Distribution

Let us saying we are interested to know the amount of observations that are located in 2 standard deviations around the mean. This means $k = 2$:

$$1 - \frac{1}{2^2} = 0.75$$

which means that 75% of the data lies between 2 standard deviations of the distribution.



Negatively-Skewed Distribution

Activity:

Determine the amount of observations located around 3 standard deviations from the mean assuming a non-normal distribution.

Solution:

When the distribution is not normal, we use Chebshev's theorem. Since we want to know how many observations are located around 3 standard deviations from the mean, we set $k = 3$:

$$1 - \frac{1}{3^2} = 0.889$$

which means 88.9% of the data is located within 3 standard deviations.

3.4 Ethics in Reporting Results

Statistical analysis involves making choices about how data are summarised and presented. These choices can influence how readers interpret the results. It is therefore important that statistics are reported in an **objective, transparent, and honest** manner.

Different measures of central tendency emphasise different features of the data. For example, the **mean** is affected by extreme observations, whereas the **median** is more resistant to them. The measure that **best represents the characteristics of the data** should be reported, rather than selecting a measure simply because it produces a more favourable

result.

The same principle applies to **measures of dispersion**. Different measures of spread may communicate different information about the variability of the data.

Where appropriate, reporting multiple descriptive statistics can provide a more complete and transparent representation of the data. Statistical results should be communicated clearly and should not deliberately hide inconvenient observations, exaggerate patterns, or create a misleading impression.

Statistical reporting should therefore provide an accurate representation of the data, even when the results do not support the conclusion that the researcher, business, or organisation might prefer.

3.5 Problem Set

1) There are a total of 46 bus stops in Bahrain. The distances between each consecutive pair of stops, measured in hundreds of metres, are given below.

11	4	10	4	9	3	8	10	3	14	1	6
4	3	5	2	2	5	4	2	3	2	5	5
1	1	2	7	8	10	2	3	7	5	4	3
2	1	1	2	1	1	2	1	2	1		

(a) Why can these observations be considered a **population**?

(b) What is the **mean distance** between consecutive bus stops?

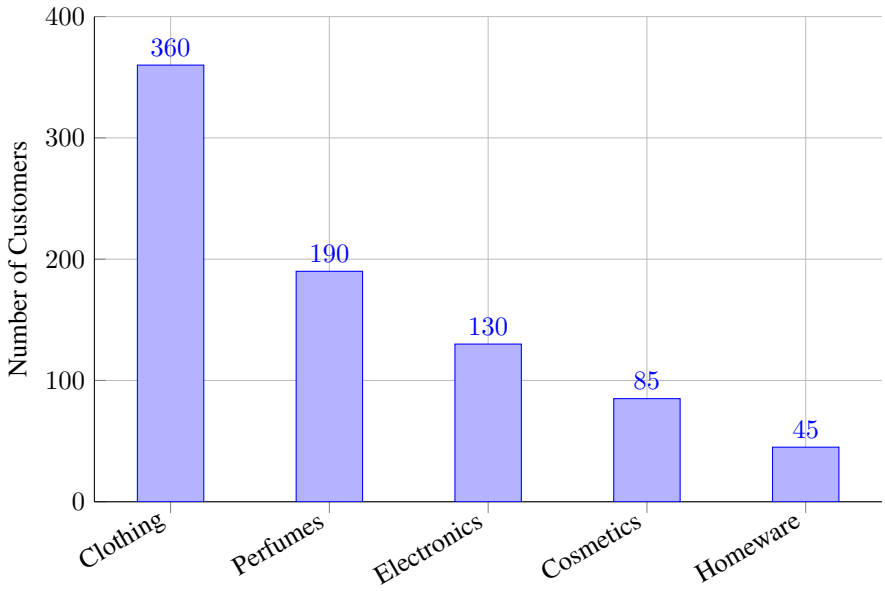
2) The number of hours per day that a sample of STC customers use their phones is shown below.

4.1	3.7	4.3	4.2	5.5	5.1
4.2	5.1	4.2	4.6	5.2	3.8

What is the arithmetic mean number of hours per day used? Why is this not considered a population dataset?

3) Bahrain City Center surveyed customers during a promotional event and recorded the main type of product purchased.

The results are shown below.



Which category represents the mode of the dataset?

4) The General Directorate of Traffic in Bahrain records all the number of traffic citations issued during each month of the year. The monthly figures are shown below.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Citations	19	17	22	18	28	34	45	39	38	44	34	10

Calculate the **population variance** σ^2 of the number of monthly traffic citations.

5) A sample of five part-time employees working in Akhbar Al Khaleej was selected. Their hourly wages, measured in Bahraini Dinars (BD), are shown below.

12	20	16	18	19

Calculate the **sample variance** s^2 and the **sample standard deviation** s of the hourly wages.

3.6 Solution

1) (a) The observations are considered the population of bus stop data because the question explicitly states that it includes all 46 bus stops in Bahrain. It would be considered a sample if the question had said "we have data on sample of 10 bus stops in Bahrain...".

(b) The population mean is computed as:

$$\begin{aligned}\mu &= \frac{\sum_{i=1}^{N} x_i}{N} \\ &= \frac{192}{46} \\ &\approx 4.17\end{aligned}$$

2) Remember that this is a sample dataset, so we use $\bar{x}$ rather than μ:

$$\begin{aligned}\bar{x} &= \frac{\sum_{i=1}^{n} x_i}{n} \\ &= \frac{4.1 + 3.7 + 4.3 + 4.2 + 5.5 + 5.1 + 4.2 + 5.1 + 4.2 + 4.6 + 5.2 + 3.8}{12} \\ &= \frac{54}{12} \\ &= 4.5\end{aligned}$$

In order to become a population dataset, STC would need to measure the hourly usage of mobile phones for *all* their customers, not just a sample of 12 people.

3) The clothing store represents the mode since it has the highest frequency. That is, it is the category that has the highest repetition in our dataset.

4) We first calculate the population mean:

$$\begin{aligned}
\mu &= \frac{\sum_{i=1}^{N} x_i}{N} \\
&= \frac{19 + 17 + 22 + 18 + 28 + 34 + 45 + 39 + 38 + 44 + 34 + 10}{12} \\
&= \frac{348}{12} \\
&= 29
\end{aligned}$$

Then we compute the population variance:

$$\begin{aligned}
\sigma^2 &= \frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N} \\
&= \frac{(19-29)^2 + (17-29)^2 + (22-29)^2 + (18-29)^2}{12} \\
&\quad + \frac{(28-29)^2 + (34-29)^2 + (45-29)^2 + (39-29)^2}{12} \\
&\quad + \frac{(38-29)^2 + (44-29)^2 + (34-29)^2 + (10-29)^2}{12} \\
&= \frac{100 + 144 + 49 + 121 + 1 + 25 + 256 + 100 + 81 + 225 + 25 + 361}{12} \\
&= \frac{1488}{12} \\
&= 124
\end{aligned}$$

5) We first compute the sample average:

$$\begin{aligned}\bar{x} &= \frac{\sum_{i=1}^{n} x_i}{n} \\ &= \frac{12 + 20 + 16 + 18 + 19}{5} \\ &= \frac{85}{5} \\ &= 17\end{aligned}$$

Then we compute the sample variance:

$$\begin{aligned}s^2 &= \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1} \\ &= \frac{(12-17)^2 + (20-17)^2 + (16-17)^2 + (18-17)^2 + (19-17)^2}{5-1} \\ &= \frac{25 + 9 + 1 + 1 + 4}{4} \\ &= \frac{40}{4} \\ &= 10\end{aligned}$$

The sample standard deviation is therefore:

$$\begin{aligned}s &= \sqrt{s^2} \\ &= \sqrt{10} \\ &= 3.16\end{aligned}$$

Chapter 4

Describing Data: Displaying and Exploring Data

4.1 Introduction

In this chapter, we are interested in illustrating our numerical ways of measuring a dataset. We will learn about several visualisation methods, such as the dot plot, box plot, scatter diagrams and the correlation coefficient.

4.2 Dot Plot

Consider the number of cars serviced by a Toyota and Lexus showroom for each working day of September 2026:

Toyota Showroom

Day	Cars Serviced
1	23
2	33
3	27
4	28
5	39
6	26
⋮	
24	30

Lexus Showroom

Day	Cars Serviced
1	31
2	35
3	44
4	36
5	34
6	37
⋮	
24	33

We are interested in illustrating this data in a way that highlights its distribution. One way to do this is by using a **dot plot**.

This is constructed by placing each observation on a horizontal number line. A dot is placed above the corresponding value for each observation in the dataset. If a value occurs more than once, the dots are stacked vertically, so the height of each stack represents the frequency of that value.

For the Toyota and Lexus showroom:

Figure 4.1: Toyota Showroom

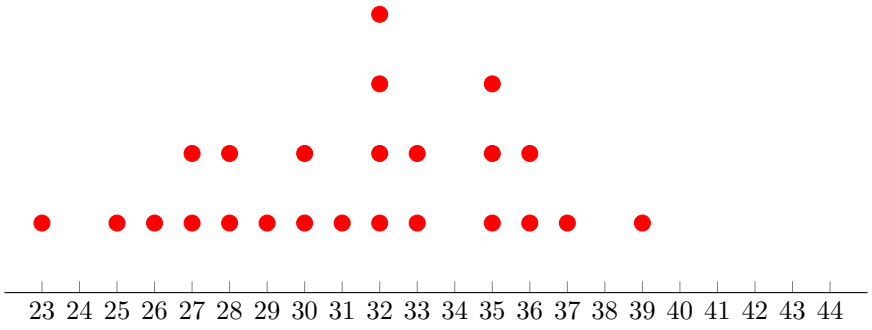
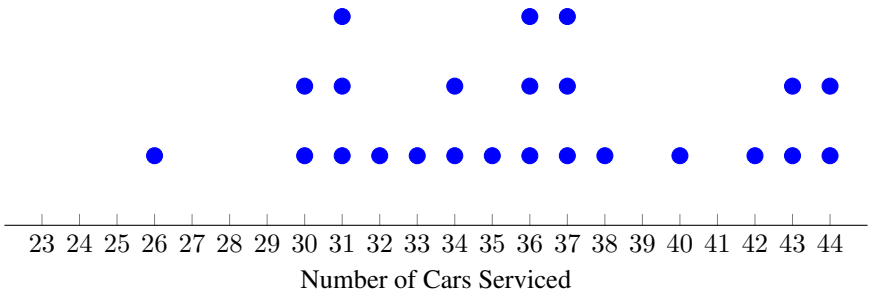


Figure 4.2: Lexus Showroom



The dot plots suggest that the Lexus showroom generally services more cars per day than the Toyota showroom, as its distribution is centred further to the right. The Lexus data is also somewhat more dispersed, indicating greater day-to-day variation in the number of cars serviced.

We can also analyse this numerically using some descriptive statistics:

Table 4.1: Descriptive Statistics

Showroom	n	**Mean**	**Variance**	**Standard Deviation**	**Median**	**Min**	**Max**
Toyota	24	31.29	16.91	4.11	32	23	39
Lexus	24	35.83	24.58	4.96	36	26	44

The visualisation is supported by the numerical descriptives. The Lexus showroom has a larger average as well as a wider variance and standard deviation.

4.3 Measures of Position

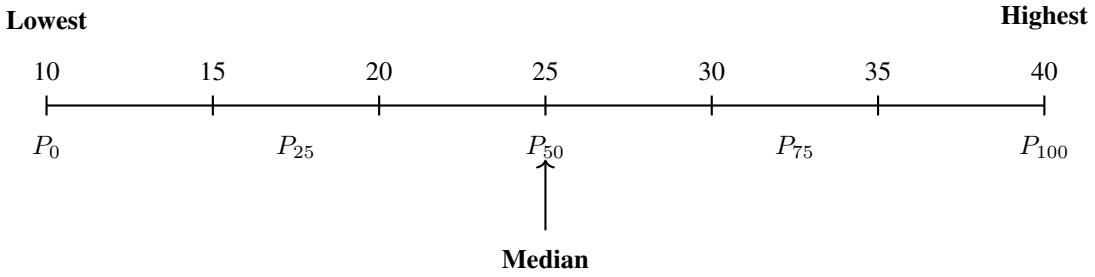
Consider a small ordered dataset:

$$10, \quad 15, \quad 20, \quad 25, \quad 30, \quad 35, \quad 40$$

Suppose we are interested in finding the median. The data is already arranged from lowest to highest and there are 7 observations. The median is therefore 25.

The median describes a position within the distribution. More specifically, we say that the median is **the 50th percentile**. This is because from its position, approximately 50% of the observations lie below it.

So if the median identifies the point with 50% of the data, then we can generalise this to other percentiles, such as 25% or 75% of our data.



More generally, the kth percentile P_k is the data value that indicates the position below which approximately $k\%$ of observations lie below it. For example, the data value of 40 is at the 100th percentile. This means 100% of observations are at or below it.

To identify the location L of the kth percentile p, we use the below formula:

$$L_p = \frac{k}{100}(n+1)$$

Activity:

You have a dataset of salespeople from the Toyota showroom and you record their commission received from their car sales:

Table 4.2: Salesperson Commissions from the Toyota Showroom

Salesperson	Commission (BD)
1	1,460
2	1,471
3	1,637
4	1,721
5	1,758
6	1,787
7	1,940
8	2,038
9	2,047
10	2,054
11	2,097
12	2,205
13	2,287
14	2,311
15	2,406

Identify the data values which represent the 25th, 50th and 75th percentile.

Solution:

The data is already sorted from smallest to largest. To identify the location, we use the below formula:

$$L_p = \frac{k}{100}(n+1)$$

There are $n = 15$ observations, which means for each percentile P_{25}, P_{50} and P_{75}:

$$L_{25} = \frac{25}{100}(15+1) = 4\text{th Observation}$$

$$L_{50} = \frac{50}{100}(15+1) = 8\text{th Observation}$$

$$L_{75} = \frac{75}{100}(15+1) = 12\text{th Observation}$$

These identify the positions. Therefore:

$$P_{25} = 1721$$

$$P_{50} = 2038$$

$$P_{75} = 2205$$

You should note that when using the location formula, you may not always get a whole number. In this scenario, you must use the following technique:

Interpolation Rule: If the calculated position L falls between two observations, use the decimal part of L to interpolate between them.

Suppose

$$L = j + d,$$

where j is the whole-number part of L and d is the decimal part. Then,

$$P = x_{(j)} + d\left(x_{(j+1)} - x_{(j)}\right),$$

where $x_{(j)}$ is the value at the lower position and $x_{(j+1)}$ is the value at the next position.

For example, if $L = 2.25$,

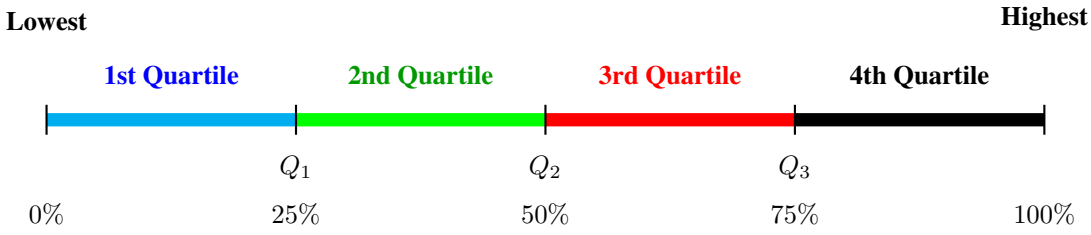
Implying that $L = j + d = 2 + 0.25$. This means:

$$P = x_{(2)} + 0.25\left(x_{(3)} - x_{(2)}\right).$$

The interpolated value does not need to be an actual observation in the dataset.

See **Problem Set Question 1 and 2** for an example of this.

A *percentile* divides the data into 100 equal parts. That is: $P_1, P_2, P_3..., P_{100}$. You should also be aware that we can also express this in **quartiles**. This means we divide the dataset into 4 equal parts:



Note that $Q_1 = P_{25}$, $Q_2 = P_{50}$ and $Q_3 = P_{75}$. Notice that even though we label 3 quartiles, the data is divided into 4 parts.

You should know that an interesting descriptive statistics is the **Interquartile Range** (IQR). This is simply the values where:

$$IQR = Q_3 - Q_1$$

You can also determine whether a data point is an outlier when it satisfies either of these conditions:

- 1.5 x IQR above Q3
- 1.5 x IQR below Q1

Activity:

Consider again our data on sales commissions:

Table 4.3: Salesperson Commissions from the Toyota Showroom

Salesperson	Commission (BD)
1	1,460
2	1,471
3	1,637
4	1,721
5	1,758
6	1,787
7	1,940
8	2,038
9	2,047
10	2,054
11	2,097
12	2,205
13	2,287
14	2,311
15	2,406

Identify the data values which are the first, second and third quartiles. Use this to calculate the IQR.

Solution:

Remember that $Q_1 = P_{25}$, $Q_2 = P_{50}$ and $Q_3 = P_{75}$. There are also $n = 15$ observations. Essentially, the answer is unchanged compared to the previous activity:

$$L_{25} = \frac{25}{100}(15 + 1) = 4\text{th Observation}$$

$$L_{50} = \frac{50}{100}(15 + 1) = 8\text{th Observation}$$

$$L_{75} = \frac{75}{100}(15 + 1) = 12\text{th Observation}$$

These identify the positions. Therefore:

$$Q_1 = P_{25} = 1721$$

$$Q_2 = P_{50} = 2038$$

$$Q_3 = P_{75} = 2205$$

The IQR is therefore:

$$IQR = 2205 - 1721 = 484$$

4.4 Box Plot

We can visualise the quartiles of a dataset by illustrating it into a box plot. This will allow us to better understand the distribution of a dataset.

In order to construct a box plot, you need the following details:

- Q_1.
- Q_2.
- Q_3.
- The maximum value of the dataset.
- The minimum value of the dataset.

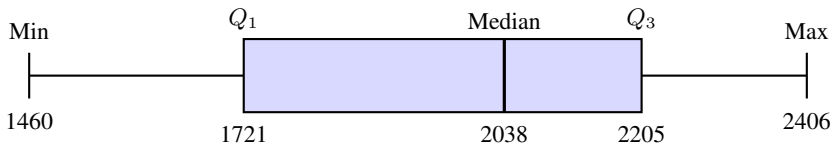
From our previous activity we can identify this:

- $Q_1 = 1721$.
- $Q_2 = 2038$.

- $Q_3 = 2205$.
- $\max = 2406$.
- $\min = 1460$.

We then draw the box plot as follows:

Figure 4.3: Box Plot



It is also common practice to denote the outlier separately away from the minimum or maximum with an asterisks (*) sign.

4.5 Scatter Diagram

A scatter diagram is usually used when trying to understand the correlation of two variables. This is when we have two quantitative variables, one listed on the vertical axis and the other on the horizontal axis.

A scatter plot is a graphical technique showing the relationship between two interval- or ratio-scaled variables.

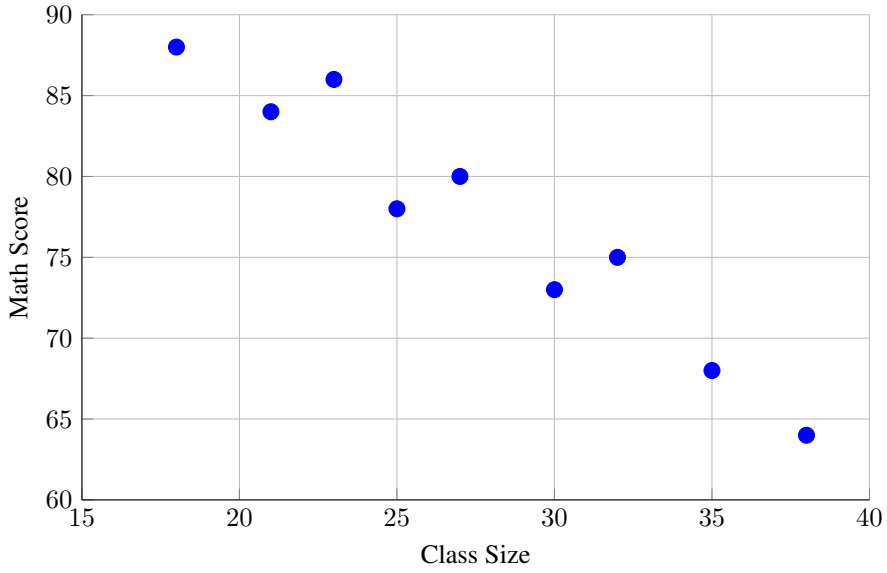
Consider this sample dataset of 9 classes in the Quantitative Business Methods module of Bahrain Polytechnic. We collect data on the following two variables:

Table 4.4: Class Characteristics and Mathematics Scores

School	Class Size	Teacher Salary (BD)	Math Score
Class 1	18	1,480	88
Class 2	21	1,350	84
Class 3	23	1,420	86
Class 4	25	1,180	78
Class 5	27	1,250	80
Class 6	30	1,050	73
Class 7	32	1,120	75
Class 8	35	950	68
Class 9	38	880	64

Suppose we are interested in the relationship between the size of a classroom and the average math score. One way to see this is to have the values of each variable on the axes. For example, lets have the math score on the vertical axis and the class size on the horizontal axis:

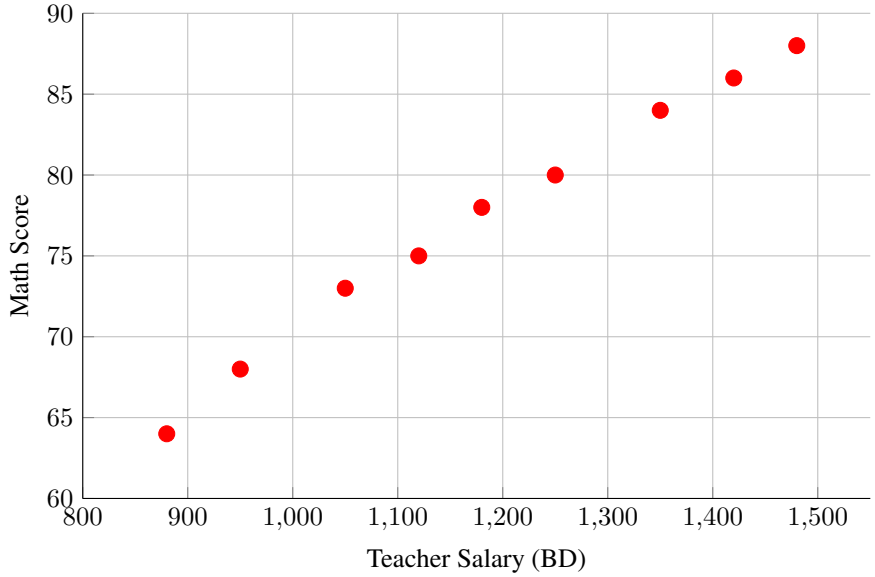
Figure 4.4: Class Size and Mathematics Scores



From the visual, we can see that there is a negative relationship between these two variables. This suggests that the larger a class room becomes, the lower they score in the math exam.

Similarly, let us say we want to plot the relationship between math scores and the salary of the teacher leading the class:

Figure 4.5: Teacher Salary and Mathematics Scores

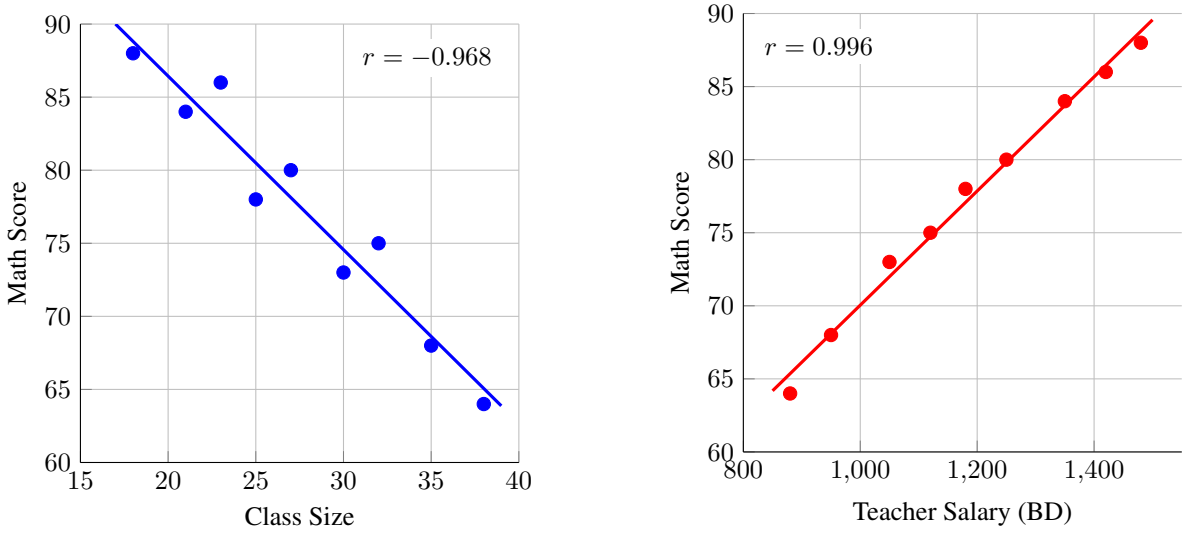


Now we see a positive relationship between math scores and the amount of money a teacher is paid. This might mean that a higher paid teacher can achieve better performance in math scores for the class they teach.

4.6 Correlation Coefficient

When variables are related to each other, as they are clearly shown in the scatter plot, then we can express this visually using a correlation graph:

Figure 4.6: Relationships between School Characteristics and Mathematics Scores



(a) Class Size and Math Score

(b) Teacher Salary and Math Score

The correlation coefficient r is a number that ranges between -1 and +1.

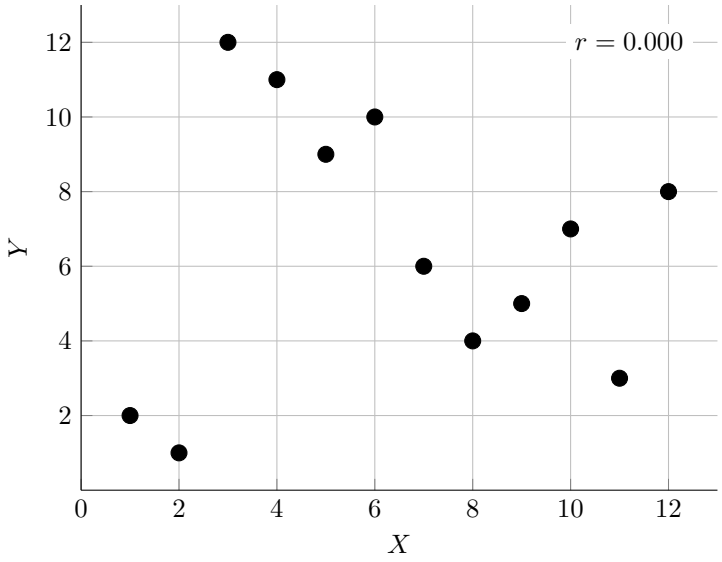
- When the correlation r is very close to -1, we say the variables are **strongly** negatively correlated.
- When the correlation r is very close to +1, then we say it is **strongly** positively correlated.
- When the correlation r is close to 0, then we say there is no relationship.

We can see in the plot that $r = -0.968$ for math and class sizes whereas it is $r = 0.996$ for the math and teacher salary. This is what we expected from our analysis earlier.

The correlation coefficient measures the direction and strength of the relationship between two variables.

Activity:

Consider the below scatter plot between variables Y and X:



What is the relationship between the two variables?

Solution:

There is no relationship between Y and X. This is seen from the scatter plot itself which does not consistently show any positive or negative relation. Moreover, the $r = 0$, so there is no correlation between them.

4.7 Contingency Table

A scatter plot is useful when analysing the relationship between two quantitative variables. However, if we are interested in the relationship between **categorical** variables, then we could consider using a **contingency table**.

A contingency table is used to examine the frequency distribution of two categorical variables. The categories may be nominal or ordinal.

Suppose the CEO of Batelco wants to know about the performance of their branches across Bahrain. Data on customer satisfaction and branch location is given below:

Table 4.5: Customer Satisfaction by Location

Satisfaction	Manama	Riffa	Muharraq	Isa Town	Total
Satisfied	25	20	19	26	90
Dissatisfied	27	20	26	17	90
Total	**52**	**40**	**45**	**43**	**180**

It seems that across all branches, customers are equally happy as they are unhappy. We can see this by looking at the last column under **Total**:

$$\frac{\text{Satisfied}}{\text{Total}} = \frac{90}{180} = 0.5$$

therefore, 50% of all customers surveyed are happy with their service across all branches.

> **Activity:**
>
> Between each branch, which location has the best customer satisfaction?
>
> **Solution:**
>
> You need to divide the satisfaction of each branch by their total. We can see that the Isa Town branch has the highest satisfaction:
>
> $$\frac{26}{43} = 0.64$$
>
> meaning that 60.4% of customers were happy with the service.

4.8 Ethics in Statistics

Statistical results should be presented in an **objective, transparent, and honest** manner. Choices about how data are summarised and displayed can influence how readers interpret the results.

- **Outliers** should be reported rather than silently removed. Observations should only be excluded when there is a

clear and defensible reason for doing so.

- The **shape and skewness** of a distribution should be considered when reporting descriptive statistics. For example, reporting only the mean for a highly skewed distribution may give a misleading impression of the data.
- Graphical displays, such as **box plots and dot plots**, should represent the relevant dataset rather than a selected subset chosen to produce a particular visual impression.
- Researchers should report results that accurately represent the data, even when the findings do not support the conclusion they expected or preferred.

Artificial intelligence can assist with statistical analysis and the communication of results, but its outputs should be used carefully.

- AI should not be used to "make the data look better" by altering, removing, or selectively presenting observations in order to produce a preferred result.
- Statistical summaries, calculations, and interpretations generated by AI should be **checked against the original data** rather than accepted without verification.
- The researcher remains responsible for ensuring that the final analysis accurately and transparently represents the underlying data.

4.9 Problem Set

1) Consider the following ordered dataset:

5.24	6.02	6.67	7.30	7.59	7.99	8.03	8.35	8.81	9.45
9.61	10.37	10.39	11.86	12.22	12.71	13.07	13.59	13.89	15.42

Calculate the following measures of position:

(a) The first quartile, Q_1.

(b) The median, Q_2.

(c) The third quartile, Q_3.

Use the percentile position formula

$$L_K = \frac{K}{100}(n+1),$$

as well as the interpolation method.

2) Consider the following test scores for 10 students:

42	50	55	60	64	68	72	78	84	90

Derive the 50th and 10th percentile.

4.10 Solution

1) There are $n = 20$ observations.

(a) **First Quartile (Q_1)**

Since $Q_1 = P_{25}$,

$$L_{25} = (20 + 1)\left(\frac{25}{100}\right)$$
$$= 5.25$$

Therefore, Q_1 lies 0.25 of the way between the 5th and 6th observations:

$$x_{(5)} = 7.59, \qquad x_{(6)} = 7.99.$$

Using linear interpolation,

$$Q_1 = 7.59 + 0.25(7.99 - 7.59)$$
$$= 7.59 + 0.10$$
$$= \boxed{7.69}$$

(b) **Median (Q_2)**

Since $Q_2 = P_{50}$,

$$L_{50} = (20 + 1)\left(\frac{50}{100}\right)$$
$$= 10.5$$

Therefore, the median lies halfway between the 10th and 11th observations:

$$x_{(10)} = 9.45, \qquad x_{(11)} = 9.61.$$

Thus,

$$Q_2 = 9.45 + 0.5(9.61 - 9.45)$$
$$= \boxed{9.53}$$

(c) **Third Quartile (Q_3)**

Since $Q_3 = P_{75}$,

$$L_{75} = (20 + 1)\left(\frac{75}{100}\right)$$
$$= 15.75$$

Therefore, Q_3 lies 0.75 of the way between the 15th and 16th observations:

$$x_{(15)} = 12.22, \qquad x_{(16)} = 12.71.$$

Using linear interpolation,

$$Q_3 = 12.22 + 0.75(12.71 - 12.22)$$
$$= 12.22 + 0.3675$$
$$= \boxed{12.5875 \approx 12.59}$$

2) There are $n = 10$ observations and the data are already arranged from smallest to largest.

50th Percentile

$$L_{50} = (n + 1)\frac{50}{100}$$
$$= (10 + 1)\frac{50}{100}$$
$$= 5.5$$

The 50th percentile lies halfway between the 5th and 6th observations, which are 64 and 68.

$$\begin{aligned} P_{50} &= 64 + 0.5(68 - 64) \\ &= 66 \end{aligned}$$

Therefore,

$$\boxed{P_{50} = 66}$$

10th Percentile

$$\begin{aligned} L_{10} &= (n+1)\frac{10}{100} \\ &= (10+1)\frac{10}{100} \\ &= 1.1 \end{aligned}$$

The 10th percentile lies between the 1st and 2nd observations, which are 42 and 50.

$$\begin{aligned} P_{10} &= 42 + 0.1(50 - 42) \\ &= 42.8 \end{aligned}$$

Therefore,

$$\boxed{P_{10} = 42.8}$$

Chapter 5

A Survey of Probability Concepts

5.1 Introduction to Probability

Consider a coin toss of a Bahraini coin. We know that there are only two possible **outcomes**: heads or tails.

Formally, we denote this as an experiment with two possible outcomes:

$$S = \{H, T\}$$

Suppose we want to know the probability that the coin will show heads. We will formally denote this as **event** A:

$$A = \{\text{Coin shows heads}\} = \{H\}$$

We know that there are 2 possible states: heads or tails. Since only one of them can be shown, then the probability of event A occurring is simply:

$$P(A) = \frac{1}{2}$$

Activity

We know that when we toss a coin, the possible outcomes are:

$$S = \{H, T\}$$

Determine the possible outcomes for the two experiments:

(a) Rolling a six-sided die.

(b) Tossing two coins.

Solution:

(a) When you roll a die, there are only 6 possible outcomes:

$$S = \{1, 2, 3, 4, 5, 6\}$$

(b) When you toss two coins, there are 4 possible outcomes:

$$S = \{(H, T), (H, H), (T, H), (T, T)\}$$

In general, when all possible outcomes are **equally likely**, the classical probability of an event A occurring is:

$$P(A) = \frac{\text{Number of outcomes in event } A}{\text{Total number of possible outcomes}}.$$

So if $S = \{H, T\}$ and event $A = \{H\}$, then $P(A)$ is equal to:

$$P(A) = \frac{1}{2}$$

Activity:

Determine the following probabilities:

(a) Rolling an even number from a six-sided die.

(b) Tossing two coins and getting the same side.

Solution:

(a) We know that for a six-sided die:

$$S = \{1, 2, 3, 4, 5, 6\}$$

which means we have a total of 6 possible outcomes. The event A is when the die reveals an even number:

$$A = \text{Even number from die roll} = \{2, 4, 6\}$$

which is 3 of the possible 6 total outcomes. Therefore:

$$P(A) = \frac{3}{6} = \frac{1}{2}$$

(b) When we toss two coins, we know that:

$$S = \{(H, T), (H, H), (T, H), (T, T)\}$$

which means we have a total of 4 possible outcomes. The event A is when both coins show the same side.

$$A = \text{Both coins show same side} = \{(H, H), (T, T)\}$$

which is 2 of the possible 4 outcomes. Therefore:

$$P(A) = \frac{2}{4} = \frac{1}{2}$$

5.2 Approaches to Assigning Probabilities

When we are dealing with simple experiments, such as tossing a coin or rolling a die, it is relatively easy to list and count all the possible outcomes. However, as experiments become more complicated, listing every possible outcome individually can become difficult.

In such cases, we can use different **counting methods** to determine the number of possible outcomes more efficiently. These methods can also help us determine the number of outcomes associated with a particular event.

The three counting methods we will consider are:

- **Multiplication**
- **Permutations**
- **Combinations**

5.2.1 Multiplication

Consider again the outcomes if you toss a coin. The coin can only give you heads H or tails T:

$$S_1 = \{H, T\}$$

Now suppose you toss 2 coins. The possible outcomes now become:

$$S_2 = \{(H, T), (H, H), (T, H), (T, T)\}$$

If you toss 3 coins, there possible outcomes become:

$$S_3 = \{(H, H, H), (H, H, T), (H, T, H), (H, T, T),$$
$$(T, H, H), (T, H, T), (T, T, H), (T, T, T)\}.$$

In general, you can determine the combinations by just multiplying the total outcomes of each coin. Since each coin can only be heads or tails:

(a) One coin - 2 outcomes.

(b) Two coins - $2 \cdot 2 = 4$ outcomes.

(c) Three coins - $2 \cdot 2 \cdot 2 = 8$ outcomes.

Activity:

What are the total number of outcomes for the below experiments:

(a) Tossing 4 coins.

(b) Throwing 2 dices.

Solution:

(a) A coin can either be heads or tails. If a person tosses 4 coins, then:

$$2 \cdot 2 \cdot 2 \cdot 2 = 16 \text{ combinations}$$

(b) A die can have 6 outcomes. Therefore if a person throws two dices, the amount of combinations will be:

$$6 \cdot 6 = 36 \text{ combinations}$$

5.2.2 Combinations

A **combination** is used when we want to select a number of objects from a larger group and the **order of the selection does not matter**.

Consider a class with four students:

$$S = \{A, B, C, D\}.$$

A teacher needs to select 2 students to represent the class. They do this by randomly selecting any 2 from the 4 people in the classroom.

The possible combinations of class representatives are:

$\{A, B\}$	$\{B, A\}$	$\{C, A\}$	$\{D, A\}$
$\{A, C\}$	$\{B, C\}$	$\{C, B\}$	$\{D, B\}$
$\{A, D\}$	$\{B, D\}$	$\{C, D\}$	$\{D, C\}$

A crucial point to see here is that the **order does not matter** here. For example, $\{A, B\} = \{B, A\}$.

Rather than listing every possible committee, we can calculate the number of possible selections using the **combination rule**:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!},$$

where:

$$n = \text{the total number of objects available,}$$

$$r = \text{the number of objects we want to select.}$$

In our example, there are four students available and we want to randomly select two of them. Therefore:

$$n = 4, \qquad r = 2.$$

Applying the combination rule:

$$\binom{4}{2} = \frac{4!}{2!(4-2)!}$$

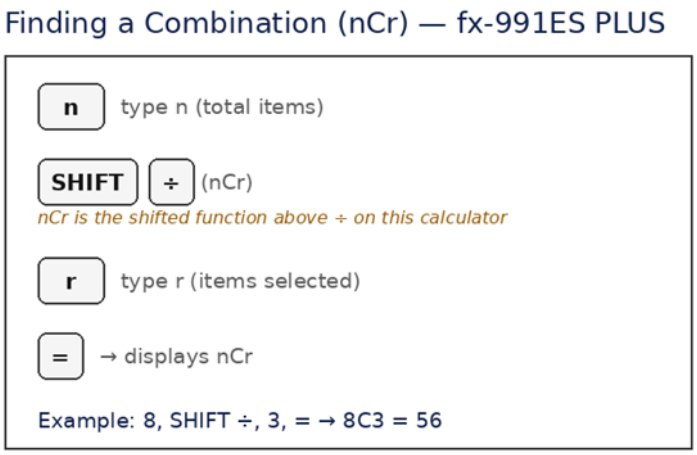
$$= \frac{4!}{2!2!}$$

$$= \frac{24}{2 \times 2}$$

$$= 6.$$

Therefore, there are **6 different committees** that can be formed by selecting two students from a group of four.

In your calculator, you can perform a combination calculation as described below:



5.2.3 Permutations

A **permutation** is used when we want to select a number of objects from a larger group and the **order of the selection matters**.

Consider again a class with four students:

$$S = \{A, B, C, D\}.$$

Suppose we want to select two students. However, this time the two students will have different positions: one student will be the **President** and the other will be the **Vice-President**.

For example, consider the following outcome:

$$(A, B).$$

This means that student A is the President and student B is the Vice-President. Now consider:

$$(B, A).$$

This represents a different outcome because student B is now the President and student A is the Vice-President. Therefore:

$$(A, B) \neq (B, A).$$

Unlike our previous committee example, changing the order of the students creates a different outcome. Therefore in this case, the **order matters**.

We can list all the possible outcomes:

(A, B)	(B, A)	(C, A)	(D, A)
(A, C)	(B, C)	(C, B)	(D, B)
(A, D)	(B, D)	(C, D)	(D, C)

In each ordered pair, the first student is the President and the second student is the Vice-President. All 12 outcomes are therefore different.

Rather than listing every possible outcome, we can calculate the number of possible arrangements using the **permutation rule**:

$$_nP_r = \frac{n!}{(n-r)!}$$

where just as before:

$$n = \text{the total number of objects available},$$

$$r = \text{the number of objects being selected.}$$

In our example, there are four students available and two positions that need to be filled from selection:

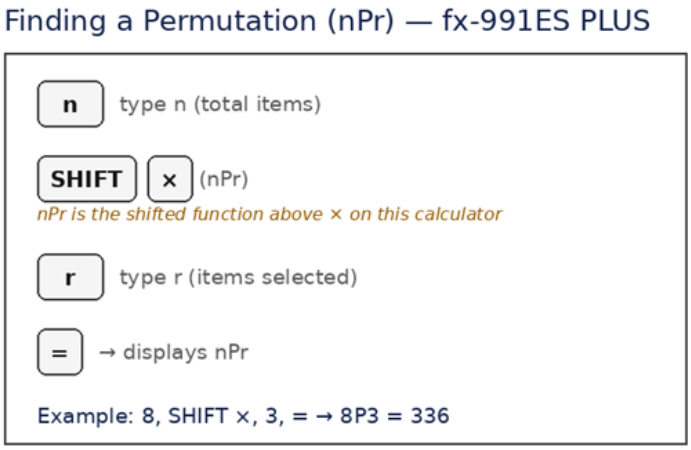
$$n = 4, \qquad r = 2.$$

Applying the permutation rule:

$$
\begin{aligned}
{}_4P_2 &= \frac{4!}{(4-2)!} \\
&= \frac{4!}{2!} \\
&= \frac{24}{2} \\
&= 12.
\end{aligned}
$$

Therefore, there are **12 different ways** of selecting a President and Vice-President from the four students.

You can perform a permutation calculation using a calculator as described below:



5.3 Other Probability Methods

Before we proceed, you should be aware of the other probability methods which are **empirical** and **subjective probability**.

Empirical Probability

The probability of an event is based on a collection of observations or data.

This is based on relative frequencies:

$$\frac{\text{Number of times an event occurs}}{\text{Total number of observations}}$$

Activity:

Suppose you have data on a company for the past 20 years. In each year, you know whether they made a profit or a loss.

From the data, the company is shown to have made a loss in 4 years, and a profit in 16 years.

What is the probability that the company makes a loss?

Solution:

Going by the formula, the probability that the company makes a loss using the empirical method is:

$$= \frac{\text{Number of times an event occurs}}{\text{Total number of observations}}$$

$$\frac{4}{20} = 0.2$$

Subjective Probability

This is a probability that is assigned by an individual based on whatever information is available, when there is little or no prior experience to draw on.

5.4 Notation

To better understand this section, we need to introduce some notation from set theory, which is the concept of **union** and **intersection**.

5.4.1 Union

Consider a dice which has 6 sides. When rolling this dice, we should it to show one of the six possible outcomes:

$$S = \{1, 2, 3, 4, 5, 6\}.$$

all of which have an equally likely chance of occurring, which is $1/6$.

Suppose we define two events from this dice roll A and B:

$$A = \text{An even number} = \{2, 4, 6\}$$

$$B = \text{A number greater than 3} = \{4, 5, 6\}$$

We are interested in the **union** of events A and B. This means we want to list out the numbers[a] that result from combining A and B. We formally write this as:

$$A \cup B = \{2, 4, 5, 6\}$$

Notice how we do not double count the numbers. Meaning, even if 4 appears in both events, its union only includes it once. We also generally refer the term union $\cup$ with the word **or**:

[a] This is called *elements* in set theory.

$$A \cup B = A \text{ or } B.$$

5.4.2 Intersection

Suppose now we want to know the **intersection** of events A and B. This means we want to list out the numbers are in both A and B. We formally write this as:

$$A \cap B = \{4, 6\}$$

Both 4 and 6 are numbers which are even and are also greater than 3. You can also easily see the overlapping numbers by looking at the numbers inside A and B.

We generally refer the term intersection $\cap$ with the word **and**:

$$A \cap B = A \text{ and } B.$$

We will use both of these notations when analysing the probabilities of different events.

5.5 Rules of Addition

You should keep in mind two formulas which we will use which refer to the **rules of addition**.

Suppose we are interested in the **probability** that a number from A **or** B can occur. We calculate this by denoting:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B),$$

We will apply this formula when differentiating between events that are mutually exclusive and those that are not.

5.5.1 Mutually Exclusive Events

Let us begin by considering the case of **mutually exclusive events**. Two events are mutually exclusive if they cannot occur at the same time.

Consider again the experiment of rolling a fair six-sided die:

$$S = \{1, 2, 3, 4, 5, 6\}.$$

Suppose we define the following two events:

$$A = \text{The die shows a 2} = \{2\}$$

$$B = \text{The die shows a 5} = \{5\}.$$

When the die is rolled once, it cannot show both a 2 and a 5 at the same time. Therefore, events A and B are **mutually exclusive**.

Using the notation introduced earlier, we can see that the two events have no intersection:

$$A \cap B = \varnothing,$$

where $\varnothing$ represents an **empty set**. In other words, there are no outcomes that belong to both A and B and so $P(A \cap B) = 0$.

Suppose we are interested in the probability that the die shows a 2 **or** a 5. Using our notation for a union, we are interested in:

$$P(A \cup B).$$

Since the events are mutually exclusive, we can simply add their individual probabilities:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = P(A) + P(B)$$

In our example, we know that:

$$P(A) = \frac{1}{6}, \qquad P(B) = \frac{1}{6}.$$

$$P(A \cup B) = P(A) + P(B)$$

$$= \frac{1}{6} + \frac{1}{6}$$

$$= \frac{2}{6}$$

Therefore, the probability that the die shows either a 2 or a 5 is $\frac{2}{6}$.

5.5.2 Non-Mutually Exclusive Events

Let us now consider the case of **non-mutually exclusive events**. Two events are non-mutually exclusive if they can occur at the same time. In other words, the two events have some outcomes in common.

Consider again the experiment of rolling a fair six-sided die:

$$S = \{1, 2, 3, 4, 5, 6\}.$$

Suppose we define the following two events:

$$A = \text{The die shows an even number} = \{2, 4, 6\}$$

$$B = \text{The die shows a number greater than 3} = \{4, 5, 6\}.$$

Notice that the outcomes 4 and 6 belong to both events. Therefore, events A and B are **non-mutually exclusive**.

Using the notation introduced earlier, the intersection of the two events is:

$$A \cap B = \{4, 6\}.$$

Since there are two outcomes in the intersection, and each outcome has a probability of $1/6$, we have:

$$P(A \cap B) = P(A \text{ and } B)$$

$$= \frac{2}{6}$$

Now suppose we are interested in the probability that the die shows an even number **or** a number greater than 3. Using our notation for a union, we are interested in:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{3}{6} + \frac{3}{6} - \frac{2}{6}$$

$$= \frac{4}{6}$$

$$= \frac{2}{3}.$$

Therefore, the probability that the die shows an even number **or** a number greater than 3 is $\frac{2}{3}$.

5.6 The Rules of Multiplication

The rules of multiplication are used when we are interested in the probability of two or more events occurring together. However, we need to distinguish between **independent** and **dependent** events.

5.6.1 Independent Events

Two events are said to be **independent** when the occurrence of one event does not affect the probability of the other event occurring.

Consider an experiment where we roll two fair six-sided dice. Let us define the following two events:

$$A = \text{The first die shows a 3}$$

$$B = \text{The second die shows a 6.}$$

We know that the probability of each event is:

$$P(A) = \frac{1}{6}, \qquad P(B) = \frac{1}{6}.$$

Events A and B are **independent**. The number shown by the first die does not affect the number that can be shown by the second die. For example, even if the first die shows a 3, the probability that the second die shows a 6 remains $1/6$.

When two events are independent, the probability that both events occur can be calculated by multiplying their individual probabilities:

$$P(A \cap B) = P(A) \times P(B).$$

Applying this rule to our example:

$$P(A \cap B) = P(A) \times P(B)$$

$$= \frac{1}{6} \times \frac{1}{6}$$

$$= \frac{1}{36}.$$

Therefore, the probability that the first die shows a 3 **and** the second die shows a 6 is $\frac{1}{36}$.

5.6.2 Dependent Events

Two events are said to be **dependent** when the occurrence of one event affects the probability of another event occurring.

Consider again the experiment of rolling a fair six-sided die:

$$S = \{1, 2, 3, 4, 5, 6\}.$$

We define the following two events:

$$A = \text{The die shows a number greater than } 3 = \{4, 5, 6\}$$

$$B = \text{The die shows an even number} = \{2, 4, 6\}.$$

Now suppose you throw the die. You are told by somebody that the number on the die is **greater than 3**, but they do not tell you which number. Basically, you know that event A has occurred, but you do not know which number it is.

Our possible outcomes are therefore restricted to:

$$\{4, 5, 6\}.$$

After this has happened, you are now asked to determine the probability **that the die shows an even number**. That is, you need to determine the probability of event B **given that** event A has already occurred.

Formally, this is denoted as:

$$P(B|A) = \frac{2}{3}$$

which is true, given that the only two numbers that are even is 4 and 6 out of the three possible outcomes.

Now let us step back to the point before the die has been rolled. We are interested in determining the probability of seeing an even number **given that** the number greater than 3. To do this, we apply the following multiplicative formula:

$$P(A \cap B) = P(A) \cdot P(B|A)$$

Given that:

$$P(A) = \frac{3}{6}$$

$$P(B|A) = \frac{2}{3}$$

therefore, the probability of this **dependent event** from occurring is:

$$P(A \cap B) = P(A) \cdot P(B|A)$$

$$= \frac{3}{6} \cdot \frac{2}{3}$$

$$= \frac{2}{6} = \frac{1}{3}$$

5.7 Contingency Table

The Sales Manager of Batelco is studying the patterns for placing international calls based on the age of their customers. They conduct a survey and collect the following results for September 2026:

Table 5.1: International Calls by Customer Age

Customer	Age	Age Classification	International Calls
Customer 1	24	Young	2
Customer 2	42	Adult	1
Customer 3	67	Senior	0
Customer 4	31	Adult	3
Customer 5	19	Young	1
Customer 6	58	Adult	2
		⋮	
Customer 500	72	Senior	1

The results of the survey are then expressed in a contingency table:

Table 5.2: International Calls by Age Classification

Calls Placed	Young	Adult	Senior	Total
0	15	50	10	75
1	25	100	75	200
2	55	60	60	175
3	5	15	30	50
Total	**100**	**225**	**175**	**500**

Activity:

Determine the following probabilities:

(a) The probability of a person placing 3 calls.

(b) The probability of a person placing 1 call **or** less.

(c) The probability of a person placing 3 calls **or** is a senior.

(d) The probability of a person placing 3 calls **and** they are a senior.

(e) The probability of a person placing 3 calls **given that** they are a senior.

Solution:

(a) There are 50 people in the dataset who placed 3 calls out of a total of 500.

$$P(\text{3 Calls}) = \frac{\text{Number of customers who placed 3 calls}}{\text{Total number of customers}}$$

$$= \frac{50}{500}$$

$$= 0.10.$$

(b) We need to include the probability that the person either place 1 call **or** 0 calls.

$$P(\text{1 Call or Less}) = P(\text{0 Calls}) + P(\text{1 Call})$$

$$= \frac{75}{500} + \frac{200}{500}$$

$$= \frac{275}{500}$$

$$= 0.55.$$

(c) Remember that placing a call or being a senior are **not mutually exclusive**. That is, it is certainly possible for a person who placed 3 calls to also be a senior and so we need to subtract this intersection.

$$P(\text{3 Calls or Senior}) = P(\text{3 Calls}) + P(\text{Senior}) - P(\text{3 Calls and Senior})$$

$$= \frac{50}{500} + \frac{175}{500} - \frac{30}{500}$$

$$= \frac{195}{500}$$

$$= 0.39.$$

(d) We need to determine amongst all individuals, the probability that they placed 3 calls **and** they are a senior. We can see this intersection includes 30 people.

$$P(\text{3 Calls and Senior}) = \frac{\text{Number of seniors who placed 3 calls}}{\text{Total number of customers}}$$

$$= \frac{30}{500}$$

$$= 0.06.$$

(e) In this regard, "being a senior" is already an event that has occurred. So we are just trying to determine the probability that this senior placed 3 calls.

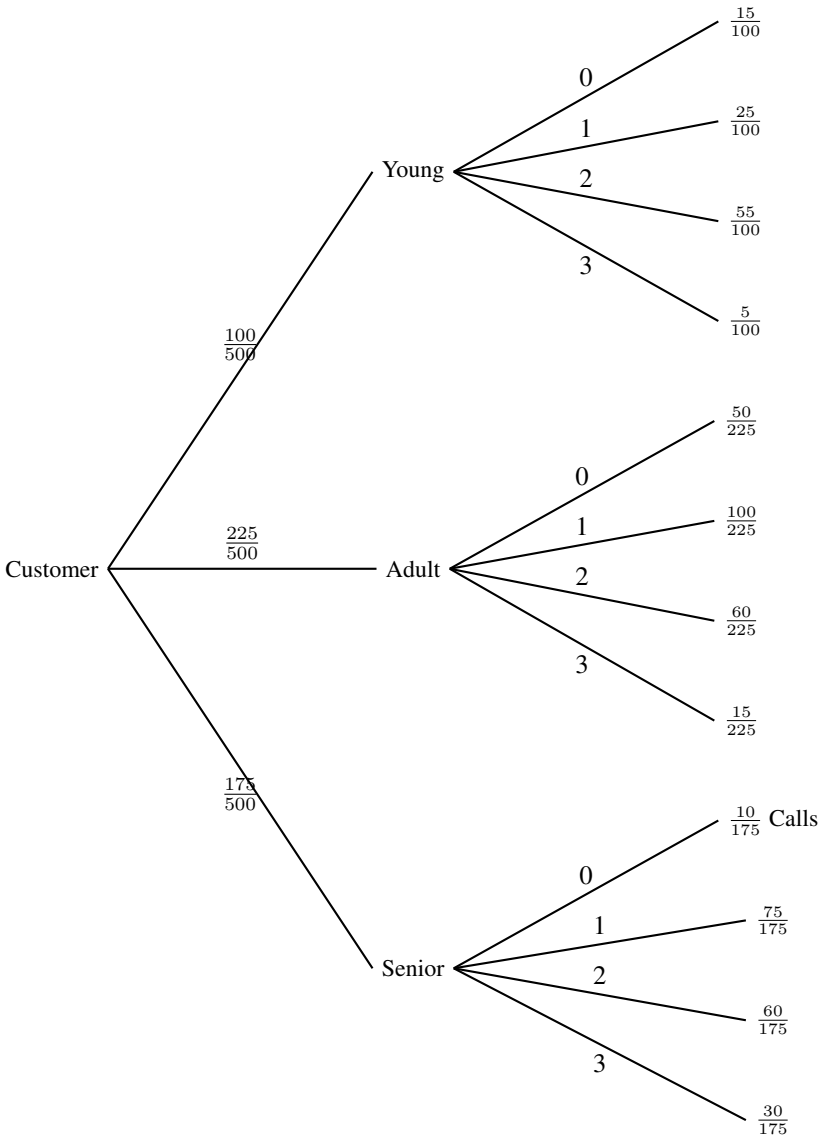
$$P(\text{3 Calls} \mid \text{Senior}) = \frac{\text{Number of seniors who placed 3 calls}}{\text{Total number of seniors}}$$

$$= \frac{30}{175}$$

$$= 0.171.$$

We can also visualise the probability outcomes using a Tree Diagram:

Figure 5.1: Probability Tree for International Calls by Age Classification



5.8 The Complement Rule

The **complement** of an event A consists of all outcomes where event A does **not** occur. We denote the complement of A as A^c.

Consider rolling a fair six-sided die:

$$S = \{1, 2, 3, 4, 5, 6\}.$$

Suppose we define event A as rolling an even number:

$$A = \{2, 4, 6\}.$$

The complement of A contains all the outcomes where A does not occur. Therefore:

$$A^c = \{1, 3, 5\}.$$

Since an event must either occur or not occur, the probabilities of an event and its complement must add to 1:

$$P(A) + P(A^c) = 1.$$

Therefore, the **Complement Rule** is:

$$\boxed{P(A^c) = 1 - P(A)}.$$

In our example:

$$P(A^c) = 1 - P(A)$$

$$= 1 - \frac{3}{6}$$

$$= \frac{3}{6}$$

$$= \frac{1}{2}.$$

Therefore, the probability of **not** rolling an even number is $\frac{1}{2}$.

5.9 Ethics and Reporting Probability

Probability is used to describe uncertainty and should therefore be reported carefully and transparently. Consider the below ideas when incorporating ethical judgements of your probability analysis:

- Present probability results honestly and meaningfully, not as false certainties.
- Do not selectively report only favorable probabilities.
- Be transparent about assumptions behind conditional probability calculations.

Ethical Use of AI

- AI-generated risk estimates should not replace expert human judgment in high-stakes decisions.

5.10 Problem Set

1) A company is developing a new product and offers customers several options for customising their order. Customers can choose from:

- 3 different product sizes,
- 4 different package designs, and
- 2 different delivery options.

If a customer must select one option from each category, how many different product orders are possible?

2) A marketing agency has produced 8 different video segments for an advertising campaign. Due to time constraints, the final advertisement can include only 3 of these segments. The order in which the segments appear in the advertisement is important.

How many different arrangements of 3 video segments can the agency create from the 8 available segments?

3) A consulting firm has 7 employees available for a new project. The firm needs to select 3 employees to form a project team. All members of the team will have the same role.

How many different project teams can the firm form?

4) A **stock index** measures the performance of a group of companies traded in the stock market. One example is the S&P 500, which tracks the performance of many large companies in the United States.

If the value of the index is lower at the end of a year than it was at the beginning of the year, we will describe the index as being **down** for that year.

Over the previous 20 years, suppose that the S&P 500 was down in 4 of those years.

Using this historical information, estimate the probability that the index will be down in a given year.

5) A manufacturing company produces metal components that must meet a specified size requirement. As part of its quality control process, 4,000 components are inspected and classified as either **undersized**, **within specification**, or **oversized**.

The results of the inspection are shown below:

Classification	**Event**	**Number of Components**	**Probability**
Undersized	A	100	0.025
Within Specification	B	3,600	0.900
Oversized	C	300	0.075
Total		**4,000**	**1.000**

Determine:

(a) The probability that a randomly selected component is either **undersized or oversized**.

(b) The probability that a randomly selected component is **within specification**.

6) A standard deck of playing cards contains **52 cards**. The cards are divided equally into four groups, called **suits**:

Hearts, Diamonds, Clubs, Spades.

Each suit contains 13 cards. Among these cards is one card called a **King**. Therefore, in the entire deck:

- There are 52 cards in total.
- There are 13 Hearts.
- There are 4 Kings, one from each suit.
- One of these cards is the **King of Hearts**.

Suppose one card is randomly selected from the deck. What is the probability that the selected card is a **King or a Heart**?

7) A survey of consumers in Bahrain found that 60% had purchased a product online during the previous month. Suppose two consumers are randomly selected from the population.

Assuming that the purchasing behaviour of one consumer is independent of the other, what is the probability that **both consumers** purchased a product online during the previous month?

8)A company receives a shipment containing 12 electronic components. Of these components, 9 are classified as **Grade A** and 3 are classified as **Grade B**.

Two components are randomly selected for inspection. The first component is not returned to the shipment before the second component is selected.

What is the probability that **both components** selected are Grade A?

5.11 Solution

1) There are three stages involved in creating an order:

$$\begin{aligned} \text{Product Size:} &\quad 3 \text{ choices} \\ \text{Package Design:} &\quad 4 \text{ choices} \\ \text{Delivery Option:} &\quad 2 \text{ choices} \end{aligned}$$

Using the **Multiplication Principle**, we multiply the number of choices available at each stage:

$$\begin{aligned} \text{Total Number of Possible Orders} &= 3 \times 4 \times 2 \\ &= 24. \end{aligned}$$

Therefore, the company can offer customers **24 different possible product orders**.

2) Since the order in which the video segments appear matters, we use a **permutation**.

There are 8 video segments available, of which 3 must be selected and arranged. Therefore:

$$\begin{aligned} n &= 8, \\ r &= 3. \end{aligned}$$

Using the permutation rule:

$$
\begin{aligned}
{}_nP_r &= \frac{n!}{(n-r)!} \\
{}_8P_3 &= \frac{8!}{(8-3)!} \\
&= \frac{8!}{5!} \\
&= 8 \times 7 \times 6 \\
&= 336.
\end{aligned}
$$

Therefore, the marketing agency can create **336 different arrangements** of 3 video segments.

3) Since the order in which the employees are selected does not matter, we use a **combination**.

There are 7 employees available, of which 3 must be selected. Therefore:

$$
\begin{aligned}
n &= 7, \\
r &= 3.
\end{aligned}
$$

Using the combination rule:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

$$\binom{7}{3} = \frac{7!}{3!(7-3)!}$$

$$= \frac{7!}{3!4!}$$

$$= 35.$$

Therefore, the consulting firm can form **35 different project teams**.

4) Let event A represent a year in which the stock index is down.

We can estimate the probability using the relative frequency with which the event occurred in the historical data:

$$P(A) = \frac{\text{Number of years the index was down}}{\text{Total number of years observed}}$$

$$= \frac{4}{20}$$

$$= 0.20.$$

Therefore, based on the historical data, the estimated probability of the stock index being down in a given year is 0.20, or 20%.

5) **(a)** Since a component cannot be both undersized and oversized, events A and C are mutually exclusive. Therefore:

$$P(A \cup C) = P(A) + P(C)$$

$$= 0.025 + 0.075$$

$$= 0.10.$$

Therefore, the probability that a randomly selected component is either undersized or oversized is 0.10, or 10%.

(b) A component that is within specification is the complement of a component that is either undersized or oversized. Therefore:

$$P(B) = 1 - P(B^c)$$

$$= 1 - P(A \cup C)$$

$$= 1 - 0.10$$

$$= 0.90.$$

Therefore, the probability that a randomly selected component is within specification is 0.90, or 90%.

6) Let:

$A =$ The selected card is a King,

$B =$ The selected card is a Heart.

There are 4 Kings in the deck, so:

$$P(A) = \frac{4}{52}.$$

There are 13 Hearts in the deck, so:

$$P(B) = \frac{13}{52}.$$

However, these two events **overlap**. One card, the King of Hearts, is both a King and a Heart. Therefore:

$$P(A \cap B) = \frac{1}{52}.$$

Using the addition rule for non-mutually exclusive events:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{4}{52} + \frac{13}{52} - \frac{1}{52}$$

$$= \frac{16}{52}$$

$$\approx 0.3077.$$

Therefore, the probability of randomly selecting a card that is either a King or a Heart is approximately 0.3077, or 30.77%.

7) Let:

$$A = \text{The first consumer purchased a product online},$$

$$B = \text{The second consumer purchased a product online}.$$

We know that:

$$P(A) = 0.60, \qquad P(B) = 0.60.$$

Since we assume that the two events are **independent**, we can use the multiplication rule:

$$P(A \cap B) = P(A) \times P(B)$$

$$= 0.60 \times 0.60$$

$$= 0.36.$$

Therefore, the probability that **both consumers** purchased a product online during the previous month is 0.36, or 36%.

8) Let:

$$A_1 = \text{The first component selected is Grade A},$$

$$A_2 = \text{The second component selected is Grade A}.$$

For the first selection, there are 9 Grade A components out of 12 components:

$$P(A_1) = \frac{9}{12}.$$

If the first component selected is Grade A, there are now only 11 components remaining, of which 8 are Grade A. Therefore:

$$P(A_2 \mid A_1) = \frac{8}{11}.$$

The first selection therefore changes the probability of the second selection. The two events are **dependent**.

Using the multiplication rule for dependent events:

$$P(A_1 \cap A_2) = P(A_1) \times P(A_2 \mid A_1)$$

$$= \frac{9}{12} \times \frac{8}{11}$$

$$= \frac{72}{132}$$

$$\approx 0.545.$$

Therefore, the probability that both components selected are Grade A is approximately 0.545, or 54.5%.

Chapter 6

Discrete Probability Distributions

6.1 Introduction

In the previous chapters, we discussed that when the values of a variable can be counted, it is called **discrete** (coin toss, dice roll, the number of students in a class room). When a variable can take any range of values, then it is known as **continuous** (temperature, distance, bank balance).

In this chapter, we will be focused on mapping out the **probability distribution** of discrete variables. The two important distributions will be the Binomial and Poisson distributions.

6.2 Probability Distribution

Consider the case of a fair coin toss. We know that the sample space is described by the two outcomes:

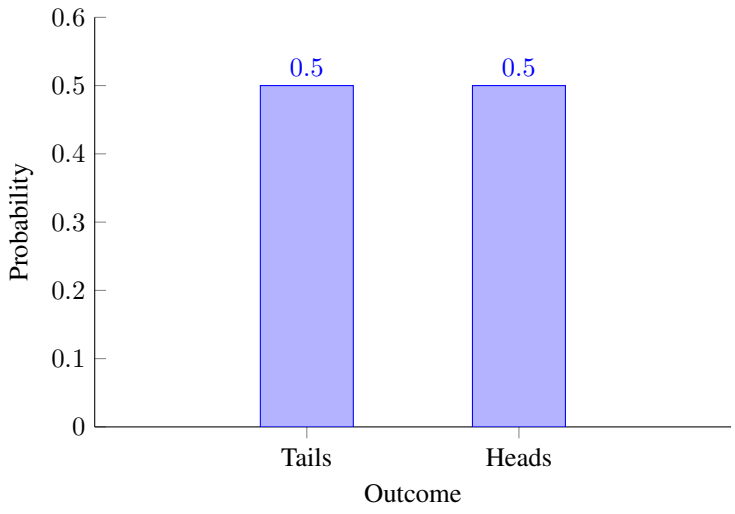
$$S = \{H, T\}$$

and its associated probabilities were:

$$P(H) = \frac{1}{2} \qquad P(T) = \frac{1}{2}$$

Graphically, we can illustrate the probability in a graph:

Figure 6.1: Probability Distribution for One Fair Coin Toss



Notice that we are using qualitative notation for heads H, tails T. In this chapter we are interested in using a general quantitative notation so we can incorporate all kinds of examples.

Suppose we toss the coin again, but this time we define the **random variable** X as:

$$X = \begin{cases} 0 & \text{if Tails occurs,} \\ 1 & \text{if Heads occurs.} \end{cases}$$

that is, the researcher simply notates 0 as the coin showing tails and 1 if the coin shows heads. This means X can take only two values:

$$X = \{0, 1\}$$

and its associated probabilities are:

$$P(X = 0) = \frac{1}{2} \qquad P(X = 1) = \frac{1}{2}$$

The way you read this is that the probability that the random variable X is equal to 0 (meaning that it is tails) is $\frac{1}{2}$. Similarly, the probability that the random variable X is equal to 1 (meaning that is heads) is $\frac{1}{2}$.

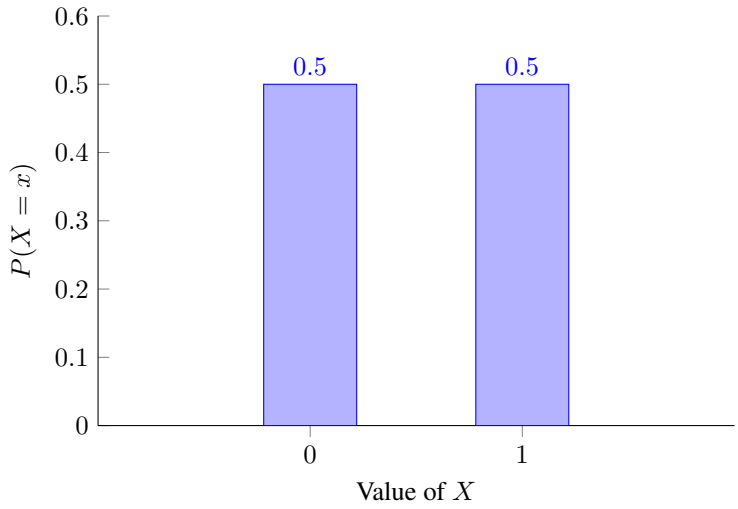
We can more easily read this in a table:

x	$P(X = x)$
0	0.5
1	0.5

This is exactly the same coin toss experiment as before. The difference is that we have now assigned a numerical value to each possible outcome.

We can illustrate the **probability distribution** of the random variable X:

Figure 6.2: Probability Distribution of X, where $X = 0$ for Tails and $X = 1$ for Heads



Let us now extend the example where we toss two coins. We know that the sample space will now become:

$$S = \{HH, HT, TH, TT\}$$

Let us now define the random variable X as being **the number of heads** that the coins show. For each possible

outcome, we can see that the number of heads will be:

$$TT \to 0$$

$$HT \to 1$$

$$TH \to 1$$

$$HH \to 2$$

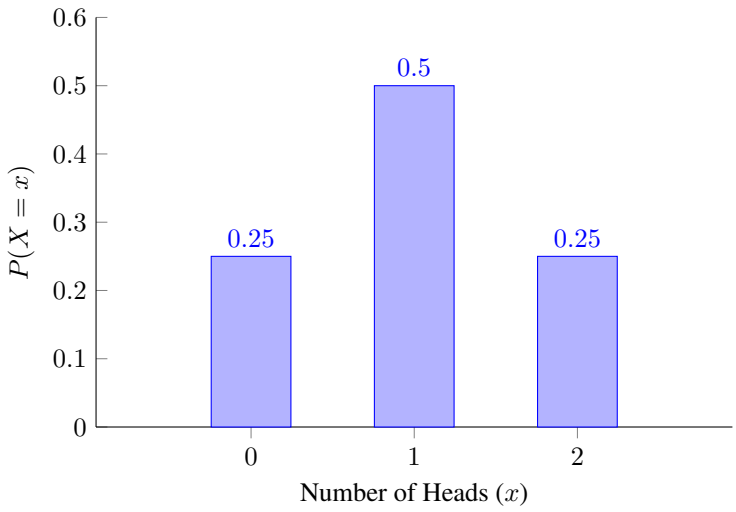
Formally, the random variable X can take the following values:

$$X = \{0, 1, 2\}$$

and based on its frequency, the associated probabilities of each value is:

$$P(X = 0) = \frac{1}{4}, \qquad P(X = 1) = \frac{2}{4}, \qquad P(X = 2) = \frac{1}{4}$$

Figure 6.3: Probability Distribution of the Number of Heads from Two Coin Tosses



Activity:

Consider the case of tossing three coins.

(a) Define the sample space S.

(b) Let X represent the number of heads the coins show. Describe the frequency of X for each outcome.

(c) What are the possible values that X can take?

(d) Based on its frequency, what are the associated probabilities of each value of X?

(e) Illustrate the probability distribution of X.

Solution:

(a) When three coins are tossed, there are $2^3 = 8$ possible outcomes. The sample space is:

$$S = \{(H, H, H), (H, H, T), (H, T, H), (H, T, T),$$
$$(T, H, H), (T, H, T), (T, T, H), (T, T, T)\}.$$

(b) Let X represent the **number of heads** observed when the three coins are tossed. We can determine the value of X associated with each outcome:

Outcome	**Number of Heads (X)**
(H, H, H)	3
(H, H, T)	2
(H, T, H)	2
(H, T, T)	1
(T, H, H)	2
(T, H, T)	1
(T, T, H)	1
(T, T, T)	0

(c) The possible values that the random variable X can take are:

$$x = 0, 1, 2, 3.$$

That is, when three coins are tossed, we can observe zero, one, two, or three heads.

(d) Since the coins are fair, all eight outcomes in the sample space are equally likely. We can therefore calculate the probability associated with each value of X using:

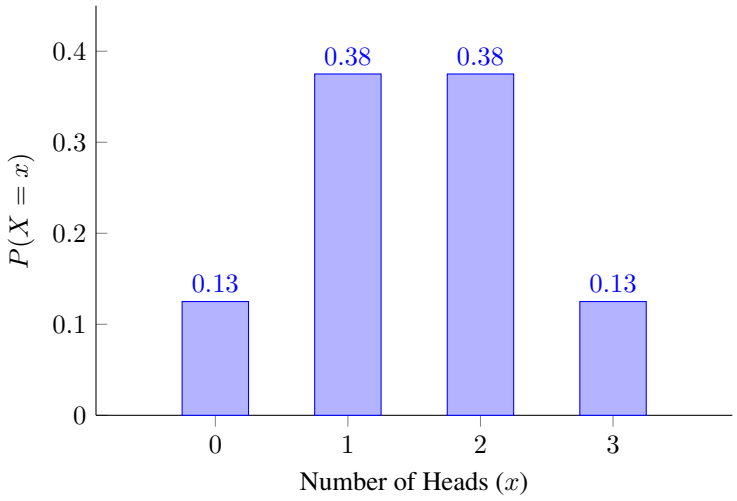
$$P(X = x) = \frac{\text{Frequency of } x}{\text{Total number of possible outcomes}}.$$

Therefore:

$$\begin{aligned} P(X = 0) &= \frac{1}{8} = 0.125, \\ P(X = 1) &= \frac{3}{8} = 0.375, \\ P(X = 2) &= \frac{3}{8} = 0.375, \\ P(X = 3) &= \frac{1}{8} = 0.125. \end{aligned}$$

(e) The probability distribution of X can be illustrated graphically:

Figure 6.4: Probability Distribution of the Number of Heads from Three Coin Tosses



6.3 The Mean and the Expectations Operator

In the previous chapter, we introduced the **mean** as a measure of central tendency. We can apply a similar idea to a probability distribution using the concept of an **expected value**.

> The expected value describes the long-run average value that we would expect from a random variable if an experiment were repeated many times.

Consider the example of rolling a fair six-sided die. Let the random variable X represent the number shown on the die. The possible values of X are:

$$x = 1, 2, 3, 4, 5, 6.$$

Since the die is fair, each value has the same probability:

$$P(X = x) = \frac{1}{6}.$$

The probability distribution of X is therefore:

x	$P(X = x)$
1	$\frac{1}{6}$
2	$\frac{1}{6}$
3	$\frac{1}{6}$
4	$\frac{1}{6}$
5	$\frac{1}{6}$
6	$\frac{1}{6}$

To find the mean of a probability distribution, we multiply each possible value of X by its associated probability and add the results together.

In general, the **expected value** of a random variable is:

$$E(X) = \sum xP(X = x)$$

where E is known as the **expectation operator**.

For the die example, we can write this as:

$$\mu = E(X) = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right)$$

$$= \frac{1+2+3+4+5+6}{6}$$

$$= \frac{21}{6}$$

$$= 3.5.$$

The expected value (mean) of the die roll is therefore 3.5.

Notice that it is impossible to roll a 3.5 on a six-sided die. The expected value does **not** necessarily represent an outcome that can actually occur. Instead, it represents the average value we would expect over a very large number of repeated experiments.

6.4 Variance and Standard Deviation

Just as before, we are also interested in the variance and standard deviation of a particular distribution.

We have already established that the expected value of a fair six-sided die is:

$$\mu = E(X) = 3.5.$$

The **variance** of a discrete random variable measures the expected squared distance of its possible values from the mean.

$$\mathrm{Var}(X) = \sum (x - \mu)^2 P(X = x)$$

Therefore:

$$\begin{aligned}\mathrm{Var}(X) = (1 - 3.5)^2 \left(\frac{1}{6}\right) + (2 - 3.5)^2 \left(\frac{1}{6}\right) \\ + (3 - 3.5)^2 \left(\frac{1}{6}\right) + (4 - 3.5)^2 \left(\frac{1}{6}\right) \\ + (5 - 3.5)^2 \left(\frac{1}{6}\right) + (6 - 3.5)^2 \left(\frac{1}{6}\right).\end{aligned}$$

Therefore:

$$\begin{aligned}\mathrm{Var}(X) &= \frac{6.25 + 2.25 + 0.25 + 0.25 + 2.25 + 6.25}{6} \\ &= \frac{17.5}{6} \\ &= 2.917.\end{aligned}$$

The variance is expressed in squared units. As we learned previously, we can take the square root of the variance to obtain the **standard deviation**:

$$\sigma = \sqrt{\mathrm{Var}(X)}$$

$$= \sqrt{2.917}$$

$$= 1.708.$$

The standard deviation tells us how dispersed the possible values of the random variable are around its expected value. In this example, the possible values of the die roll are distributed around the expected value of 3.5 with a standard deviation of approximately 1.708.

An Alternative Formula for Variance

The variance of a discrete random variable can also be calculated using the expectation operator:

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

We already know that:

$$E(X) = 3.5.$$

We can calculate $E(X^2)$ by multiplying each squared value of X by its associated probability:

$$\begin{aligned} E(X^2) &= 1^2\left(\frac{1}{6}\right) + 2^2\left(\frac{1}{6}\right) + 3^2\left(\frac{1}{6}\right) \\ &\quad + 4^2\left(\frac{1}{6}\right) + 5^2\left(\frac{1}{6}\right) + 6^2\left(\frac{1}{6}\right) \end{aligned}$$

$$= \frac{1+4+9+16+25+36}{6}$$

$$= \frac{91}{6}$$

$$\approx 15.167.$$

Therefore:

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= 15.167 - (3.5)^2$$

$$= 15.167 - 12.25$$

$$\approx 2.917.$$

6.5 Binomial Distribution

Now that we understand what probability distributions are, we can work with the famous kinds of discrete probability distributions.

A random variable X is said to follow a **binomial distribution** if it satisfies these four basic conditions:

1. There is a fixed number of trials n.
2. Each trial has only two possible outcomes, 0 or 1[a].
3. The probability of the outcome occurring is a constant π.
4. The trials n are independent.

If this is satisfied, we denote the random variable X as:

$$X \sim \text{Bin}(n, \pi)$$

Activity:

Consider the experiment of tossing a two coins. Explain how this generates a random variable X that is drawn from a binomial distribution.

[a]This is conventionally called failure or success.

Solution:

The coin toss is either $X = 1$ if heads or $X = 0$ if tails. This experiment satisfies the binomial distribution conditions:

1. There is a fixed number of trials. Tossing two coin means $n = 2$.
2. Each trial only has two possible outcomes, heads or tails (X = 1 or 0).
3. The probability of seeing the event is fixed. This is true because $\pi = 0.5$ in all cases.
4. The trials are independent. The coins being tossed have nothing to do with each other.

Formally, we define the variable X as:

$$X \sim \text{Bin}(2, 0.5)$$

When a random variable satisfies the binomial distribution conditions, then we can apply a general formula to extract the probability of X showing a particular number x:

$$P(X = x) = nCx \cdot \pi^x (1 - \pi)^{n-x}$$

The nCx component is simply the combination rule that we discussed in the previous chapter, but we instead use the values for n and x. The π component represents the probability of the event occurring.

Lets consider again the basic example of two coin tosses. We know that the sample space is:

$$S = \{HH, HT, TH, TT\}$$

If the random variable X measures the number of heads, then we know that the probability distribution is represented by[b]:

[b] Read section 6.2 if you misremember this part.

$$P(X=0)=\frac{1}{4}, \qquad P(X=1)=\frac{2}{4}, \qquad P(X=2)=\frac{1}{4}$$

However since the coin toss generates an the random variable $X \sim \text{Bin}(2, 0.5)$, we can instead just input the values for n and x to recover this answer.

For the formula:

$$P(X=x) = nCx \cdot \pi^x (1-\pi)^{n-x}$$

Given that $X = \{0, 1, 2\}$ and $X \sim \text{Bin}(2, 0.5)$, we have:

$$\begin{aligned} n &= 2, \\ \pi &= 0.5, \\ 1-\pi &= 0.5. \end{aligned}$$

For $X = 0$:

$$\begin{aligned} P(X=0) &= 20(0.5)^0(1-0.5)^{2-0} \\ &= 1(1)(0.5)^2 \\ &= 0.25. \end{aligned}$$

For $X = 1$:

$$\begin{aligned} P(X = 1) &= 2C1(0.5)^1(1 - 0.5)^{2-1} \\ &= 2(0.5)(0.5) \\ &= 0.50. \end{aligned}$$

For $X = 2$:

$$\begin{aligned} P(X = 2) &= 2C2(0.5)^2(1 - 0.5)^{2-2} \\ &= 1(0.5)^2(1) \\ &= 0.25. \end{aligned}$$

which is identical to our answer from before.

Activity

Consider the case of three coins. We already derived the probability distribution from before:

$$\begin{aligned} P(X = 0) &= \frac{1}{8} = 0.125, \\ P(X = 1) &= \frac{3}{8} = 0.375, \\ P(X = 2) &= \frac{3}{8} = 0.375, \\ P(X = 3) &= \frac{1}{8} = 0.125. \end{aligned}$$

Since $X \sim \text{Bin}(3, 0.5)$, verify that these answer are accurate when using the formula:

$$P(X = x) = nCx \cdot \pi^x(1 - \pi)^{n-x}$$

Solution:

Given that $X = \{0, 1, 2, 3\}$ and $X \sim \text{Bin}(3, 0.5)$, we have:

$$\begin{aligned} n &= 3, \\ \pi &= 0.5, \\ 1 - \pi &= 0.5. \end{aligned}$$

We can input each possible value of X into the binomial probability formula.

For $X = 0$:

$$\begin{aligned} P(X = 0) &= 3C0(0.5)^0(1 - 0.5)^{3-0} \\ &= 1(1)(0.5)^3 \\ &= 0.125. \end{aligned}$$

For $X = 1$:

$$\begin{aligned} P(X = 1) &= 3C1(0.5)^1(1 - 0.5)^{3-1} \\ &= 3(0.5)(0.5)^2 \\ &= 0.375. \end{aligned}$$

For $X = 2$:

$$\begin{aligned} P(X = 2) &= 3C2(0.5)^2(1 - 0.5)^{3-2} \\ &= 3(0.5)^2(0.5) \\ &= 0.375. \end{aligned}$$

For $X = 3$:

$$\begin{aligned} P(X = 3) &= 3C3(0.5)^3(1 - 0.5)^{3-3} \\ &= 1(0.5)^3(1) \\ &= 0.125. \end{aligned}$$

When a random variable is drawn from a binomial distribution, we can define its mean and variance as follows:

$$\text{Mean} = n\pi$$

$$\text{Variance} = n\pi(1 - \pi)$$

6.6 Poisson Distribution

Suppose we are interested in the analysing the probability of the below events:

- The number of customers arriving at a BBK branch within 1 hour.
- The number of international calls made in 1 day.
- The number of traffic violations in Riffa in October.

Notice how the events are restricted by hour, day or month. Basically, when a discrete random variable X represents the number of times an event occurs within a specific interval, we say that X is drawn from a **Poisson distribution**[c].

Formally, we denote this as:

$$X \sim \text{Po}(\mu)$$

[c] To be clear, the Poisson distribution can be used for **any interval**, such as time, space, distance, area, volume, etc. For example, we may be interested in the number of defects within one meter of fabric.

where μ represents the average number of occurrences of that event within the specified interval.

Consider an example where a Batelco call centre receives an average 4 calls every 10 minutes. Let us define the random variable X as:

$$X = \text{Number of calls received in 10 minutes}$$

Even though only 4 calls are observed on average within 10 minutes, it is theoretically possible that X can take an infinite amount of (non-negative integer) values:

$$X = \{0, 1, 2, 3, 4, 5, 6, 7, 8, ...\}$$

If the Batelco manager believes that X is drawn from a Poisson distribution, then it they can calculate the probability of any number of calls that might happen within the 10 minute period given the average.

The probability of observing exactly $X = x$ occurrences is:

$$P(X = x) = \frac{e^{-\mu}\mu^x}{x!}$$

For example, the Batelco manager might be interested in calculating the probability of seeing 7 calls within 10 minutes **given that** the average is 4. In this case:

$$P(X = 7) = \frac{e^{-4}4^x}{7!}$$

The Poisson distribution is appropriate under certain assumptions:

- We count the number of occurrences within a specified interval.
- Occurrences happen independently of one another.

- The average rate of occurrence is approximately constant over the interval.
- Occurrences happen individually rather than simultaneously.

Under general conditions, it is possible to show that the Poisson distribution has an identical mean and variance as described below:

$$\mu = n\pi$$

$$\sigma^2 = n\pi$$

Notice how this is the same as the mean of the binomial distribution.

6.7 Ethics

With the ethical uses of statistics:

- Be transparent about distribution assumptions used (e.g., independence).
- Convey expected values honestly, including the variability around them.
- Do not use probability models to create or perpetuate unfair outcomes.

With the ethical uses of using AI:

- AI may apply distribution assumptions (independence, fixed trials) that do not hold, so assumptions must be verified and disclosed.
- AI probability outputs alone should not justify decisions that create or perpetuate unfair outcomes.

6.8 Problem Set

1) The number of new cars sold on a Saturday has the following probability distribution:

x	$P(X = x)$
0	0.10
1	0.30
2	0.40
3	0.20

(a) What type of probability distribution is this?

(b) How many cars do you expect to sell on a Saturday?

(c) What are the variance and standard deviation?

2) A manager for Bahrain City Center analysed a sample of their shops and found that 28% of purchases were made by a credit card.

Suppose the manager randomly selects 10 observations from the sample.

(a) What is the probability that exactly 1 was made by a credit card?

(b) What is the probability distribution of the number of credit card purchases?

(c) What is the probability that 6 or more out of 10 purchases were made by credit card?

(d) What is the mean and standard deviation?

3) Suppose Gulf Air wishes to study the number of checked bags that are lost on its flights. A random sample of 500 flights is examined, and a total of 20 lost bags are recorded.

(a) What is the mean number of lost bags per flight?

(b) Using the Poisson distribution, what is the probability that a randomly selected flight has no lost bags?

6.9 Solution

1) This exercise relates to the mean and variance section of this chapter.

(a) This is a **discrete probability distribution**, since the random variable X can only take a finite number of possible values.

(b) The expected number of cars sold is:

$$E(X) = \sum xP(X = x)$$

$$= 0(0.10) + 1(0.30) + 2(0.40) + 3(0.20)$$

$$= 0 + 0.30 + 0.80 + 0.60$$

$$= 1.70.$$

Therefore:

$$\boxed{E(X) = 1.70}$$

We expect to sell an average of **1.7 cars** on a Saturday.

(c) To calculate the variance, we first calculate $E(X^2)$:

$$E(X^2) = \sum x^2 P(X = x)$$

$$= 0^2(0.10) + 1^2(0.30) + 2^2(0.40) + 3^2(0.20)$$

$$= 0 + 0.30 + 1.60 + 1.80$$

$$= 3.70.$$

We can now calculate the variance using:

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= 3.70 - (1.70)^2$$

$$= 3.70 - 2.89$$

$$= 0.81.$$

The standard deviation is the square root of the variance:

$$\sigma = \sqrt{\text{Var}(X)}$$

$$= \sqrt{0.81}$$

$$= 0.90.$$

2) Let us define X as:

$$X = \{\text{The number of purchases made with a credit card}\}$$

Each purchase is either classified as being with a credit card (X=1) or not (X=0). The question has told us that there 10 purchases selected $n = 10$ and the probability of occurrence is $\pi = 0.28$:

$$X \sim \text{Bin}(10, 0.28)$$

Since there are 10 purchases, the possible values of X are:

$$X = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

(a) We want the probability that one purchase from the 10 was made by credit card. This means:

$$P(X = 1)$$

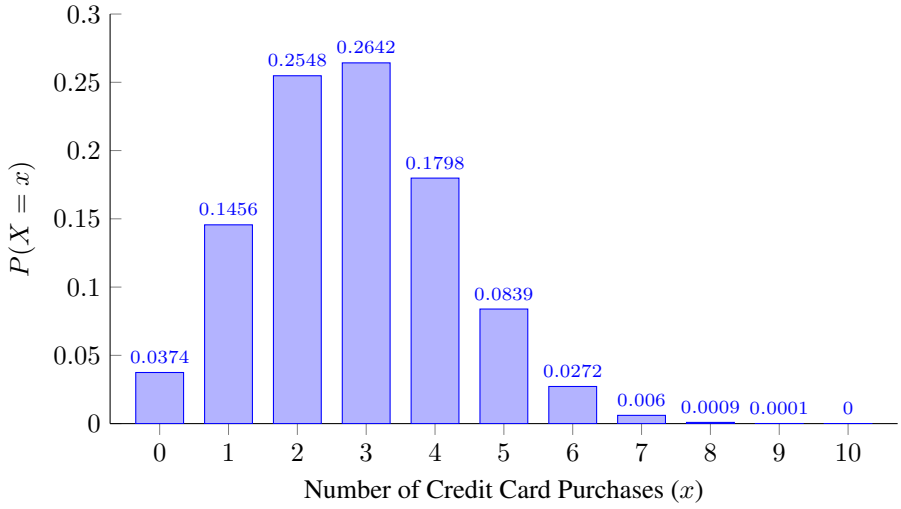
Using the binomial probability formula:

$$P(X = 1) = 10C1 \cdot 0.28^1(1 - 0.28)^{10-1}$$

$$= 0.1456$$

(b) To illustrate the probability distribution, you would need to apply the formula to each value of $X = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

Figure 6.5: Probability Distribution of Credit Card Purchases



Note that $X = 10$ is not exactly zero, but a very small number $0.000...1$.

(c) The probability that we see 6 or more purchases can be denoted as:

$$P(X \geq 6) = P(6) + P(7) + P(8) + P(9) + P(10)$$

$$= 0.027 + 0.006 + 0.0009 + 0.0001 + 0$$

$$= 0.04$$

That is, there is 4% chance of seeing 6 or more credit card purchases.

(d) The mean is given by:

$$\text{Mean} = n\pi = 10 \cdot 0.28 = 2.8$$

The variance is given by:

$$\text{Variance} = n\pi(1 - \pi) = 10 \cdot 0.28 \cdot 0.72 = 2.016$$

Therefore the standard deviation is just the square root of the variance:

$$\text{Standard Deviation} = \sqrt{2.016} = 1.42$$

3)

(a) The mean number of occurrences per flight is:

$$\begin{aligned}\lambda &= \frac{\text{Total number of lost bags}}{\text{Total number of flights}} \\ &= \frac{20}{500} \\ &= 0.04.\end{aligned}$$

(b) Let X represent the number of lost bags on a randomly selected flight.

Since the number of lost bags is assumed to follow a Poisson distribution:

$$X \sim \text{Po}(0.04).$$

The Poisson probability formula is:

$$P(X = x) = \frac{e^{-\mu}\mu^x}{x!}.$$

We want the probability that no bags are lost, so $x = 0$:

$$\begin{aligned}P(X = 0) &= \frac{e^{-0.04}(0.04)^0}{0!} \\ &= e^{-0.04} \\ &= 0.9608.\end{aligned}$$

Chapter 7

Continuous Probability Distributions

7.1 Introduction

In this chapter, we study situations where the random variable X is a continuous number and so it is drawn from a continuous probability distribution.

In order for you to calculate what values X can take, you must examine it across a range within the distribution.

7.2 Uniform Distribution

Consider an example where a passenger is waiting for a train. The passenger knows that the train always arrives at the station in 20 minutes, but they do not know how long it has been since it last arrived.

We are interested in the probability of how long a passenger must wait for the bus to arrive.

Let the random variable X be defined as **the number of minutes a passenger must wait** until the train arrives:

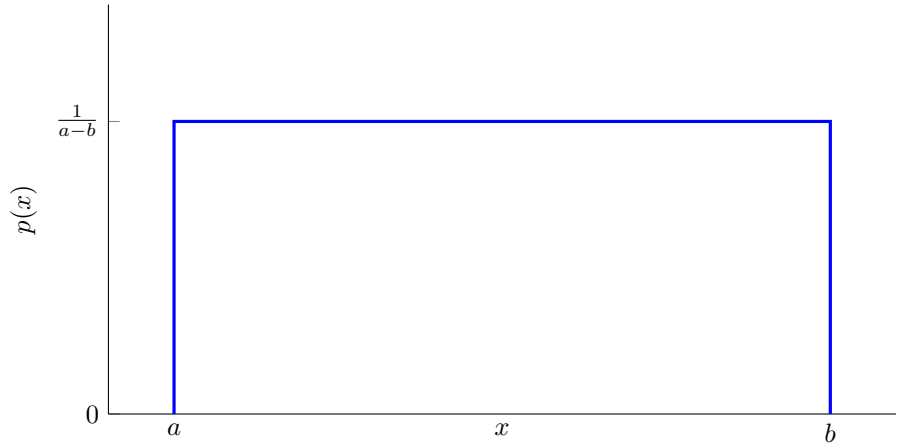
- The maximum number of minutes a passenger must wait is 20.
- The minimum number of minutes a passenger must wait is 0.

When a continuous random variable X has a lower bound a and upper bound b and that the probability of the event is equally likely on all values of X, then we say that **X** is **uniformly distributed**:

$$X \sim U(a, b).$$

We can illustrate this idea on a graph:

Figure 7.1: Uniform Distribution of Variable X



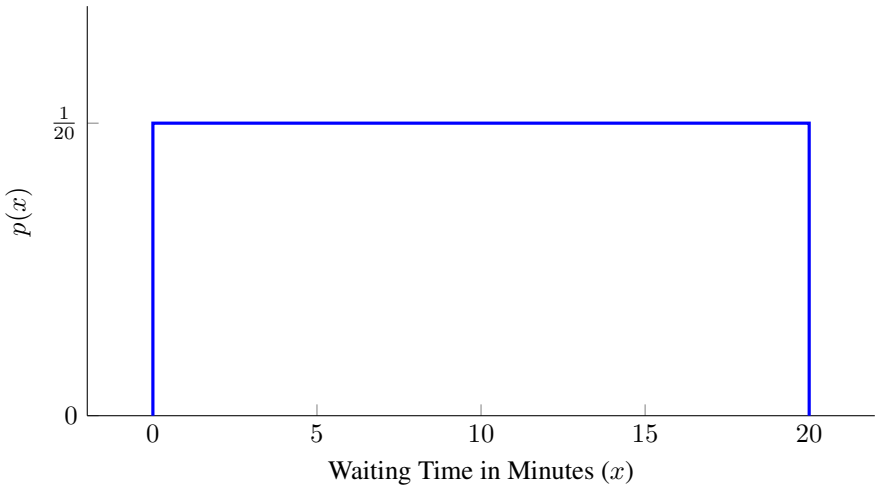
Here, the rectangle is a **uniform probability distribution**. Notice how the probability that any value of X is all the same. It is represented by the height of the rectangle, $\frac{1}{a-b}$. The length of the rectangle is the distance between b and a.

The bus stop scenario is an example of a random variable which is drawn from a uniform distribution. This is because when a passenger arrives at the station, they do not know when the train will arrive, and only that it always arrives in 20 minutes since it last left. The probability that the train arrives is equal for any number between 0 and 20 minutes.

$$X \sim U(0, 20)$$

We can illustrate this on a graph:

Figure 7.2: Uniform Distribution of Passenger Waiting Time



Suppose the passenger arrives at the station and wants to know the probability that they need to wait less than 10 minutes for the train to arrive. Formally, we are trying to find:

$$P(X < 10).$$

To do this, we are essentially trying to find the area of the rectangle[a] between 0 and 10:

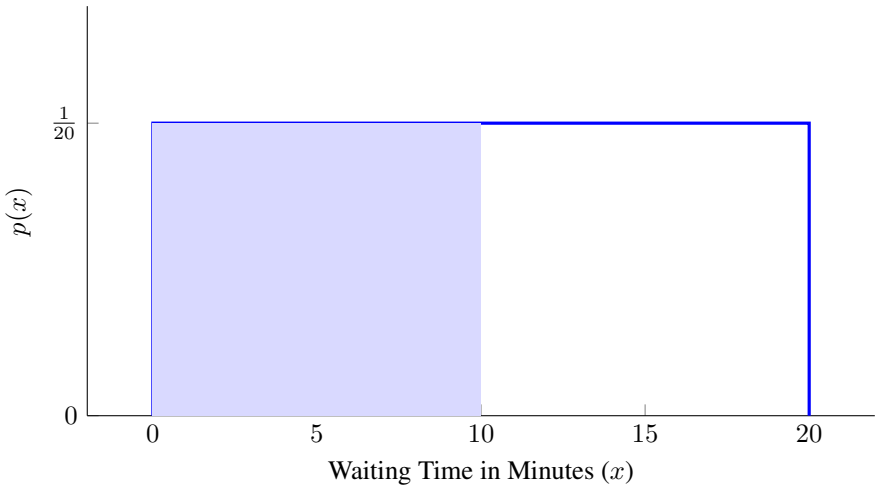
$$P(X < 10) = (10 - 0) \cdot \left(\frac{1}{20} - 0\right)$$

$$= \frac{10}{20} = 0.5$$

that is, the probability that they wait less than 10 minutes is 50%.

[a] Remember that the area of a rectangle is Width x Height.

Figure 7.3: Uniform Distribution of Passenger Waiting Time



Activity

Find the probability that the passenger waits:

(a) Less than 5 minutes.

(b) Less than 20 minutes.

Solution:

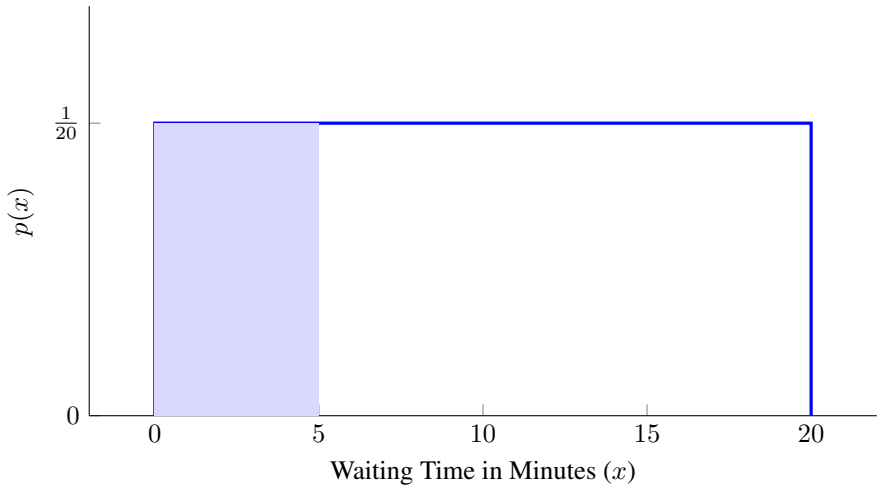
The probability that the passenger waits less than 5 minutes is:

$$P(X < 5) = (5 - 0) \cdot \left(\frac{1}{20} - 0\right)$$

$$= 0.25.$$

There is a 25% chance they have to wait less than 5 minutes:

Figure 7.4: Uniform Distribution of Passenger Waiting Time



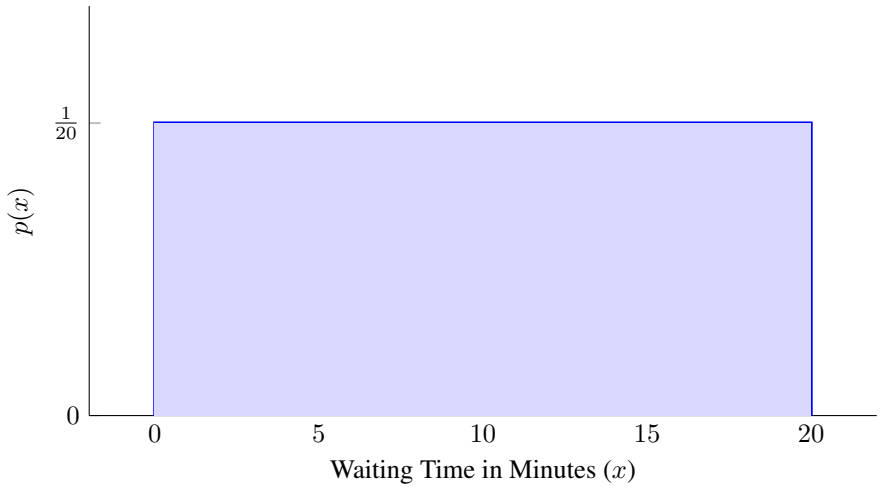
The probability that the passenger waits less than 20 minutes is:

$$P(X < 20) = (20 - 0) \cdot \left(\frac{1}{20} - 0\right)$$

$$= 1.$$

There is a 100% chance the passenger has to wait less than 20 minutes. This is true given that the train always arrives in 20 minutes.

Figure 7.5: Uniform Distribution of Passenger Waiting Time



Before leaving this section, you need to be aware of a technical aspect when we deal with continuous variables. In

particular, we cannot acquire a probability outcome when try to calculate only one exact value of X. For example, the probability that a passenger waits exactly 10 minutes is:

$$P(X = 10) = 0$$

This is because of the nature of continuous variables. In theory, the random variable X can take a infinite number of values. For example, the number of minutes a person waits is actually 10.559232312 minutes.

As a result, whenever we analyse continuous variables, we always specify it within particular ranges:

$$P(X < 10)$$

$$P(10 < X < 20)$$

$$P(X > 15)$$

that way, the entire range of values are covered within the density of the probability distribution.

7.3 Normal Distribution

Previously, we saw that when a continuous variable X is generated from a uniform distribution, then the probability of seeing a value is equally likely.

We now turn towards the most famous kind of continuous variable distribution in statistics. Suppose we generate the random variable X from a distribution with the following observations:

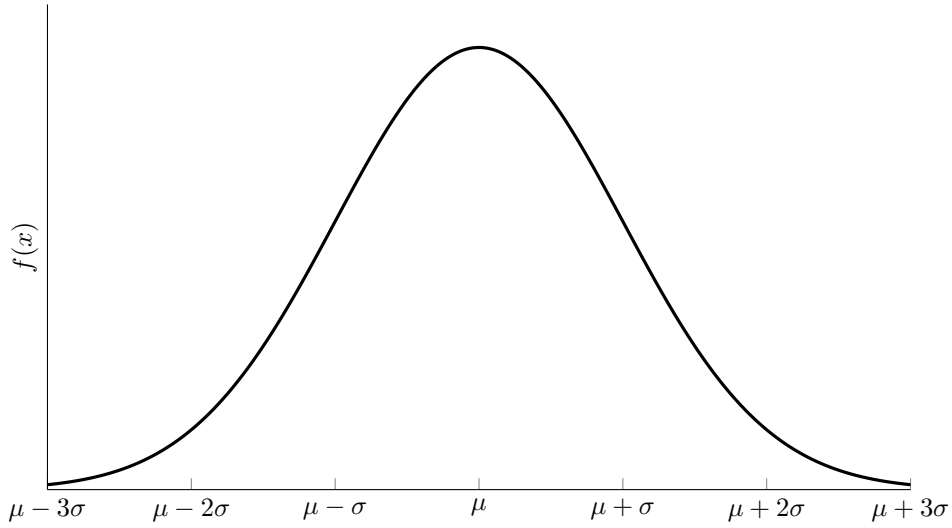
1. Most values tend to be densely concentrated around the mean.
2. Less values tend to be away from the mean.

then this is the characteristic of a **normal distribution**. Formally, we say that X is generated from a normal distribution with mean μ and variance σ^2:

$$X \sim N(\mu, \sigma^2)$$

We can illustrate the normal distribution with the canonical symmetric bell shape:

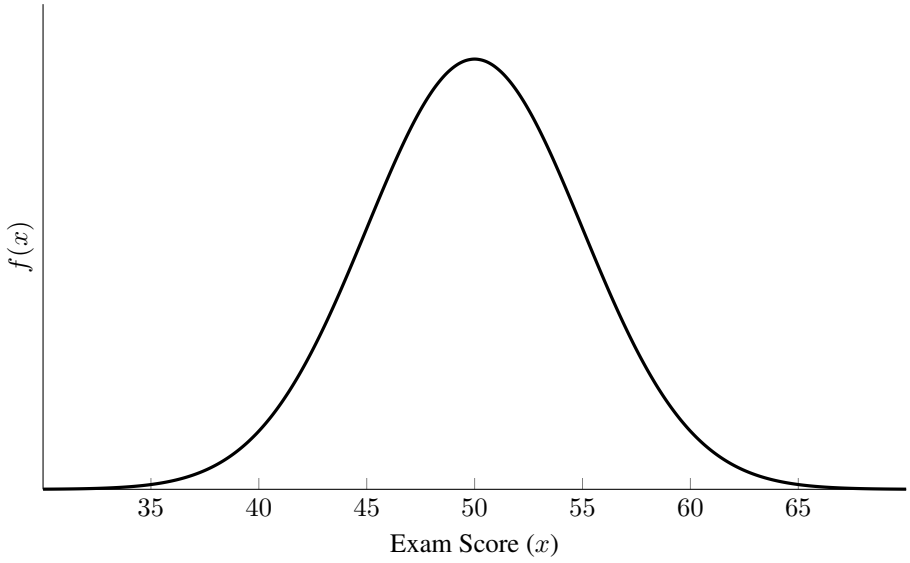
Figure 7.6: The Normal Distribution



Consider a population of students where X represents the mathematics exam score of a randomly selected student. Suppose that exam scores are normally distributed with a mean of 50 and a variance of 25:

$$\mu = 50 \qquad \sigma^2 = 25$$

Figure 7.7: The Normal Distribution

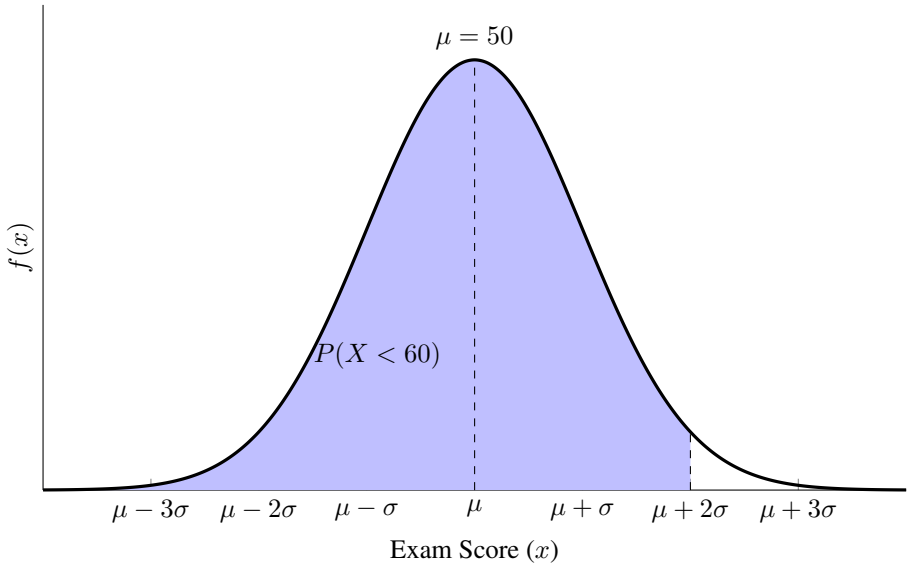


Just like we did before, we can use this information to understand the probability of observing a particular event. For example, we are interested in the probability that we observe a student that scored less than 60 in the exam:

$$P(X < 60)$$

The probability of observing this is represented by the shaded area of the normal distribution:

Figure 7.8: The Normal Distribution



The probability $P(X < 60)$ is therefore represented by the area under the normal density curve to the left of $X = 60$.

However, unlike the uniform distribution, however, calculating this area is not straightforward. The normal distribution is non-linear, so we cannot simply calculate the area using the formula for a rectangle.

To calculate normal probabilities, we can instead **standardise** the values of X and use the **standard normal distribution**.

7.4 Standardising the Normal Distribution

In order to calculate the area underneath the curve in an effective way, statisticians tend to **standardise** the normal distribution. This means that no matter what X is, so long as it is normally distributed, then we can calculate the probabilities of observing an event in the same way.

We define the following formula:

$$z = \frac{x - \mu}{\sigma}$$

When you input the values of x, μ and σ, then you will be able to measure how many **standard deviations** a particular value of x is away from the mean.

Activity:

Continuing from our exam score example:

$$X \sim N(50, 25)$$

Using the z formula, calculate how many standard deviations away the following exam scores are from the mean:

(a) $x = 60$

(b) $x = 43$

(c) $x = 50$

Solution:

Using the formula:

(a) When $x = 60$:

$$z = \frac{60 - 50}{5} = 2$$

The score of 60 is 2 standard deviations away from the mean.

(b) When $x = 43$:

$$z = \frac{43 - 50}{5} = -1.4$$

The score of 43 is -1.4 standard deviations away from the mean.

(c) When $x = 50$:

$$z = \frac{50 - 50}{5} = 0$$

The score of 50 is 0 standard deviations away from the mean.

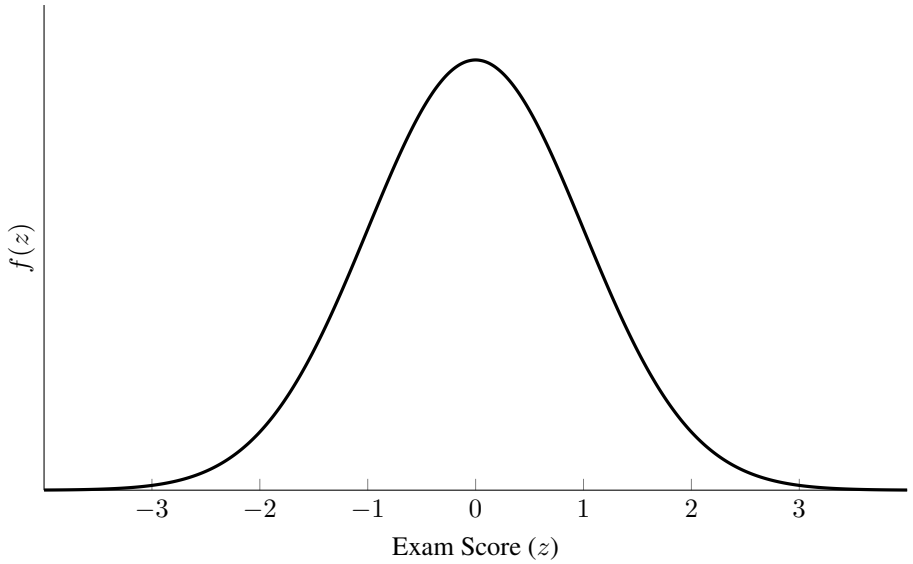
Because of this, we don't actually need to list out the values of exam scores on the horizontal axis. So long as the data follows from a normal distribution, we can just standardise the horizontal axis. Formally, the **standard normal distribution** is defined as:

$$X \sim N(0, 1)$$

That is, X has a mean $\mu = 0$ and a variance $\sigma^2 = 1$. Note that the standard deviation is the square root of the variance,

and that is also just $\sigma = 1$.

Figure 7.9: The Standard Normal Distribution



7.4.1 Case 1

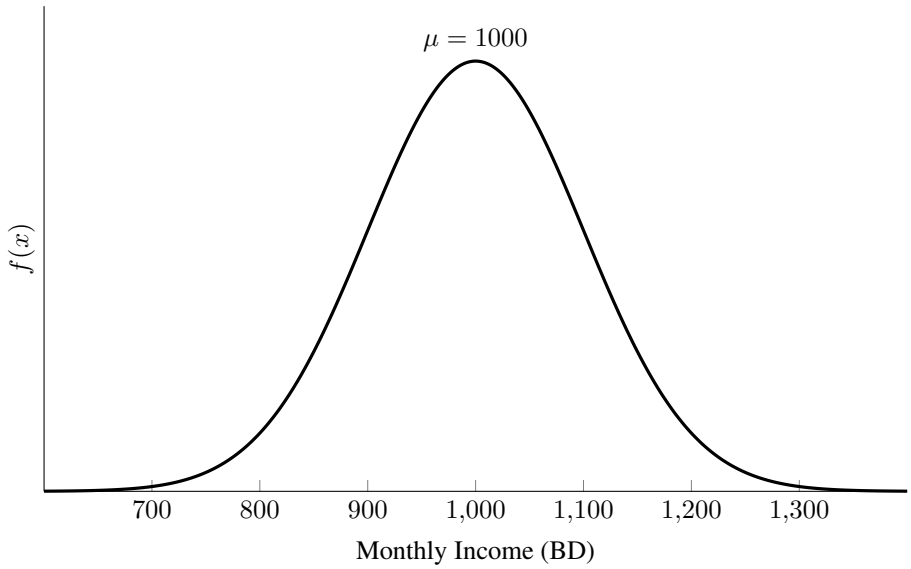
Now that we have standardised the horizontal axis, we can start with learning how to calculate the probability of observing values of the random variable X.

Consider if you have data on the income of all Bahrainis. You discover that the average monthly income $\mu = 1000$ and its variance $\sigma^2 = 10,000$:

$$X \sim N(1000, 10000)$$

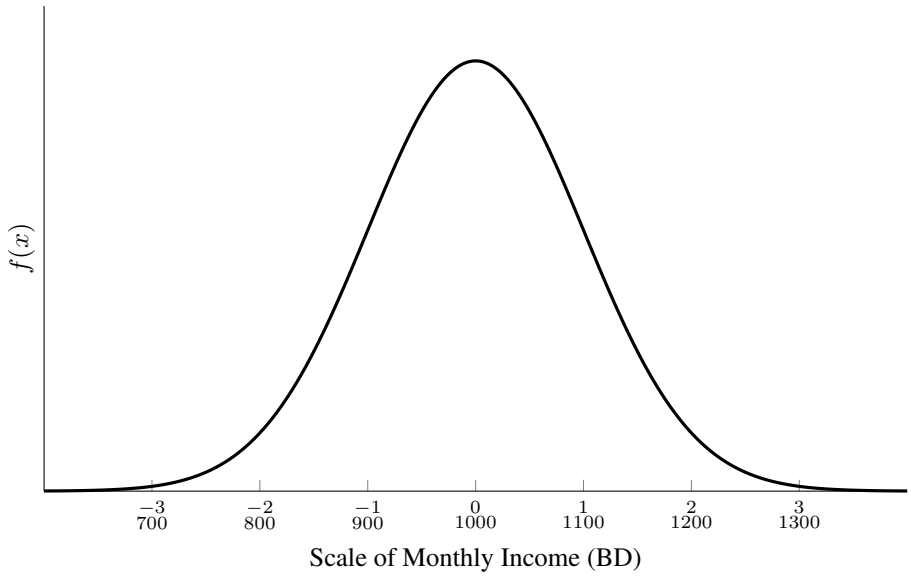
which implies that $\sigma = 100$:

Figure 7.10: Normal Distribution of Monthly Income



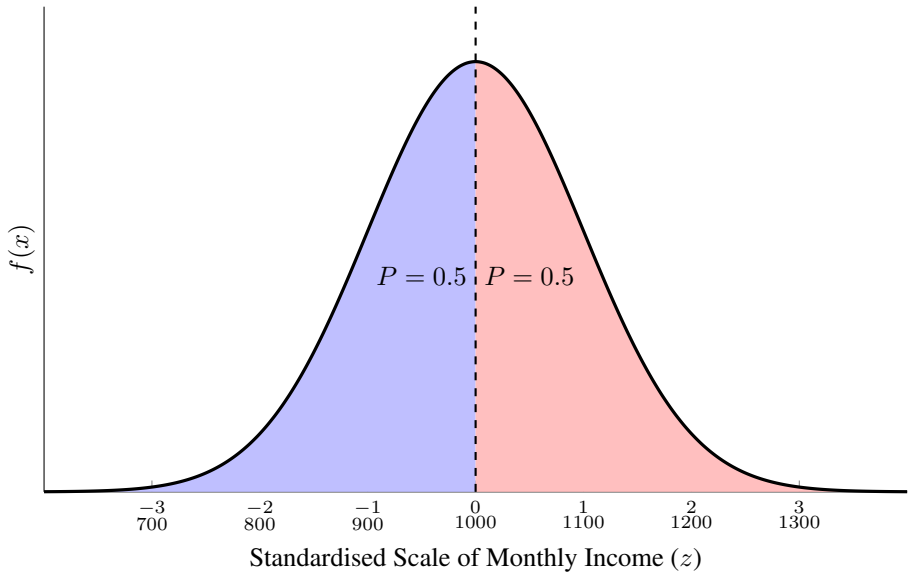
In order to easily calculate the probability, we instead work with the standardised values of X where we just measure how far away it is from the mean:

Figure 7.11: Standard Normal Distribution of Income



Let us start with a basic fact, in that we say **half** of the observations are located **under the mean** while the other **half** are located **above the mean**:

Figure 7.12: Standard Normal Distribution of Income



Let us start with a basic exercise and calculate how many standard deviations away a person is that earns 1000 BD and another person earning 1100 BD:

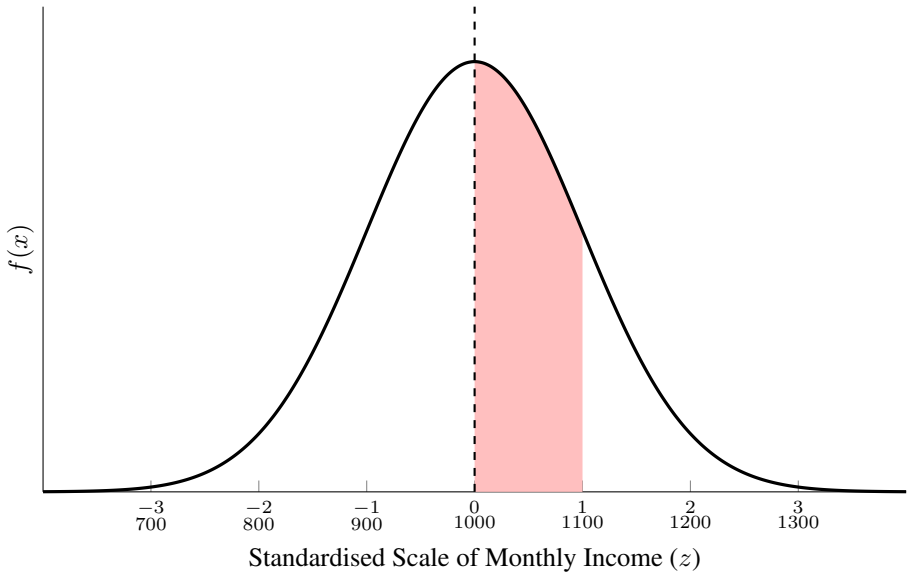
$$z = \frac{1000 - 1000}{100} = 0$$

$$z = \frac{1100 - 1000}{100} = 1$$

that is, a person earning 1000 BD is 0 standard deviations away from the mean, whereas a person earning 1,100 BD is 1 standard deviation away from the mean.

From the population, a researcher is interested in determining the probability that they observe someone who has an income **between** 1000 and 1100. Graphically, this is simplt the shaded region:

Figure 7.13: Standard Normal Distribution of Income



To answer the question, you would need to calculate the area of the red region, however this is not straight forward given that this is a non-linear function and you would need to rely on calculus.

However given that the values of X has been standardised, then so long as the distribution is normal, we can calculate the area of any region using this table:

Table 7.1: Standard Normal Distribution Table: $P(0 < Z < z)$

z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946

What each of these values represent is the area in which:

$$P(0 < Z < z)$$

In the case of finding the probability that a person earns between 1000 and 1100 BD, we are simply asking:

$$P(1000 < X < 1100) = P(0 < Z < 1)$$

To answer this, look at the first column and find the value for $z = 1$ under the column $z = 0.00$:

Table 7.2: Standard Normal Distribution Table: $P(0 < Z < z)$

z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749

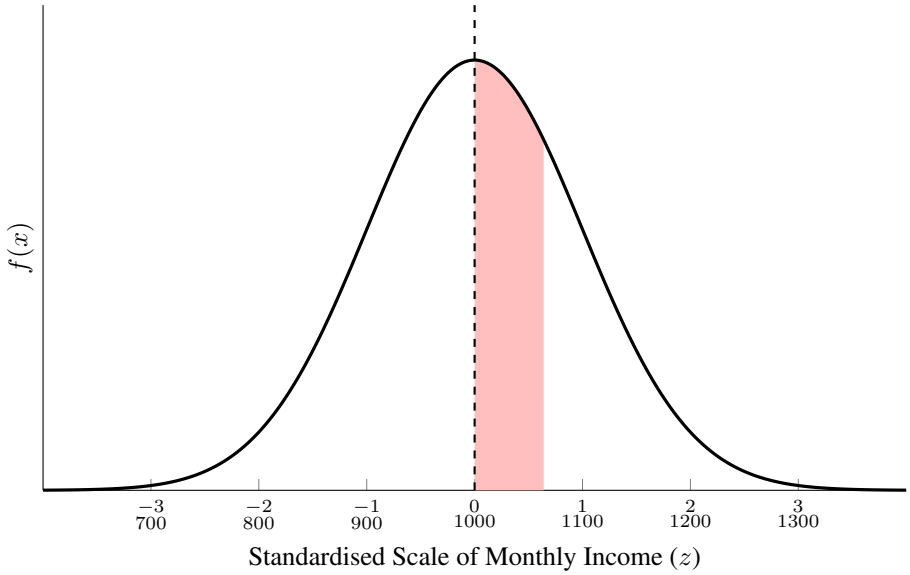
What this value means is that the probability of observing a value between $z = 0$ and $z = 1$ is 0.3413. Formally:

$$P(1000 < X < 1100) = P(0 < Z < 1) = 0.3413$$

which is 34.13%.

To understand why we only look at the first column at 0.00, let us consider another example where we try to determine the probability that a person earns between 1000 BD and 1064 BD:

Figure 7.14: Standard Normal Distribution of Income



The z values are:

$$z = \frac{1000 - 1000}{100} = 0$$

$$z = \frac{1064 - 1000}{100} = 0.64$$

Therefore, we want the probability that $P(1000 < X < 1064) = P(0 < Z < 0.64)$. Notice how in the table we do not have the exact value for $z = 0.64$. To determine the probability, we need to find the row where $z = 0.6$ and the column that says 0.04:

Table 7.3: Standard Normal Distribution Table: $P(0 < Z < z)$

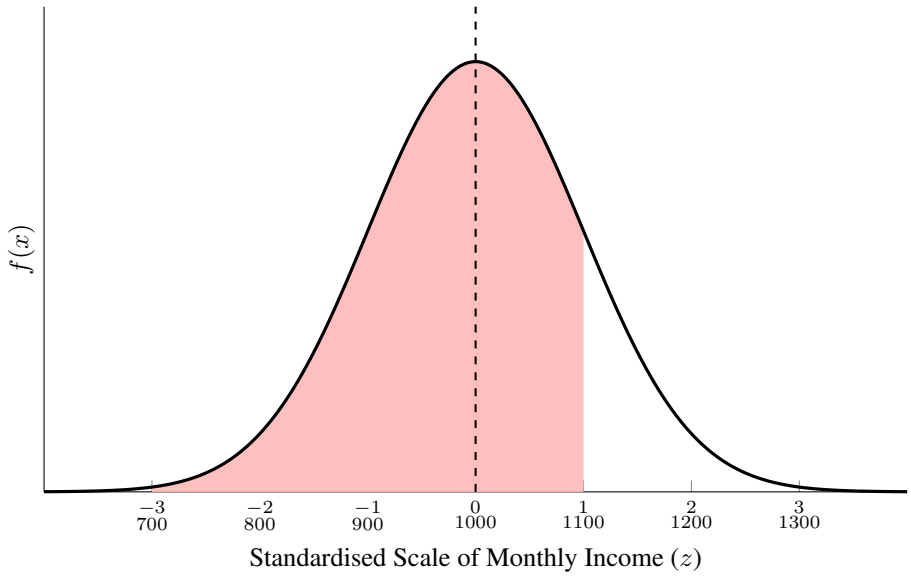
z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734

Therefore $P(0 < Z < 0.64) = 0.2389$ or 23.89%. This is how you would know which **row** and **column** to use for each case.

7.4.2 Case 2

Now suppose we are interested in determining the probability that a person earns less than 1100 BD per month. Graphically, this is simply:

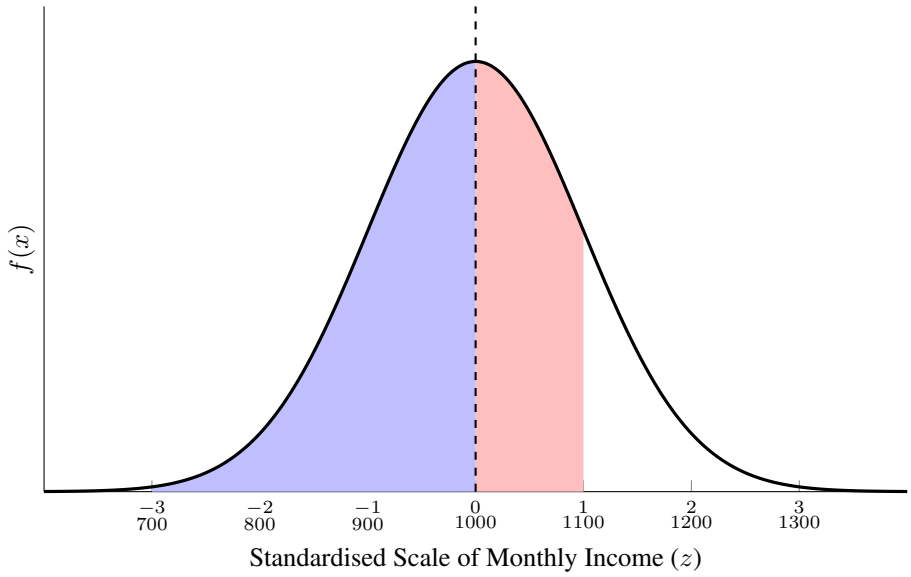
Figure 7.15: Standard Normal Distribution of Income



The problem is that the z-table only gives you the probability values for when the lower bound is always fixed at $z = 0$. Therefore, you cannot look at the table and find $P(X < 1100) = P(Z < 1)$.

However there is a way around this. Remember that we said earlier that all values less than the mean represent half of the observations. Intuitively, this means that the probability of observing **anyone** with $z < 0$ is simply $P = 0.5$:

Figure 7.16: Standard Normal Distribution of Income



where the blue region represents $P(X < 1000) = P(Z < 0) = 0.5$.

We also know what the red region is:

$$z = \frac{1100 - 1000}{100} = 1$$

which is $P(1000 < X < 1100) = P(0 < Z < 1) = 0.3413$.

Therefore, the probability of observing someone that earns less than 1100 BD is:

$$= P(X < 1000) + P(1000 < X < 1100)$$

$$= P(Z < 0) + P(0 < Z < 1)$$

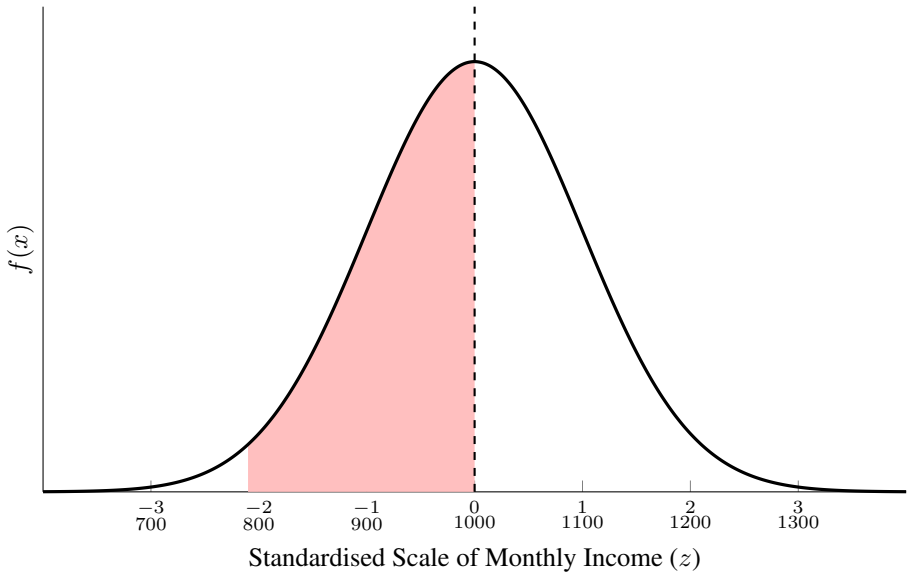
$$= 0.5 + 0.3413 = 0.8413$$

which is 84.13%.

7.4.3 Case 3

Now suppose we want to calculate the probability that a person earns between 790 BD and 1000 BD. This is similar to what we did in Case 1, but this time the value of z will be below 0:

Figure 7.17: Standard Normal Distribution of Income



Let us get the values of z:

$$z = \frac{790 - 1000}{100} = -2.1$$

$$z = \frac{1000 - 1000}{100} = 0$$

Remember that the table shows probability values for $P(0 < Z < z)$. The problem is that $z = -2.1$ is **less** than 0.

However, we can easily get around this by understanding that the normal distribution is **symmetric** around the mean. This means that all the probability values are identical for both positive and negative values of z.

Therefore, to find the probability that someone earns between 790 and 1000 BD, simply take the **absolute value** (ignore the negative sign) of the z:

$$|z| = 2.1$$

Therefore, when looking at the table:

Table 7.4: Standard Normal Distribution Table: $P(0 < Z < z)$

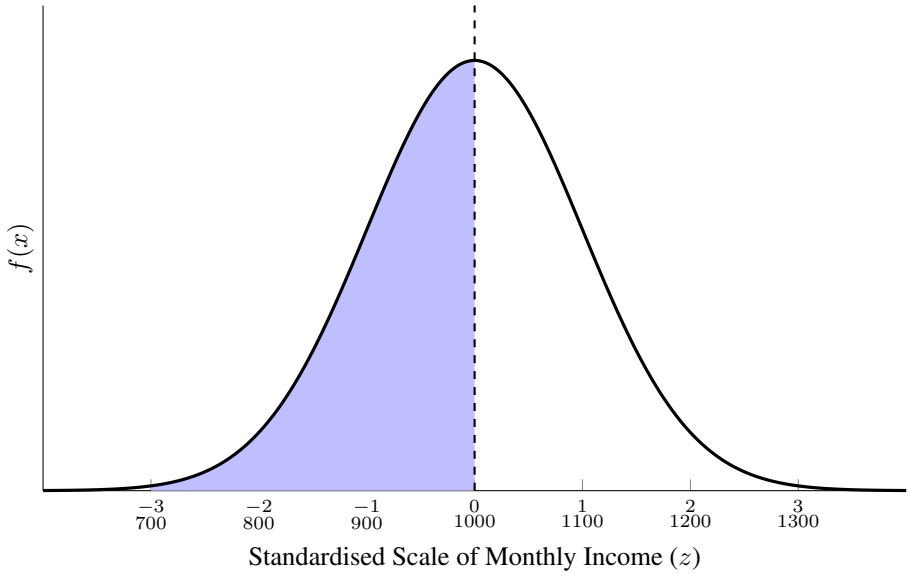
z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878

Therefore, the probability of observing someone between 790 and 1000 BD is $P(790 < X < 1000) = 0.4821$.

7.4.4 Case 4

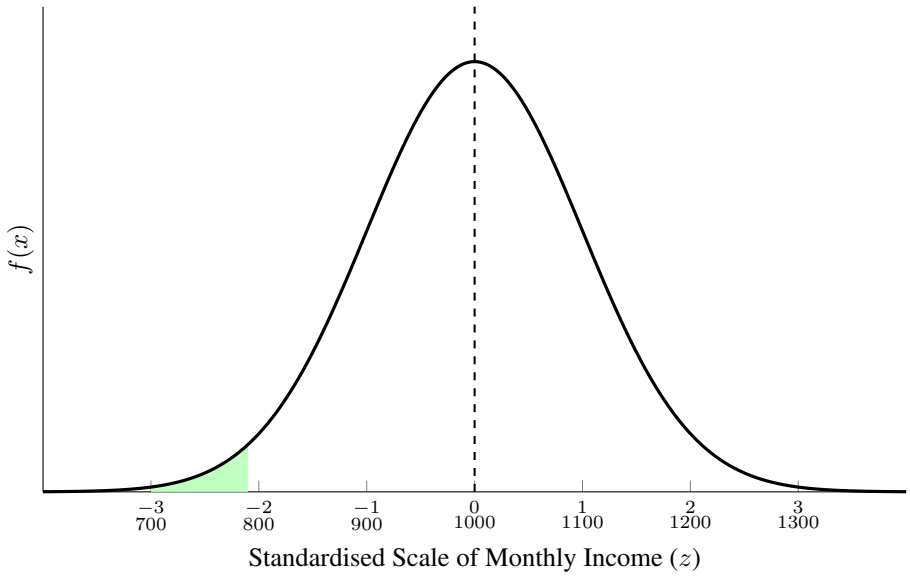
To understand how to solve this next case, remember that the probability of observing any value below the mean is simply $p = 0.5$ if the distribution is normal:

Figure 7.18: Standard Normal Distribution of Income



Suppose now we want to know the probability that we observe someone that earns less than 790 BD $P(X < 790)$. Graphically this is simply:

Figure 7.19: Standard Normal Distribution of Income

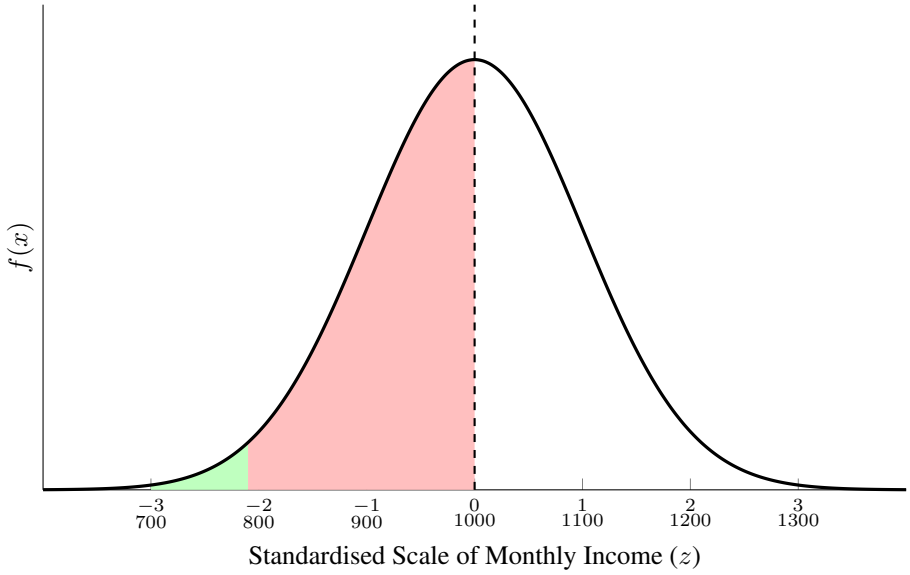


Once again, you cannot get this area from the table since it identifies the calculations only with a lower bound and an upper bound rather than just $P(X < 790)$. This means that if you were to calculate the z:

$$z = \frac{790 - 1000}{100} = -2.1$$

and so $|z| = 2.1$, then this is the area between 0 and 2.1:

Figure 7.20: Standard Normal Distribution of Income



which we know is $P(0 < Z < 2.1) = 0.4821$:

Table 7.5: Standard Normal Distribution Table: $P(0 < Z < z)$

z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878

Using this information, it is possible to determine the area of the green region. Since the probability of observing any number below the mean is $P(X < 1000) = P(Z < 0) = 0.5$, then it must mean that the green region is:

$$P(Z < 0) - P(0 < Z < 2.1)$$

$$= 0.5 - 0.48921$$

$$= 0.01079,$$

which means you have a 1.079% chance of observing a Bahraini who earns less than 790 BD.

7.4.5 Case 5

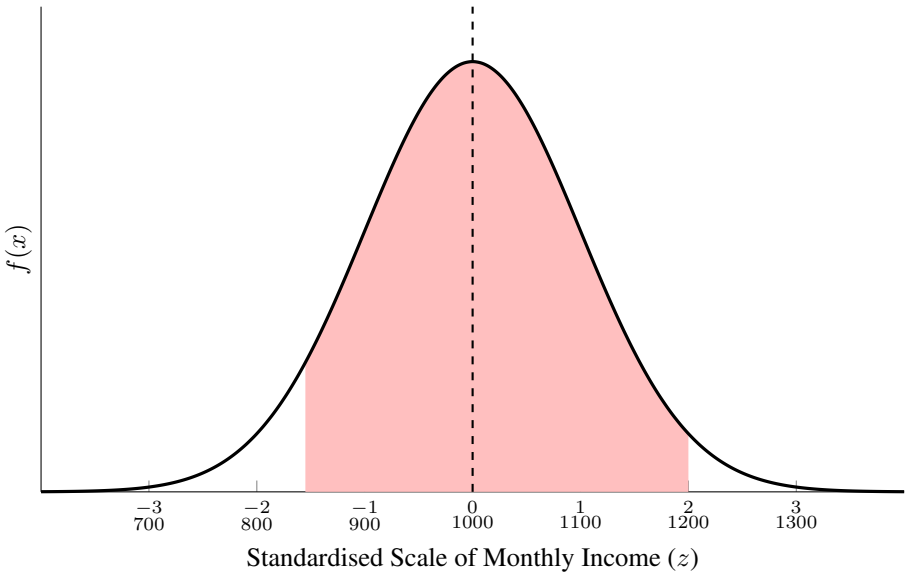
Suppose we are now interested in the probability that we observe a Bahraini who earns between $P(845 < X < 1200)$. Let us first get the z values:

$$z = \frac{845 - 1000}{100} = -1.55$$

$$z = \frac{1200 - 1000}{100} = 2$$

In the figure, we are interested in the probability of the regions between $z = -1.55$ and $z = 2$:

Figure 7.21: Standard Normal Distribution of Income



Remember that since the distribution is symmetric, we do not need to concern ourselves with the negative sign, and so we can just find for $|z| = 1.55$ and $z = 2$:

Table 7.6: Standard Normal Distribution Table: $P(0 < Z < z)$

z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842

Therefore the probability of observing someone who earns between 890 BD and 1200 BD is:

$$= P(890 < X < 1200)$$

$$= P(0 < Z < 1.55) + P(0 < Z < 2)$$

$$= 0.4394 + 0.4772$$

$$= 0.9166$$

which is a 91.66% probability.

7.4.6 Case 6

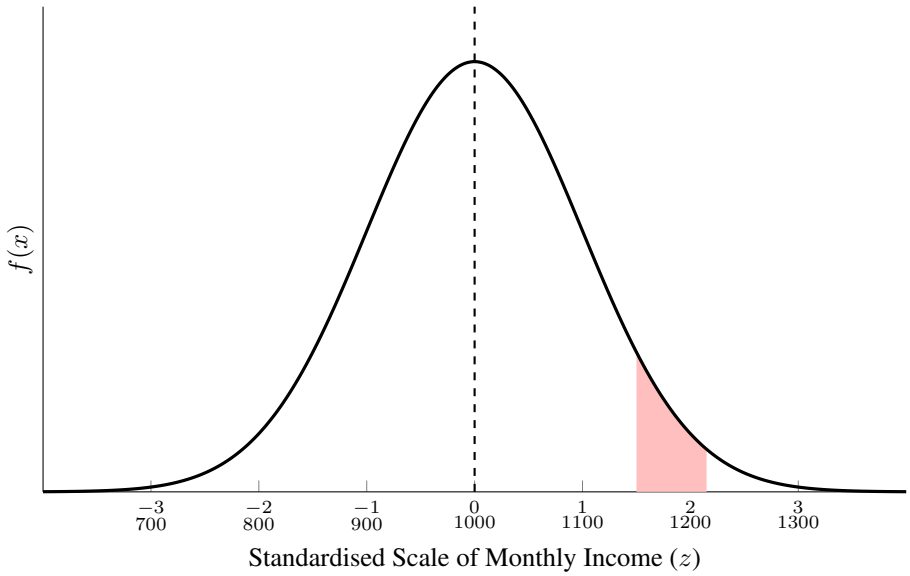
Now consider the probability of observing a Bahraini who earns between $P(1150 < X < 1215)$. Let us start by getting the z values:

$$z = \frac{1150 - 1000}{100} = 1.5$$

$$z = \frac{1215 - 1000}{100} = 2.15$$

Graphically:

Figure 7.22: Standard Normal Distribution of Income



Now remember that the problem with the z table is that it shows you probability values where the lower bound is always 0 and the upper bound is only one number when you are analysing $P(0 < Z < z)$. Therefore you cannot use the table on its own to reveal $P(1.5 < Z < 2.15)$. Instead, you can only get:

$$P(0 < Z < 1.5) = 0.4332$$

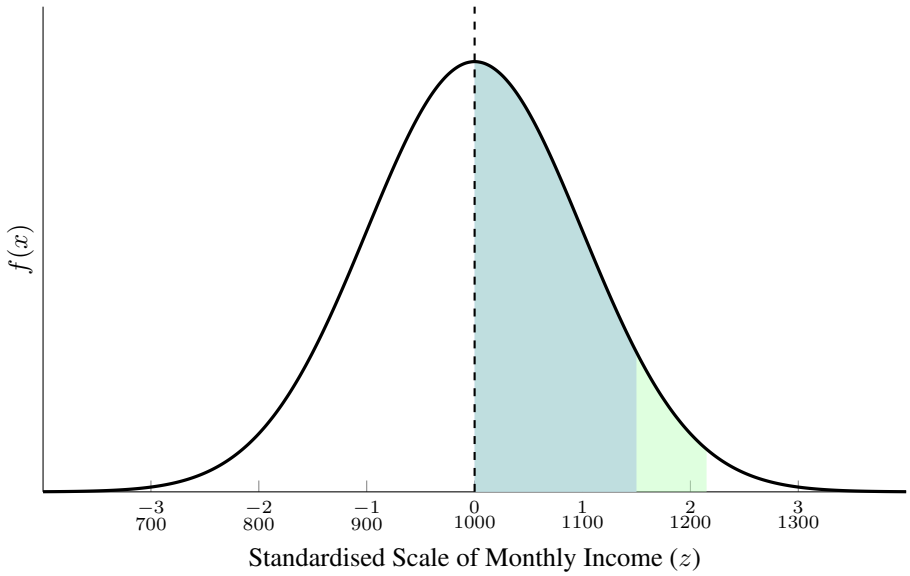
$$P(0 < Z < 2.15) = 0.4842$$

Table 7.7: Standard Normal Distribution Table: $P(0 < Z < z)$

z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878

Graphically, these regions overlap:

Figure 7.23: Standard Normal Distribution of Income



You can acquire the value of the overlapping region by just subtracting the two probabilities:

$$= P(1000 < X < 1215) - P(1000 < X < 1150)$$

$$= P(0 < Z < 1.5) - P(0 < Z < 2.15)$$

$$= 0.4842 - 0.4332$$

$$= 0.0510$$

hence the area of the shaded region represents a probability of 5.1%.

7.4.7 Case 7

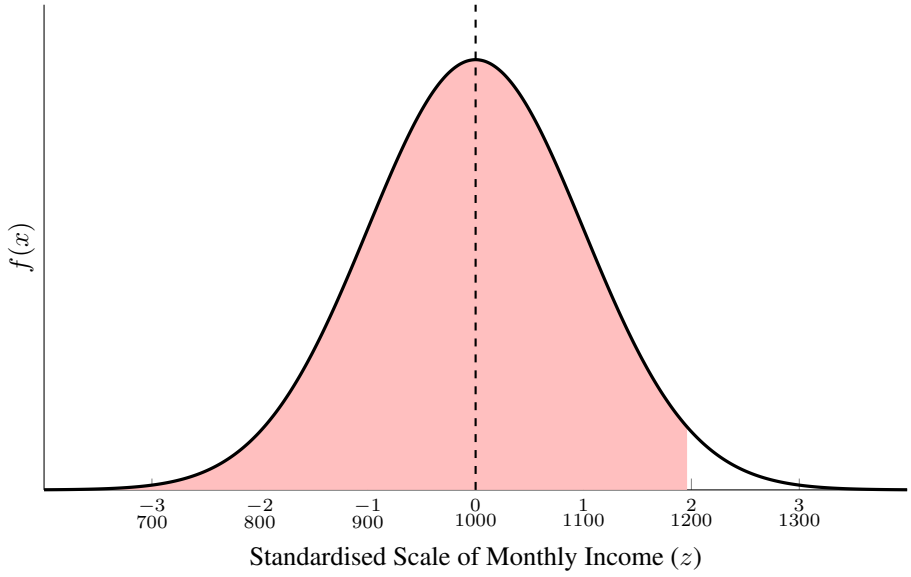
Before going through this case, you must remember the concept of a percentile. Recall that the kth percentile of a dataset is the location to which $k\%$ of data are below it.For example:

1. The 95th percentile is the value of X such that 95% of the data values are below it.

2. The 5th percentile is the value of X such that 5% of the data values are below it.

For the Bahraini income example, suppose you are interested in knowing the value of x which is the 97.5th percentile of the data. Graphically, this looks like:

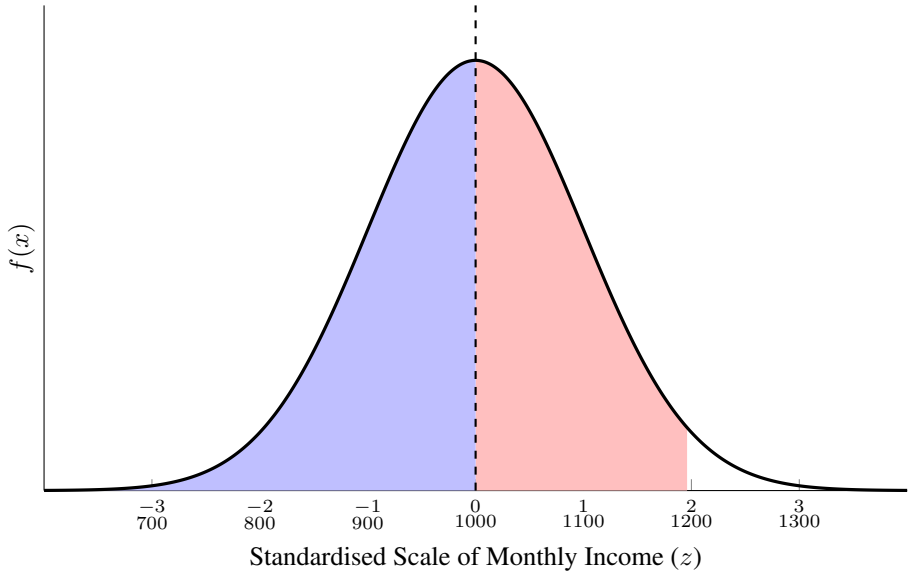
Figure 7.24: Standard Normal Distribution of Income



In other words, we want the value of income x such that we have a 97.5% chance of observing it. The graph shows you the region where $P(X < x) = 0.975$. We are interested in finding the value of this x.

Let us start by finding the z value. Remember that table only shows you values for $P(0 < Z < z)$. We know that all the values below the mean represent $P = 0.5$:

Figure 7.25: Standard Normal Distribution of Income



Therefore:

$$= P(X < x) = 0.975$$

$$= P(X < 1000) + P(1000 < X < x) = 0.975$$

$$= P(Z < 0) + P(0 < Z < z) = 0.975$$

$$= 0.5 + P(0 < Z < z) = 0.975$$

$$= P(0 < Z < z) = 0.475$$

We can now find the value of z that gives us $P(0 < Z < z) = 0.475$:

Table 7.8: Standard Normal Distribution Table: $P(0 < Z < z)$

z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.48033

Therefore, $P(0 < Z < z) = 0.475$ only when $z = 1.96$.

Now that we know the value of z which gives us the 97.5th percentile of x, we can substitute it in the z formula:

$$z = \frac{x - \mu}{\sigma} = 1.96$$

$$= \frac{x - 1000}{100} = 1.96$$

$$x = 1196$$

therefore, A Bahraini earning $x = 1196$ is the 97.5th percentile of the data.

7.5 The Empirical Rule

Recall the Empirical Rule for a normal distribution:

- Approximately 68% of observations lie within one standard deviation of the mean.
- Approximately 95% of observations lie within two standard deviations of the mean.
- Approximately 99.7% of observations lie within three standard deviations of the mean.

7.6 Ethics

For ethical uses with statistics:

- Be transparent about the assumption of normality before applying it.
- Convey z-scores and percentiles in honest, meaningful context.
- Disclose the source and currency of population parameters (μ, σ) used.

For ethical uses in AI:

- The inputs AI used (mean, SD) must reflect current, correct data sources.

7.7 Problem Set

1) The closing price of Batelco's share value is uniformly distributed with a lower bound of 20 BD and upper bound of 30 BD. Formally:

$$X \sim U(20, 30)$$

(a) More than \$27?

(b) Less than or equal to \$24?

2) Customers at Batelco experience technical difficulty with their internet cable can call a support line for technical assistance. It takes the technician between 30 seconds and 10 minutes to resolve the problem.

The distribution of this support time follows a uniform distribution.

(a) What are the values of a and b in **minutes**?

(b) What percent of the problems take more than 5 minutes to resolve?

3) The Bahraini Ministry of Finance reported an average tax income of 2573 BD with standard deviation of 150 BD from all businesses. Let us assume that the tax income X is normally distributed.

(a) What percent of the refunds are more than 2,885 BD?

(b) What percent of the refunds are more than 2,885 BD but less than 3,017 BD?

(c) What percent of the refunds are more than 2,606 BD but less than 3,017 BD?

4) Assume that the hourly operating cost of a commercial aircraft in Bahrain follows a normal distribution with a mean of 2,100 BD per hour and a standard deviation of 250 BD.

What is the operating cost for the lowest 3% of aircraft?

5) Suppose that the examination scores of students at a university in Bahrain follow a normal distribution with a mean of 60 and a standard deviation of 5.

(a) About what percent of students score between 55 and 65?

(b) About what percent of students score between 50 and 70?

(c) About what percent of students score between 45 and 75?

6) Suppose that residents of Bahrain spend an average of 270 minutes per day watching television. Assume that the distribution of daily television viewing time follows a normal distribution with a standard deviation of 23 minutes.

(a) What percent of residents watch more than 300 minutes per day?

(b) What percent of residents watch more than 220 minutes per day?

(c) What percent of residents watch between 220 and 300 minutes per day?

(d) Let us define a "binge watcher" as someone in the upper 15% of the distribution of daily television viewing time. How many minutes per day does a "binge watcher" spend watching television?

7) Suppose that the end-of-day share price of the National Bank of Bahrain (NBB), listed on the Bahrain Bourse, follows a normal distribution.

Assume that there were 240 trading days during the year. The mean closing price was BD 0.420 per share, with a standard deviation of BD 0.0225 per share.

(a) What is the probability that the end-of-day share price exceeds BD 0.450? Estimate the number of trading days during which the closing price exceeded BD 0.450.

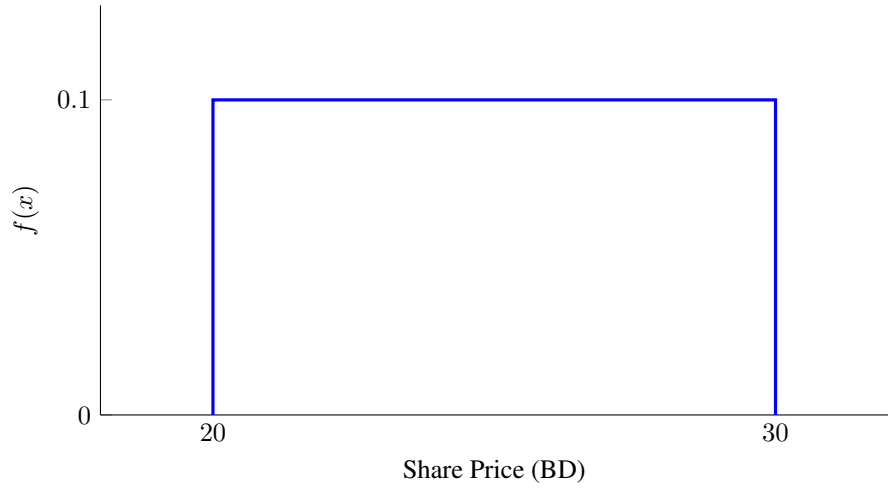
(b) What percent of trading days had a closing price between BD 0.380 and BD 0.400?

(c) What is the minimum share price for the highest 15% of end-of-day closing prices?

7.8 Solution

1) It might be helpful to illustrate the relationship on a graph. Given that X is uniformly distributed, any price of the share between the bounds is equally likely:

Figure 7.26: Uniform Distribution: $P(X > 27)$

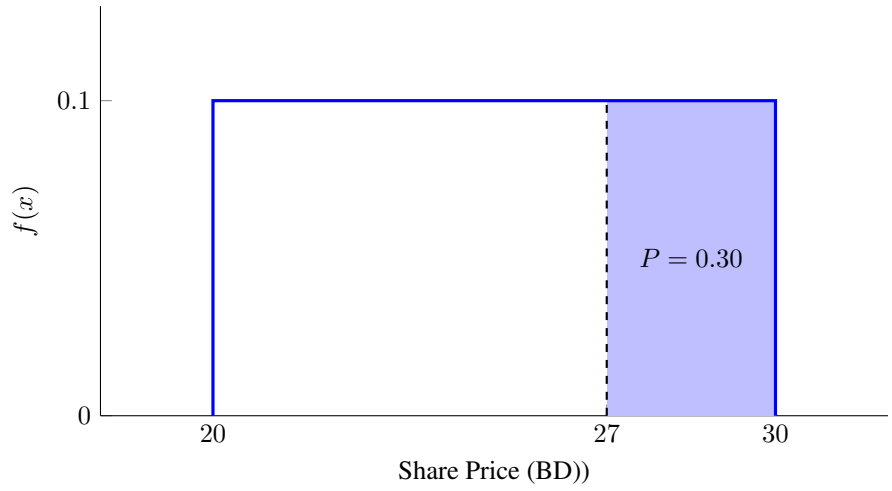


(a) Probability that the stock price is more than \$27:

$$P(X > 27) = (30 - 27) \cdot \left(\frac{1}{30 - 20} - 0 \right)$$
$$= \frac{3}{10} = 0.3$$

Therefore, there is a 30% probability that the stock price exceeds \$27.

Figure 7.27: Uniform Distribution: $P(X > 27)$

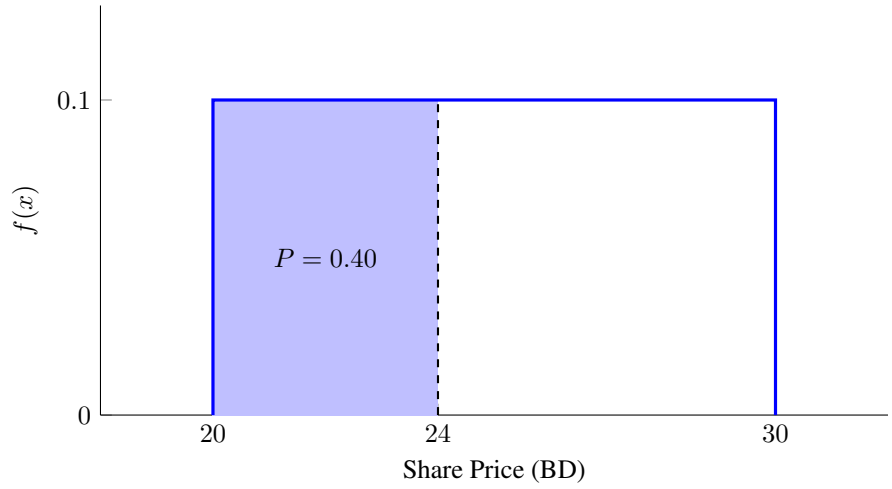


(b) Probability that the stock price is less than or equal to \$24:

$$P(X \leq 24) = (24 - 20) \cdot \left(\frac{1}{30 - 20} - 0 \right)$$
$$= \frac{4}{10} = 0.4$$

Therefore, there is a 40% probability that the stock price is less than or equal to \$24.

Figure 7.28: Uniform Distribution: $P(X \leq 24)$



2) Let X represent the time (in minutes) required to resolve a technical support problem.

(a) The minimum support time is 30 seconds. Converting this into minutes:

$$30 \text{ seconds} = \frac{30}{60} = 0.5 \text{ minutes}$$

The maximum support time is 10 minutes.

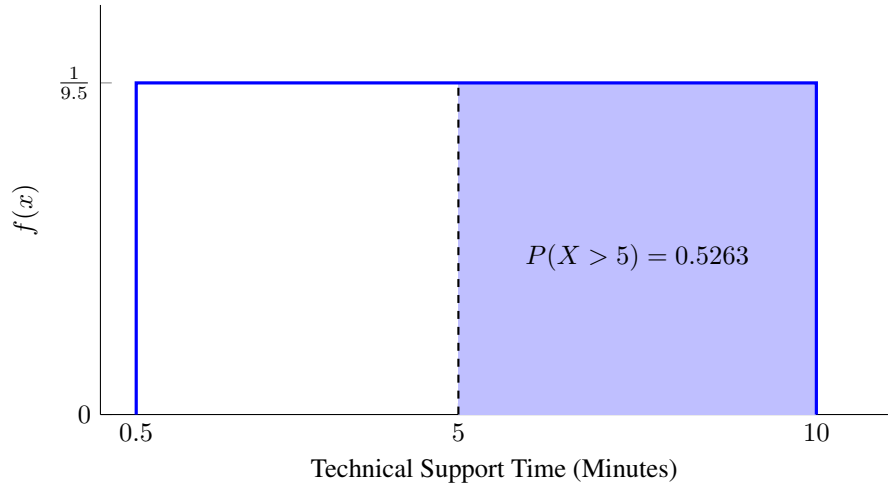
Hence, the support time X follows a uniform distribution such that:

$$X \sim U(0.5, 10)$$

(b) We want to calculate:

$$P(X > 5)$$

Figure 7.29: Uniform Distribution of Technical Support Time: $P(X > 5)$



Therefore:

$$\begin{aligned} P(X > 5) &= \frac{10 - 5}{10 - 0.5} \\ &= \frac{5}{9.5} \\ &= 0.5263 \end{aligned}$$

Therefore, approximately 52.63% of technical support problems take more than 5 minutes to resolve.

3) Let X represent the refund amount. We are given:

$$X \sim N(2573, 150^2)$$

Recall the standardisation formula:

$$z = \frac{x - \mu}{\sigma}$$

(a) **Probability that a refund exceeds 2,885 BD.**

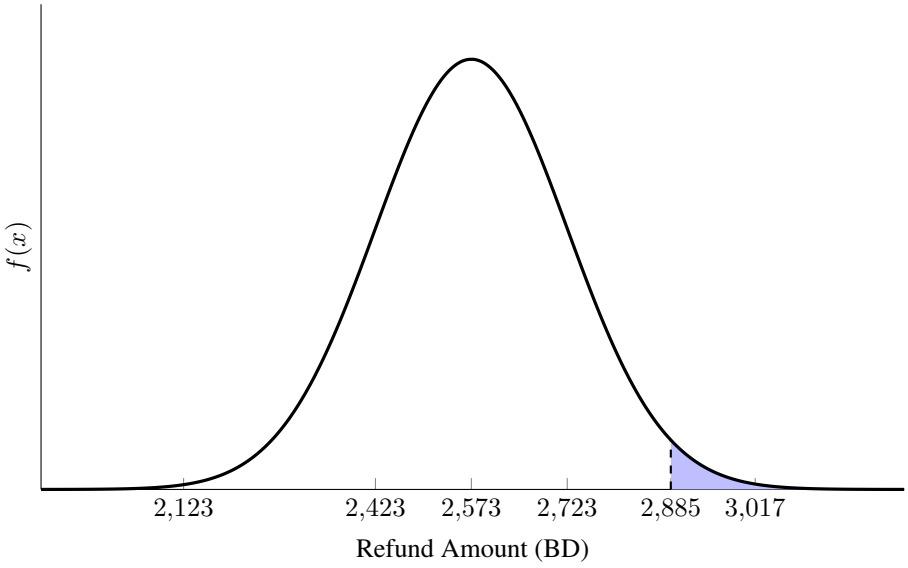
First, calculate the z-score:

$$z = \frac{2885 - 2573}{150} = 2.08$$

We want to calculate:

$$P(X > 2885) = P(Z > 2.08)$$

Figure 7.30: Normal Distribution: $P(X > 2885)$



Using the z-table:

$$P(0 < Z < 2.08) = 0.4812$$

Since the area to the right of the mean is 0.5:

$$P(Z > 2.08) = 0.5 - 0.4812$$
$$= 0.0188$$

Therefore, the probability of observing a tax income exceeding 2885 BD is 1.88%.

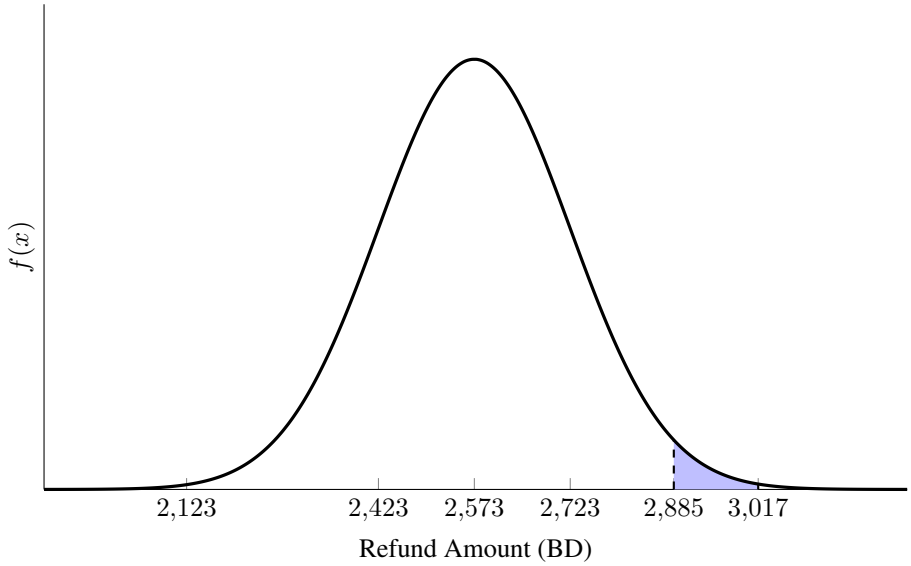
(b) **Probability that a refund is between 2,885 BD and 3,017 BD.**

First, calculate the two z-scores:

$$z_1 = \frac{2885 - 2573}{150} = 2.08$$

$$z_2 = \frac{3017 - 2573}{150} = 2.96$$

Figure 7.31: Normal Distribution: $P(2885 < X < 3017)$



We want to calculate:

$$P(2885 < X < 3017) = P(2.08 < Z < 2.96)$$

Using the z-table:

$$P(0 < Z < 2.08) = 0.4812$$

$$P(0 < Z < 2.96) = 0.4985$$

Therefore:

$$\begin{aligned} P(2.08 < Z < 2.96) &= 0.4985 - 0.4812 \\ &= 0.0173 \end{aligned}$$

which is 17.3%.

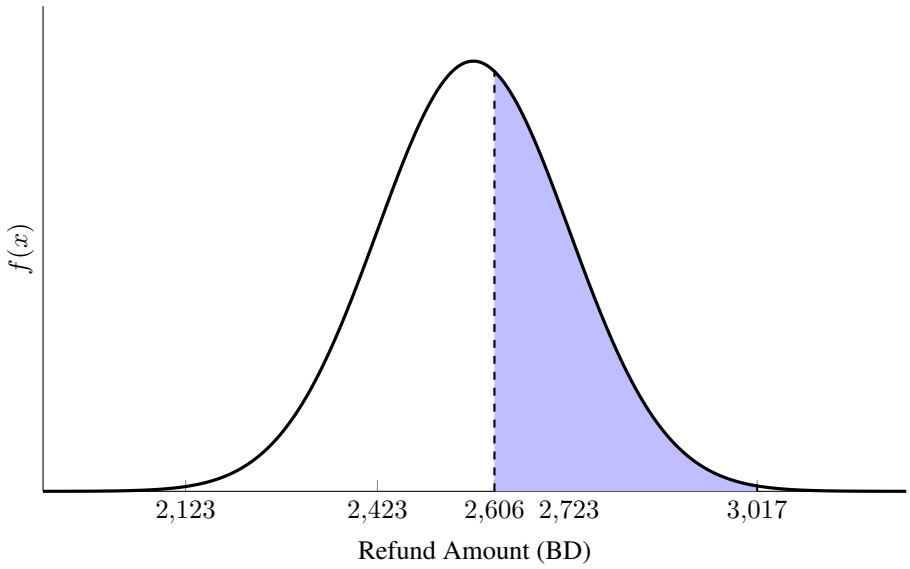
(c) **Probability that a refund is between 2,606 BD and 3,017 BD.**

First, calculate the two z-scores:

$$z_1 = \frac{2606 - 2573}{150} = 0.22$$

$$z_2 = \frac{3017 - 2573}{150} = 2.96$$

Figure 7.32: Normal Distribution: $P(2606 < X < 3017)$



We want to calculate:

$$P(2606 < X < 3017) = P(0.22 < Z < 2.96)$$

Using the z-table:

$$P(0 < Z < 0.22) = 0.0871$$

$$P(0 < Z < 2.96) = 0.4985$$

Therefore:

$$\begin{aligned} P(0.22 < Z < 2.96) &= 0.4985 - 0.0871 \\ &= 0.4114 \end{aligned}$$

which is a 41.14% probability.

4) Let X represent the hourly operating cost of an aircraft.

We are given:

$$X \sim N(2100, 250^2)$$

We want to find the value of x such that:

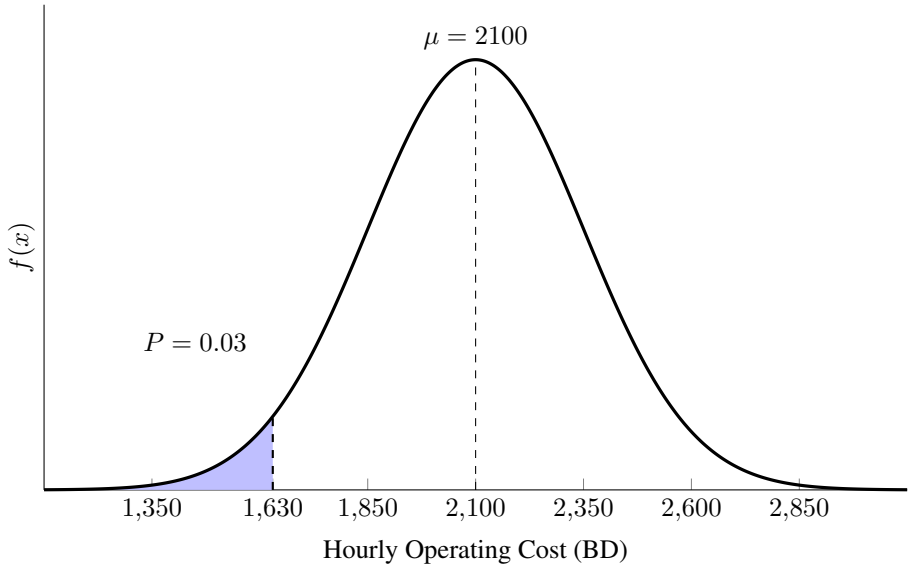
$$P(X < x) = 0.03$$

Since the normal distribution is symmetric, we know that 50% of the observations fall below the mean. Therefore, the area between the mean and the unknown value of x is:

$$0.50 - 0.03 = 0.47$$

Using the z-table, we look for the probability 0.4700, the closest value corresponds to $|z| = 1.88$. Moreover since we are interested in the lowest 3% of aircraft, the z-score must be negative:

Figure 7.33: Normal Distribution: Lowest 3% of Aircraft Operating Costs



Recall the standardisation formula:

$$z = \frac{x - \mu}{\sigma}$$

Rearranging for x:

$$x = \mu + z\sigma$$

Substituting:

$$\begin{aligned} x &= 2100 + (-1.88)(250) \\ &= 2100 - 470 \\ &= 1630 \end{aligned}$$

Therefore, the operating cost at the 3rd percentile is approximately **BD 1,630 per hour**.

In other words, approximately 3% of aircraft have hourly operating costs below BD 1,630.

5) Let X represent the examination score of a student.

We are given:

$$X \sim N(60, 5^2)$$

Recall the Empirical Rule for a normal distribution:

- Approximately 68% of observations lie within one standard deviation of the mean.
- Approximately 95% of observations lie within two standard deviations of the mean.
- Approximately 99.7% of observations lie within three standard deviations of the mean.

(a) **Percentage of students scoring between 55 and 65.**

We want to calculate:

$$P(55 < X < 65)$$

Notice that:

$$55 = 60 - 5 = \mu - \sigma$$

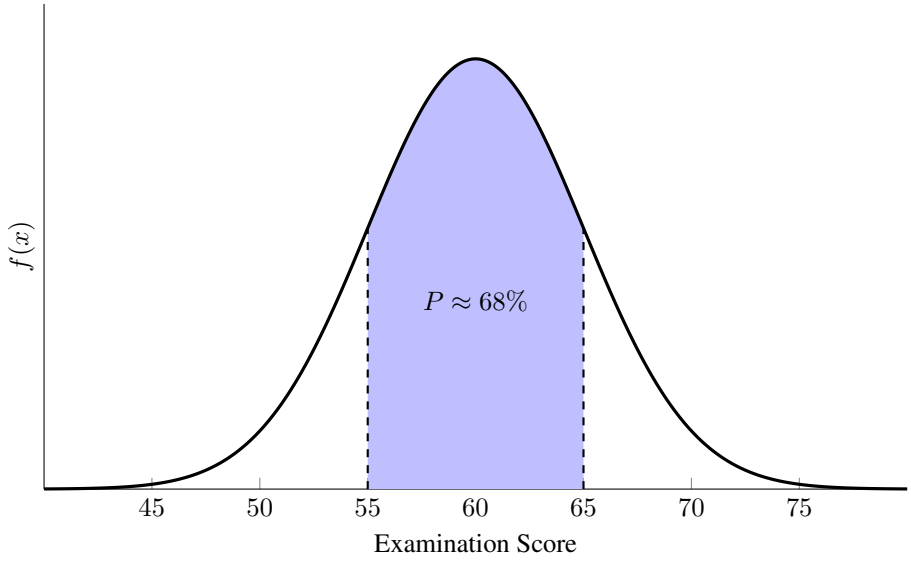
$$65 = 60 + 5 = \mu + \sigma$$

Therefore, the interval is within one standard deviation of the mean.

Using the Empirical Rule:

$$P(55 < X < 65) \approx 68\%$$

Figure 7.34: Normal Distribution: $P(55 < X < 65)$



(b) **Percentage of students scoring between 50 and 70.**

We want to calculate:

$$P(50 < X < 70)$$

Notice that:

$$50 = 60 - 2(5) = \mu - 2\sigma$$

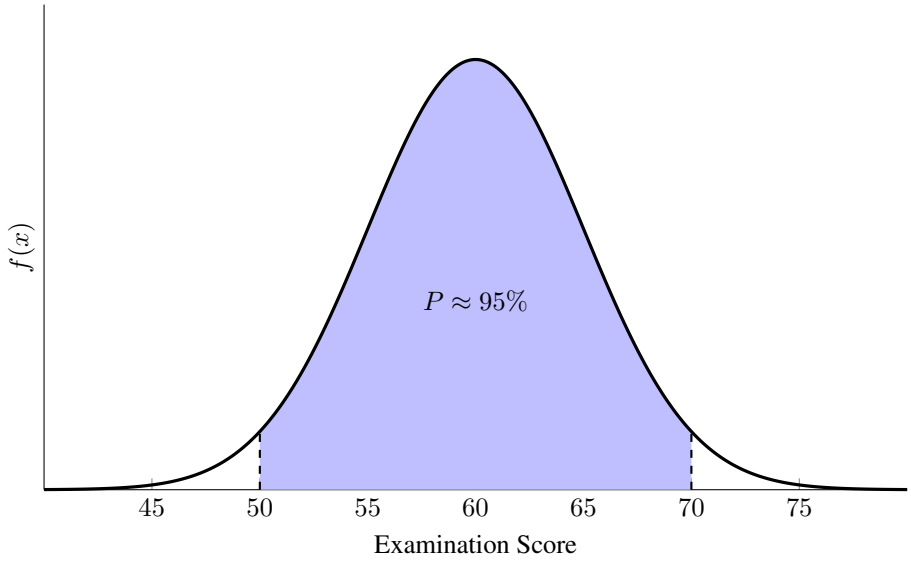
$$70 = 60 + 2(5) = \mu + 2\sigma$$

Therefore, the interval is within two standard deviations of the mean.

Using the Empirical Rule:

$$P(50 < X < 70) \approx 95\%$$

Figure 7.35: Normal Distribution: $P(50 < X < 70)$



(c) **Percentage of students scoring between 45 and 75.**

We want to calculate:

$$P(45 < X < 75)$$

Notice that:

$$45 = 60 - 3(5) = \mu - 3\sigma$$

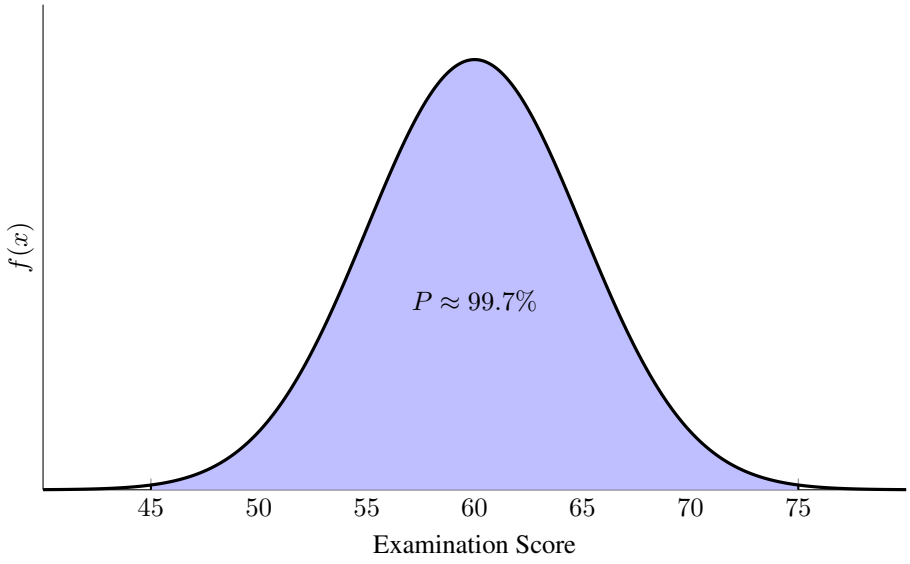
$$75 = 60 + 3(5) = \mu + 3\sigma$$

Therefore, the interval is within three standard deviations of the mean.

Using the Empirical Rule:

$$P(45 < X < 75) \approx 99.7\%$$

Figure 7.36: Normal Distribution: $P(45 < X < 75)$



6) Let X represent the number of minutes a resident spends watching television per day.

We are given:

$$X \sim N(270, 23^2)$$

Recall the standardisation formula:

$$z = \frac{x - \mu}{\sigma}$$

(a) **Percentage watching more than 300 minutes.**

We want to calculate:

$$P(X > 300)$$

First, calculate the z-score:

$$z = \frac{300 - 270}{23} = 1.30$$

Therefore:

$$P(X > 300) = P(Z > 1.30)$$

Using the z-table:

$$P(0 < Z < 1.30) = 0.4032$$

Since the area to the right of the mean is 0.5:

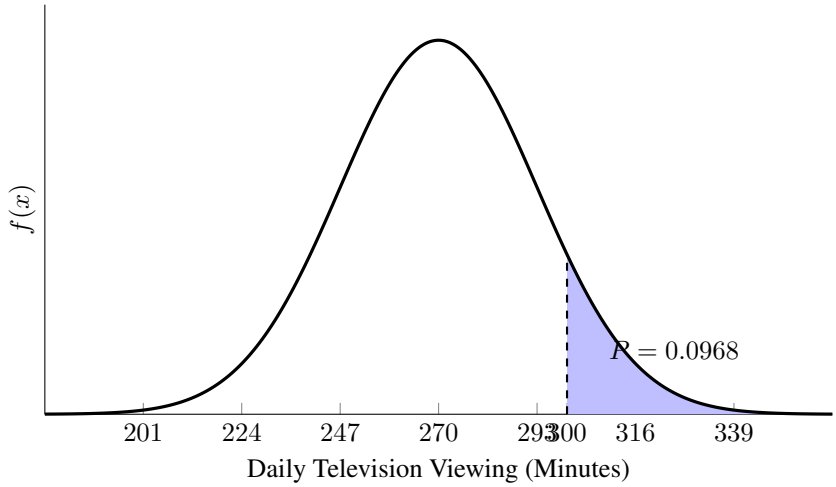
$$P(Z > 1.30) = 0.5 - 0.4032$$
$$= 0.0968$$

Therefore:

$$P(X > 300) \approx 9.68\%$$

Approximately 9.68% of residents watch more than 300 minutes of television per day.

Figure 7.37: Normal Distribution: $P(X > 300)$



(b) **Percentage watching more than 220 minutes.**

We want to calculate:

$$P(X > 220)$$

First, calculate the z-score:

$$z = \frac{220 - 270}{23} = -2.17$$

Therefore:

$$P(X > 220) = P(Z > -2.17)$$

Using the z-table:

$$P(0 < Z < 2.17) = 0.4850$$

Since the normal distribution is symmetric:

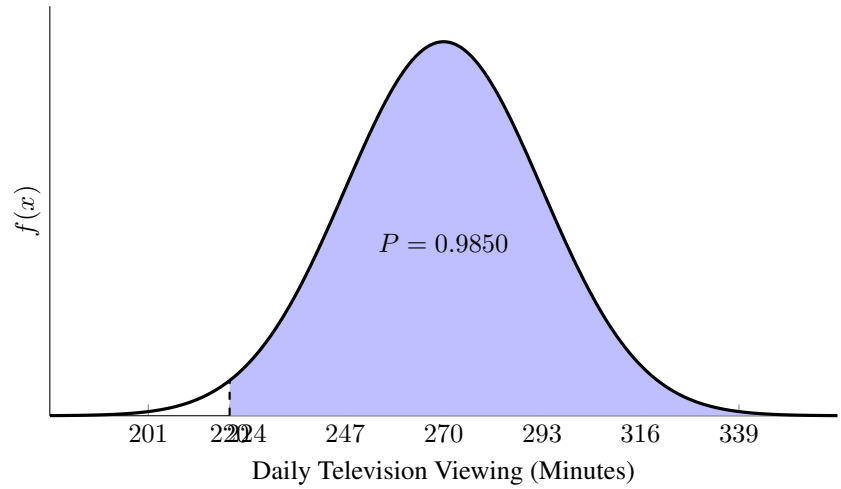
$$\begin{aligned} P(Z > -2.17) &= 0.5 + 0.4850 \\ &= 0.9850 \end{aligned}$$

Therefore:

$$P(X > 220) \approx 98.50\%$$

Approximately 98.50% of residents watch more than 220 minutes of television per day.

Figure 7.38: Normal Distribution: $P(X > 220)$



(c) **Percentage watching between 220 and 300 minutes.**

We want to calculate:

$$P(220 < X < 300)$$

First, calculate the two z-scores:

$$z_1 = \frac{220 - 270}{23} = -2.17$$

$$z_2 = \frac{300 - 270}{23} = 1.30$$

Therefore:

$$P(220 < X < 300) = P(-2.17 < Z < 1.30)$$

Using the z-table:

$$P(-2.17 < Z < 0) = 0.4850$$

$$P(0 < Z < 1.30) = 0.4032$$

Since the interval extends across the mean, we add the two probabilities:

$$P(-2.17 < Z < 1.30) = 0.4850 + 0.4032$$

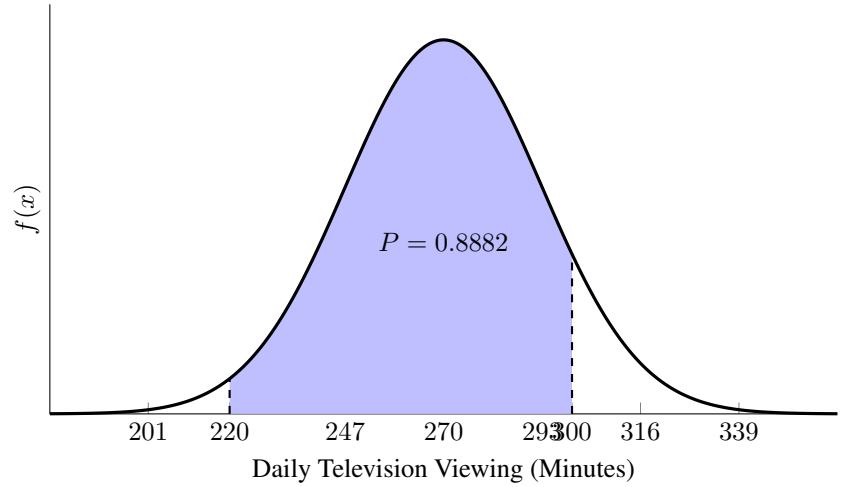
$$= 0.8882$$

Therefore:

$$P(220 < X < 300) \approx 88.82\%$$

Approximately 88.82% of residents watch between 220 and 300 minutes of television per day.

Figure 7.39: Normal Distribution: $P(220 < X < 300)$



(d) **Daily viewing time for the upper 15%.**

We want to find the value of x such that:

$$P(X > x) = 0.15$$

Since the area to the right of the mean is 0.5, the area between the mean and x is:

$$0.50 - 0.15 = 0.35$$

Using the z-table, we look for the probability closest to 0.3500.

The closest value is 0.3508, corresponding to a z-score of:

$$z = 1.04$$

Since we are interested in the upper 15%, the z-score is positive.

Recall:

$$z = \frac{x - \mu}{\sigma}$$

Rearranging for x:

$$x = \mu + z\sigma$$

Substituting:

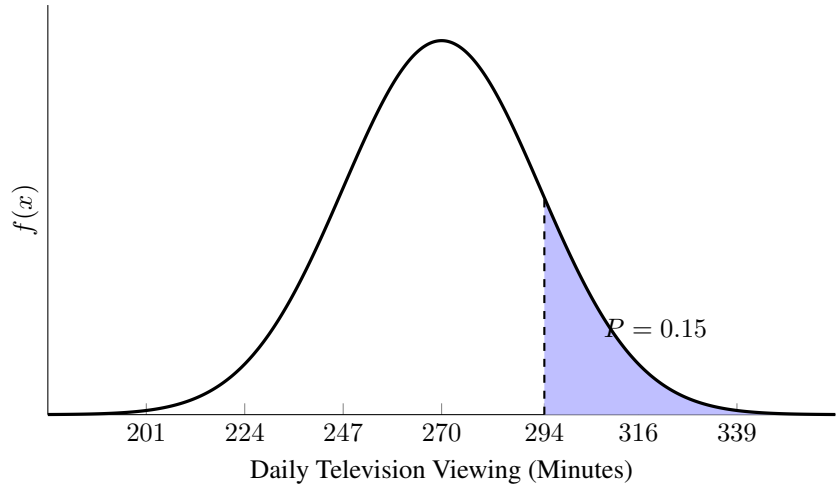
$$\begin{aligned} x &= 270 + (1.04)(23) \\ &= 270 + 23.92 \\ &= 293.92 \end{aligned}$$

Therefore, the cutoff for the upper 15% is approximately:

$$x \approx 294 \text{ minutes per day}$$

In other words, a "binge watcher" spends approximately 294 minutes or more per day watching television.

Figure 7.40: Normal Distribution: Upper 15% of Television Viewing



7) Let X represent the end-of-day share price of NBB.

We are given:

$$X \sim N(0.420, 0.0225^2)$$

Recall the standardisation formula:

$$z = \frac{x - \mu}{\sigma}$$

(a) **Probability that the share price exceeds BD 0.450.**

We want to calculate:

$$P(X > 0.450)$$

First, calculate the z-score:

$$z = \frac{0.450 - 0.420}{0.0225} = 1.33$$

Therefore:

$$P(X > 0.450) = P(Z > 1.33)$$

Using the z-table:

$$P(0 < Z < 1.33) = 0.4082$$

Since the area to the right of the mean is 0.5:

$$P(Z > 1.33) = 0.5 - 0.4082$$
$$= 0.0918$$

Therefore:

$$P(X > 0.450) \approx 9.18\%$$

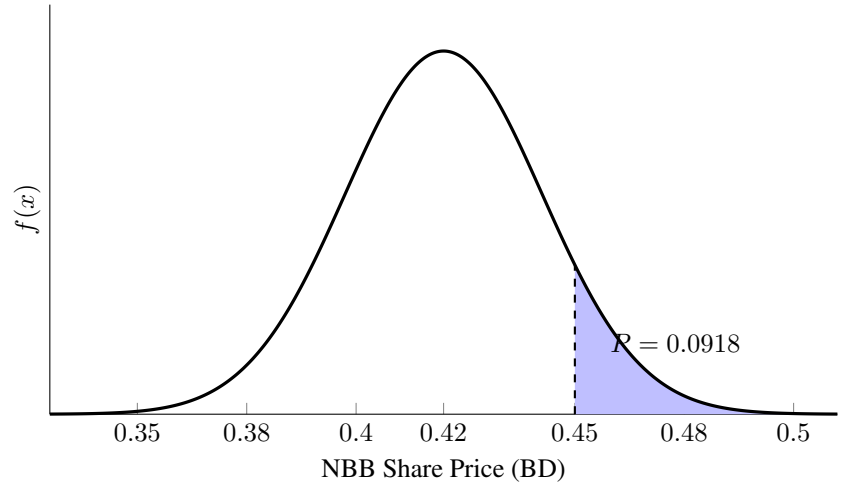
Next, we estimate the number of trading days during which the share price exceeds BD 0.450.

Since there are 240 trading days:

$$\begin{aligned}\text{Expected Number of Days} &= 240 \times 0.0918 \\ &= 22.032 \\ &\approx 22 \text{ days}\end{aligned}$$

Therefore, we expect the closing price to exceed BD 0.450 on approximately 22 trading days.

Figure 7.41: Normal Distribution: $P(X > 0.450)$



(b) **Percentage of trading days with a share price between BD 0.380 and BD 0.400.**

We want to calculate:

$$P(0.380 < X < 0.400)$$

First, calculate the two z-scores:

$$z_1 = \frac{0.380 - 0.420}{0.0225} = -1.78$$

$$z_2 = \frac{0.400 - 0.420}{0.0225} = -0.89$$

Therefore:

$$P(0.380 < X < 0.400) = P(-1.78 < Z < -0.89)$$

Using the z-table:

$$P(0 < Z < 1.78) = 0.4625$$

$$P(0 < Z < 0.89) = 0.3133$$

Since both z-scores are negative, we use the symmetry of the normal distribution.

The probability between the two z-scores is:

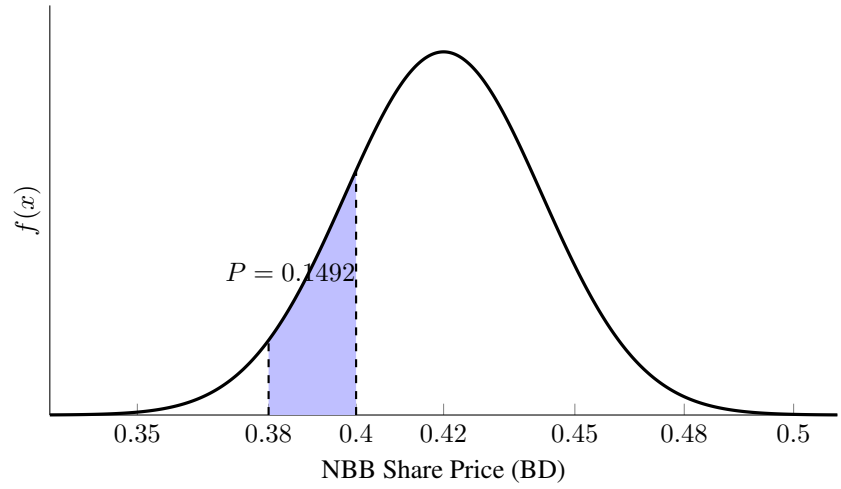
$$\begin{aligned} P(-1.78 < Z < -0.89) &= 0.4625 - 0.3133 \\ &= 0.1492 \end{aligned}$$

Therefore:

$$P(0.380 < X < 0.400) \approx 14.92\%$$

Approximately 14.92% of trading days have closing prices between BD 0.380 and BD 0.400.

Figure 7.42: Normal Distribution: $P(0.380 < X < 0.400)$



(c) **Minimum share price for the highest 15% of closing prices.**

We want to find the value of x such that:

$$P(X > x) = 0.15$$

Since the area to the right of the mean is 0.5, the area between the mean and the unknown share price is:

$$0.50 - 0.15 = 0.35$$

Using the z-table, we look for the probability closest to 0.3500.

The closest value is 0.3508, corresponding to a z-score of:

$$z = 1.04$$

Since we are interested in the highest 15%, the z-score is positive.

Recall:

$$z = \frac{x - \mu}{\sigma}$$

Rearranging for x:

$$x = \mu + z\sigma$$

Substituting:

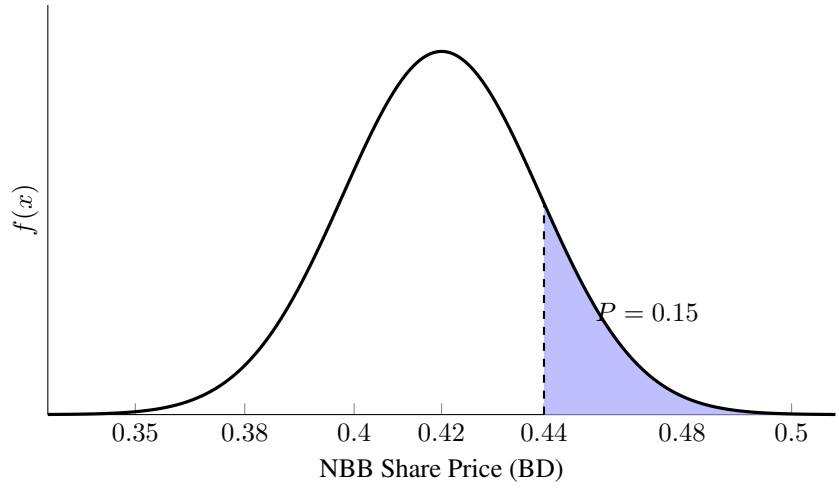
$$\begin{aligned} x &= 0.420 + (1.04)(0.0225) \\ &= 0.420 + 0.0234 \\ &= 0.4434 \end{aligned}$$

Therefore, the cutoff for the highest 15% of closing prices is approximately:

$$x \approx \text{BD } 0.4434$$

In other words, approximately 15% of trading days have closing prices above BD 0.4434.

Figure 7.43: Normal Distribution: Highest 15% of NBB Share Prices



Chapter 8

Estimation and Confidence Intervals

8.1 Introduction

Consider a dataset of the population of all 800 schools in Bahrain from which we have their math grade:

School i	**Average Mathematics Grade**	**Pass Indicator** X_i
School 1	87	1
School 2	58	1
School 3	46	0
School 4	48	0
School 5	80	1
School 6	67	1
School 7	35	0
School 8	67	1
School 9	82	1
	$\vdots$	
School 800	82	1

Since *school_data.xlsx* shows data on all schools, then this is the population dataset. Therefore the average math score and its standard deviation is:

$$\mu = 71 \qquad \sigma = 9.95$$

Since this is the data for all schools, the parameter μ represents the **true** average of the population.

Suppose however we did not have access to this population data. Instead, we only have a sample of just 5 schools:

School i	**Average Mathematics Grade** x	**Governorate**
School 342	88	Capital
School 675	68	Muharraq
School 23	56	Muharraq
School 331	48	Southern
School 662	90	Capital

Since this is a sample, we denote the average by $\bar{x}$:

$$\bar{x} = \frac{\sum x_i}{n} = \frac{88 + 68 + 56 + 48 + 90}{5}$$

$$= 70,$$

but notice that the sample average $\bar{x} = 70$ is slightly different from the population average $\mu = 71$.

We call $\bar{x}$ a **point estimate**. This is a statistic that is computed from sample data. We hope that it is a good enough estimate **of the population statistic** μ.

Now suppose we collected three additional random samples from the population of 800 schools in Bahrain, each containing only 5 schools.

The following tables present the average mathematics grades for each sample:

Table 8.1: Three Different Samples of Schools in Bahrain

Sample 1

School	**Grade** x
12	65
25	72
48	68
76	74
91	71

$\bar{x}_1 = 70$

Sample 2

School	**Grade** x
103	78
156	82
204	70
318	75
422	80

$\bar{x}_2 = 77$

Sample 3

School	**Grade** x
115	60
230	65
345	68
470	62
590	70

$\bar{x}_3 = 65$

This illustrates an important concept known as **sampling variability**: different random samples from the same population will generally produce different sample statistics. Although each sample mean provides a **point estimate** of the population mean, some estimates will be closer to the true population mean than others.

8.2 Confidence Intervals for Continuous Variables

When statisticians generate data from a sample, how confident are they that the computed point estimates are actually equal to the true population parameters? From our example earlier, we are essentially asking, "how sure are we that $\bar{x} = \mu$?".

Since $\bar{x}$ can produce different numbers based on the sample, it is like saying that $\bar{x}$ is *also* a random number, which is drawn from a distribution and which has a mean and variance.

In order for us to analyse the level of confidence we have that our point estimates like $\bar{x}$ are accurate, let us start by invoking a powerful declaration called the **central limit theorem**.

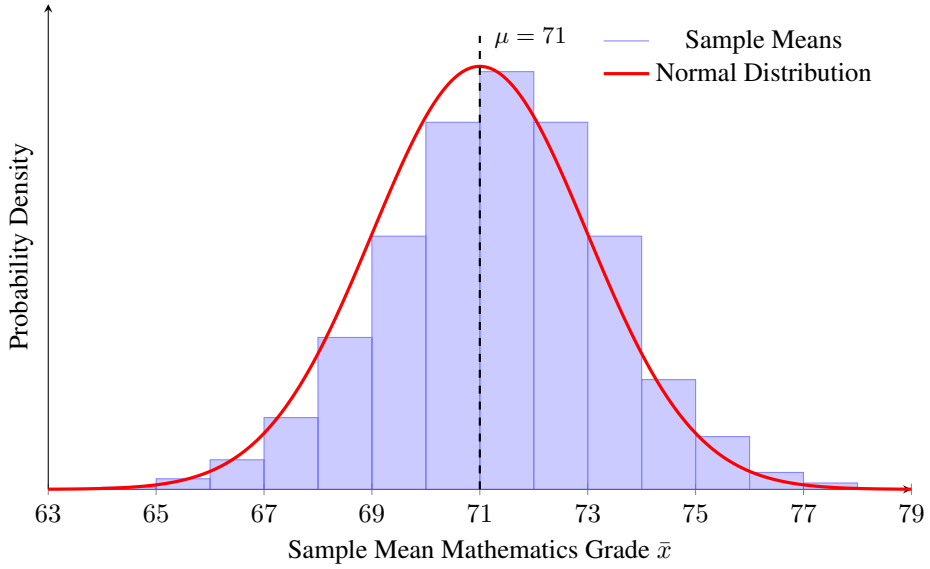
Essentially, the **central limit theorem** says that if your sample is sufficiently large, then the **distribution** of **your point estimate** is **approximately normal**:

$$\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

which is parametrised by the true population mean μ and its variance $\frac{\sigma^2}{n}$. Its standard deviation is therefore the square root of its variance:

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

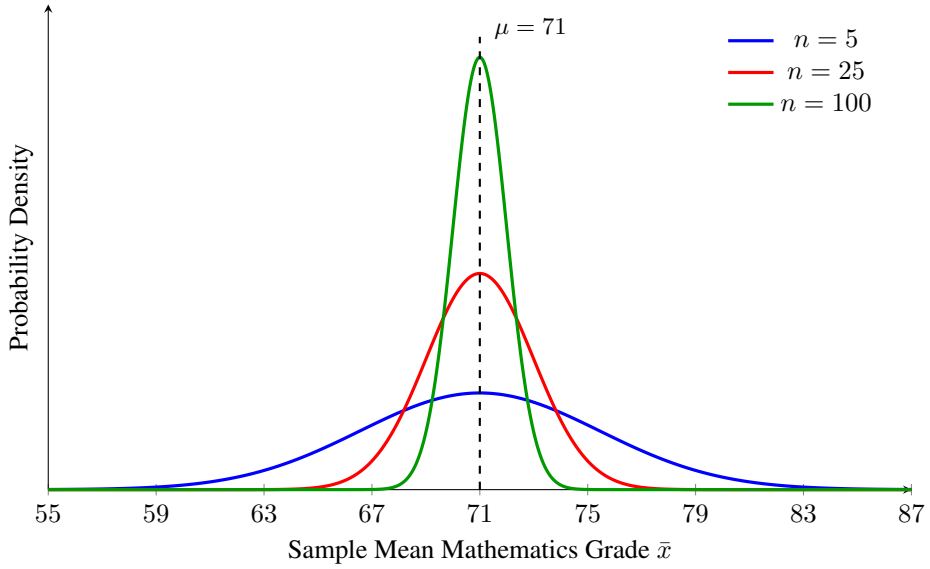
Figure 8.1: Illustrative Sampling Distribution of the Sample Mean ($n = 25$)



This images shows that if you were able to continue generating samples of $n = 25$ from the population and for each sample computed the $\bar{x}$, then the central limit theorem states that most estimates have a probability to be around the true mean μ, while the probability is much lower when it is away from μ.

An interesting result is that the standard deviation has $\sqrt{n}$ in the denominator. As n gets very large, the distribution becomes very narrow:

Figure 8.2: Sampling Distribution of the Sample Mean for Different Sample Sizes

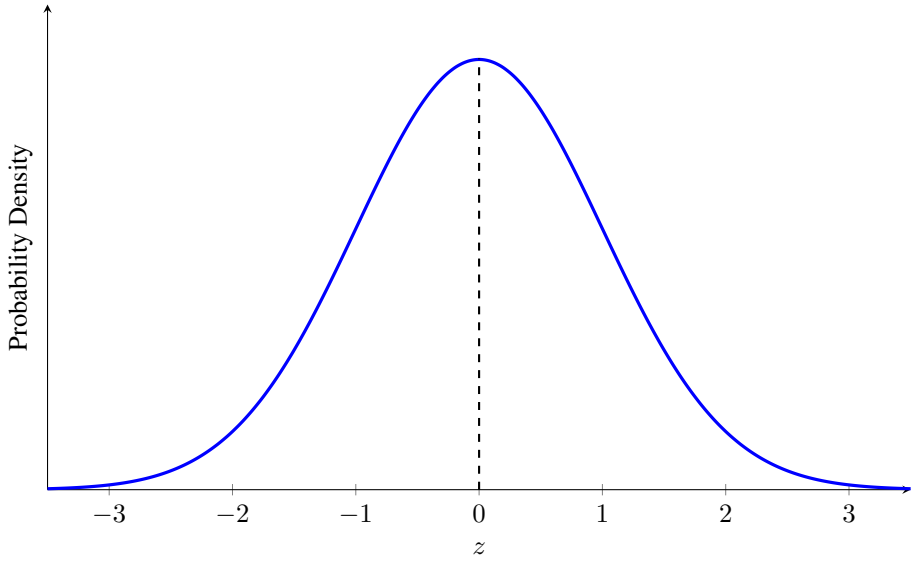


which means as your sample size gets larger, the probability of seeing $\bar{x}$ that is close to the true μ is more common.

Now just like we did in the last chapter, it does not actually matter what value is represented in the horizontal axis. So long as the shape is normally distributed, then we can standardise[a] the values of $\bar{x}$:

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Figure 8.3: Standard Normal Distribution

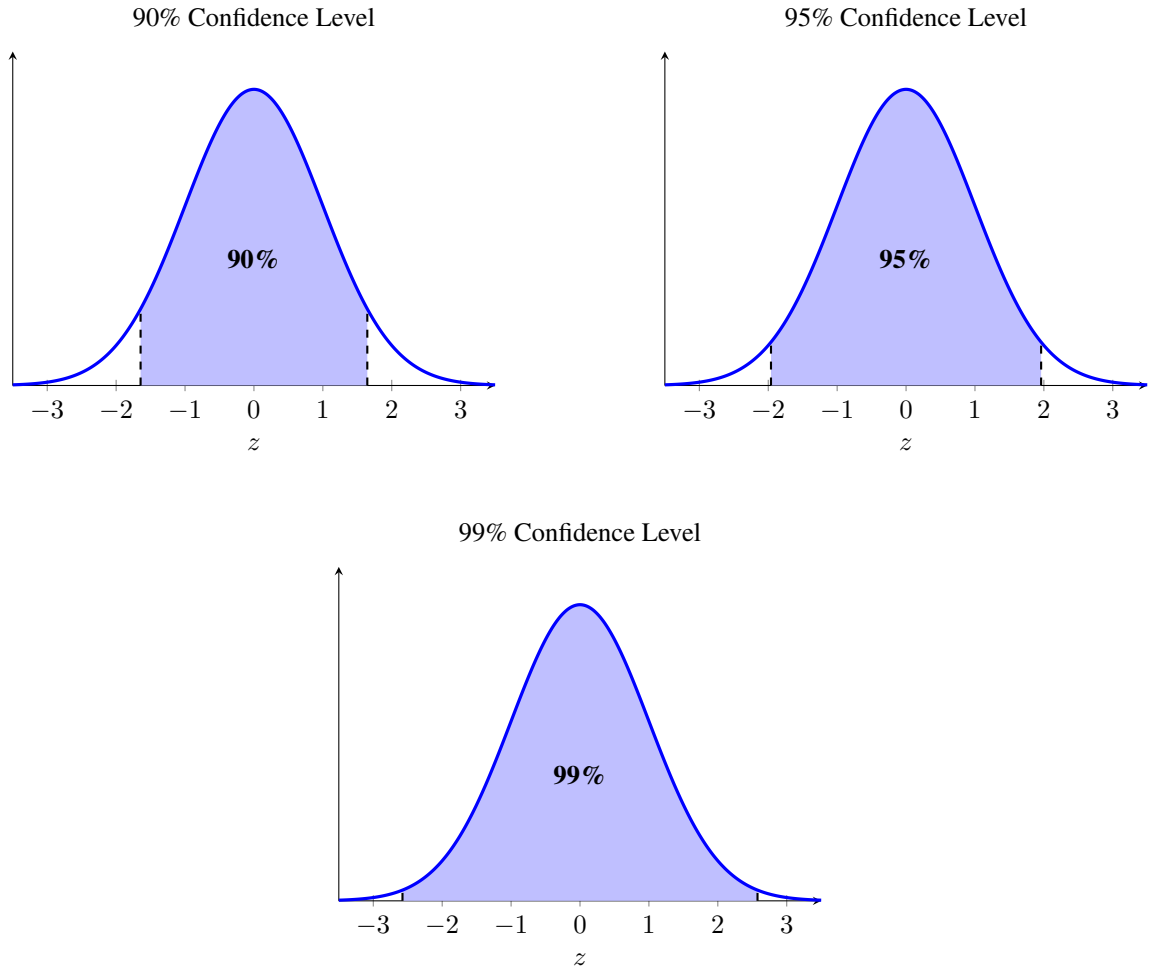


Before we continue, it is important to remember this fact:

- 90% of the distribution area is between -1.64 and +1.64 standard deviatons.
- 95% of the disitribution area is between -1.96 and +1.96 standard deviations.
- 99% of the distribution area is between -2.58 and +2.58 standard deviations.

[a] Remember that $Z = \frac{x-\mu}{\sigma}$.

Figure 8.4: Confidence Levels and the Standard Normal Distribution



8.2.1 When σ is a Known Number

Now let us start with the idea of confidence intervals. This is essentially a measure for how confident we are that the computed point estimate $\bar{x}$ is within some interval that contains the true μ. The formula is given by:

$$\bar{x} \pm z\Big(\sigma_{\bar{x}}\Big)$$

$$\bar{x} \pm z\Big(\frac{\sigma}{\sqrt{n}}\Big)$$

where z describes the **level of confidence** that interval contains μ around the estimate $\bar{x}$. Notice that $\frac{\sigma}{\sqrt{n}}$ is just the standard deviation of the estimator $\bar{x}$. The conventional levels are:

- 90% confident: z = 1.64.
- 95% confident: z = 1.96.
- 99% confident: z = 2.58.

Consider an example where a sample of $n = 25$ schools reveals that $\bar{x} = 68$ and that you are told that the population standard deviation is $\sigma = 9.95$. Let us construct the 95% confidence intervals that the population mean is within those bounds:

$$\begin{aligned} CI &= \bar{x} \pm z\left(\frac{\sigma}{\sqrt{n}}\right) \\ &= 68 \pm 1.96\left(\frac{9.95}{\sqrt{25}}\right) \\ &= 68 \pm 1.96(1.99) \\ &= 68 \pm 3.9004 \end{aligned}$$

Note that the value 3.9004 is known as the **margin of error**.

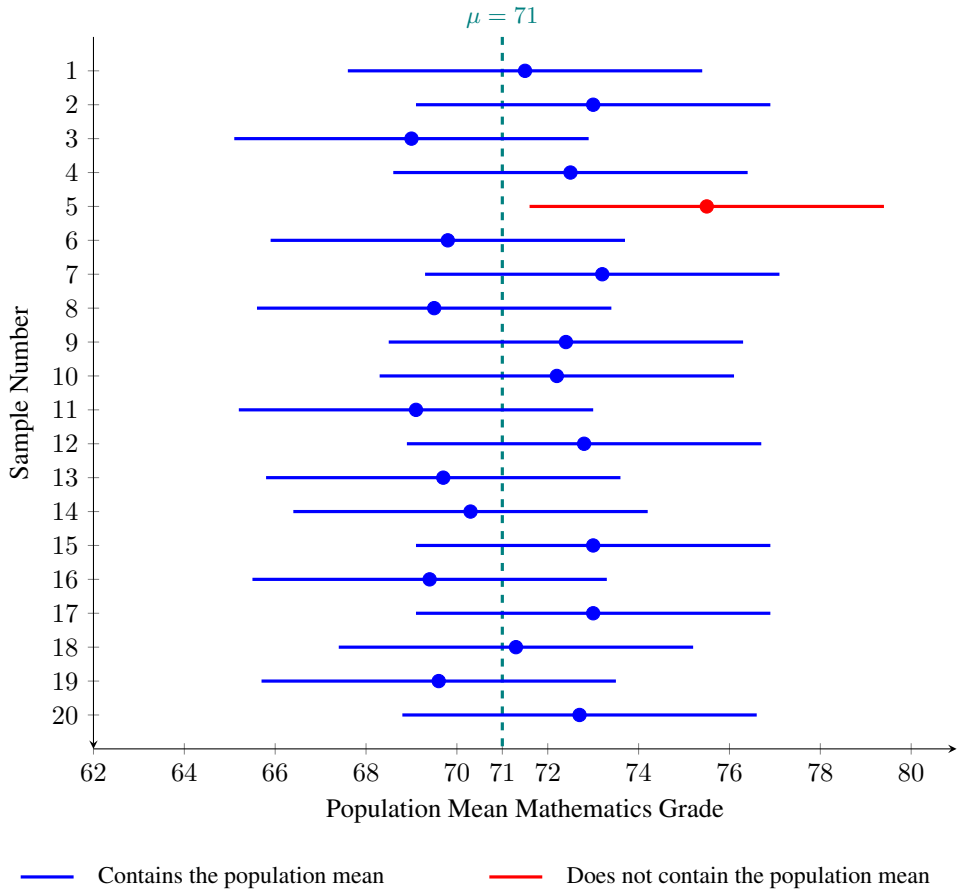
Hence, the 95% confidence interval is:

$$64.10 \leq \mu \leq 71.90$$

The way we interpret this that you are 95% confident that the true population mean mathematics grade of schools in Bahrain lies between 64.10 and 71.90.

This means that if we repeatedly collected random samples and constructed confidence intervals using this method, approximately 95% of those intervals would contain the true population mean μ.

Figure 8.5: Illustration of 20 Confidence Intervals at the 95% Confidence Level



8.2.2 When σ is Unknown

Remember that the population variance is given by:

$$\sigma^2 = \frac{\sum(x_i - \mu)}{N}$$

That meant that we could define the **normal** distribution of our estimator $\bar{x}$:

$$\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$Z = \frac{\bar{x} - \mu}{\sigma^2 n} \sim N(0, 1)$$

However, suppose that σ^2 is unknown to the statistician. This means that we cannot compute the variance of $\bar{x}$.

So in this case, statisticians should instead use the **sample** variance:

$$s^2 = \frac{\sum(x_i - \bar{x})}{n - 1}$$

The variance and standard deviation of our $\bar{x}$ therefore become:

$$s_{\bar{x}}^2 = \frac{s^2}{n}$$

$$s_{\bar{x}} = \frac{s}{\sqrt{n}}$$

However there is a consequence to doing this. If we replace the population parameter σ^2 with sample parameter s^2, then **we can no longer say** that $\bar{x}$ is normally distributed[b]. We instead have to say that $\bar{x}$ follows a **t-distribution**:

$$T = \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

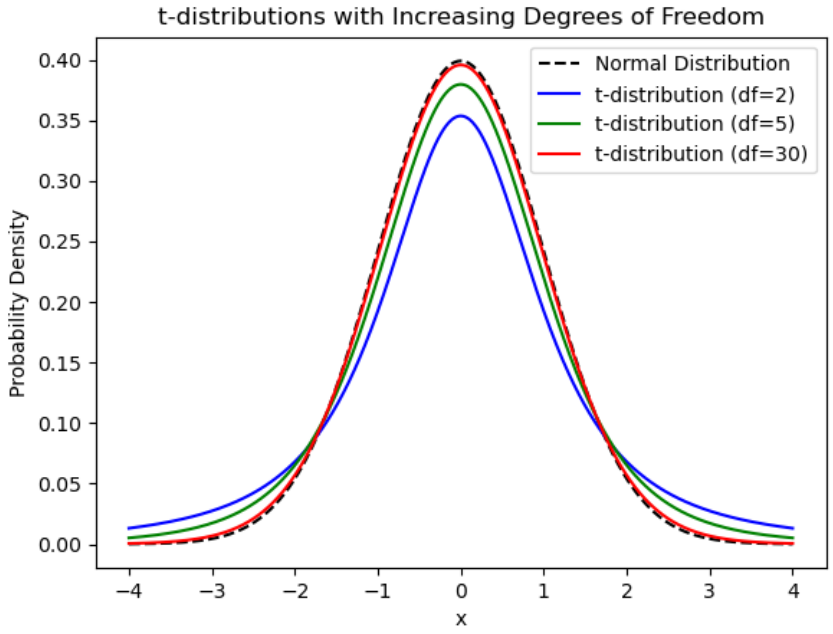
where $n - 1$ is known as the **degrees of freedom**[c].

The t distribution actually looks like a normal distribution and has the same symmetric bell-shaped properties. It is

[b] A simple explanation of this is because $\bar{x}$ is **already** a sample estimate of the population. Now if we introduce the sample s^2 then this is also **another** estimate of the population. There is going to be variation in the values of $\bar{x}$ and s^2 for different samples. Therefore, each sample will have its **own unique** distribution shape, and therefore it is not a normal distribution

[c] Just like how the normal distribution is parametrised by the mean and variance, and how the Poisson distribution is parametrised by the mean, we say that the t distribution is parametrised by the degrees of freedom $n - 1$.

only that the degree of how wide it becomes is based on the degrees of freedom:



Notice how when the degrees of freedom $n-1$ is very large, it essentially approximates the normal distribution (the red line and the dashed black line).

In order to construct confidence intervals for $\bar{x}$ when σ is unknown, we need to use the below formula:

$$\bar{x} \pm t\left(\frac{s}{\sqrt{n}}\right)$$

Previously, we were using z values, such as 1.64, 1.96 or 2.58 depending the level on confidence. We used those values because the underlying distribution of $\bar{x}$ was normal. However, because σ is unknown and we are using the sample equivalent s instead, then we are now in the t distribution and so we must use the t table instead:

Table 8.2: Critical Values of t-Distribution

Confidence Intervals					
df	**80%**	**90%**	**95%**	**98%**	**99%**
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
∞	1.282	1.645	1.960	2.326	2.576

Notice how when $df = \infty$ (meaning it is sufficiently large), its t values are the same as the z values.

Suppose you have a sample of 10 schools and that the average score of the sample was $\bar{x} = 60$ and we do not have σ but the sample standard deviation $s = 8$. We can construct the 95% confidence intervals as follows:

$$\bar{x} \pm t\left(\frac{s}{\sqrt{n}}\right)$$

$$60 \pm t\left(\frac{8}{\sqrt{10}}\right)$$

Since $n = 10$, the degrees of freedom is $n - 1 = 9$. For 95% confidence intervals, we select $t = 2.262$:

Table 8.3: Critical Values of t-Distribution

Confidence Intervals					
df	**80%**	**90%**	**95%**	**98%**	**99%**
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169

$$60 \pm t\left(\frac{8}{\sqrt{10}}\right)$$

$$60 \pm 2.262\left(\frac{8}{\sqrt{10}}\right)$$

$$60 \pm 2.262(2.530)$$

$$60 \pm 5.72$$

Hence, the 95% confidence interval is:

$$54.28 \leq \mu \leq 65.72$$

8.3 Confidence Intervals for Discrete Variables

We will now learn how to construct confidence intervals for discrete variables. Consider again the school data example:

School i	**Average Mathematics Grade**	**Pass Indicator** X_i
School 1	87	1
School 2	58	1
School 3	46	0
School 4	48	0
School 5	80	1
School 6	67	1
School 7	35	0
School 8	67	1
School 9	82	1
⋮		
School 800	82	1

Where the pass indicator is a random binary variable that takes values 0 and 1 depending on whether the grade is below or above 50 respectively.

Suppose we are interested in the average number of schools that passed. We can do this by simply summing up the individual columns:

$$\pi = \frac{\sum x_i}{n} = \frac{\sum 1+1+0+0+1+1+0+1+...+1}{800} = 0.65$$

where the average variable π represents the **proportion** of schools that passed. In this case, 65% of schools received a grade higher than 50.

The proportion is basically the average but for discrete variables that can only take values between 0 and 1. For example, a coin toss or a person's sex. However, you should note that in the previous example, we used **all** 800 schools to get the value of π. This means that $\pi = 0.65$ is the **population proportion**.

Now let us examine the proportion when we have a sample:

Table 8.4: Three Different Samples of Schools in Bahrain

Sample 1

School	**Pass** x
421	0
352	1
284	1
267	1
719	1

$p_1 = 0.8$

Sample 2

School	**Grade** x
103	1
156	1
204	0
318	0
422	1

$p_2 = 0.6$

Sample 3

School	**Grade** x
115	1
230	0
345	0
470	0
590	1

$p_3 = 0.4$

Just like when we looked at $\bar{x}$, the sample proportion p is also variable across samples. Moreover, we can similarly appeal to the central limit theorem and say that p has a[d] normal distribution, parametrised by its mean and variance:

$$p \sim N\left(p, \frac{p(1-p)}{n}\right)$$

You are not expected to know how to derive the variance of p[e], but you should know that the **standard deviation** is simply the square root of the variance:

[d] approximately

[e] Technically, we are assuming that the pass indicator variable follows a binomial distribution. If you look at this section under Chapter 6, you will see that we give you the mean and variance of the binomial distribution.

$$\sigma_p = \sqrt{\frac{p(1-p)}{n}}$$

8.3.1 When σ is a Known Number

Now that we have the standard deviation, it is possible to construct confidence intervals for our sample proportion p:

$$p \pm z\Big(\sigma_p\Big)$$

$$p \pm z\Big(\sqrt{\frac{p(1-p)}{n}}\Big)$$

where just as before, we use the z values based on the significance level:

- 90% confident: z = 1.64.
- 95% confident: z = 1.96.
- 99% confident: z = 2.58.

However, you should keep in mind that we are **only able to do this** because we are appealing to the central limit theorem. That is, only if we have **a sufficiently large sample**, are we able to let the random variable x follow a normal distribution.

When you are dealing with a discrete number, you need to therefore keep this rule in mind:

> You are only able to appeal to the central limit theorem if your sample satisfies:
>
> - $np \geq 5$.
> - $n(1-p) \geq 5$
>
> For example, suppose $n = 10$ and $p = 0.5$. This means that:

- $np = 10 \cdot 5 = 5 \geq 5$.
- $n(1-p) = 10(1-0.5) = 5 \geq 5$

then this qualifies as a large sample and so you can use the z values.

Activity:

Consider a sample dataset of $n = 2000$ people, of which 1600 are female. Let the binary variable x indicate $x = 0$ if male and $x = 1$ if female.

(a) Calculate the sample proportion p of females in the dataset.

(b) Verify if the sample satisfies the large sample conditions which allow you to appeal to the central limit theorem.

(c) Construct the 95% confidence interval.

Solution:

(a) The sample proportion:

$$p = \frac{1600}{2000} = 0.8$$

which shows that 80% of the individuals were women.

(b) The large sample conditions are:

- $np = 2000 \cdot 0.8 = 1600 \geq 5$
- $n(1-p) = 2000 \cdot (1-0.8) = 400 \geq 5$

(c) Remember that the formula is:

$$p \pm z\left(\sqrt{\frac{p(1-p)}{n}}\right)$$

- The sample proportion $\bar{p} = 0.8$.
- The 95% confidence interval implies $z = 1.96$.
- The sample size $n = 2000$.

Therefore:

$$0.8 \pm 1.96\left(\sqrt{\frac{0.8(1-0.8)}{2000}}\right)$$

Lower Limit of CI = 0.782 Upper Limit of CI = 0.818

8.4 Choosing an Appropriate Sample Size

In the previous school examples, our sample sizes were set at $n = 5$. However, there are ways that a statistician can measure what the appropriate sample size should be.

In general, we would like to have large sample sizes, but if we are practically limited (for example, if you have a limited budget to collect the data), then an appropriate rule of thumb is to apply the following formula:

For a continuous variable in extracting the mean:

$$n = \left[\frac{z \cdot \sigma}{E}\right]^2$$

For a discrete variable in extracting the proportion:

$$n = \pi(1-\pi)\left(\frac{z}{E}\right)^2$$

where for each component:

- σ is the population standard deviation.
- π is the population proportion.
- z is the conventional levels for 90%, 95% and 99% confidence level.
- E is the margin of error the researcher tolerates.

For your information, when π is not known, it is fine to set $\pi = 0.5$. Moreover, the margin of error E is a component that the researcher decides themselves.

Activity:

A government researcher wants to conduct a study on wages x of Bahrainis. Suppose that we have the following information:

- The margin of error that the ministry will tolerate is $E = 100$.
- The confidence level should be at 95%.
- The population standard deviation is known and set at $\sigma = 1000$.

Calculate the appropriate sample size required for this study to be conducted.

Solution:

The study is about wages, which is a continuous variable. In order to estimate the mean, we gather a sample size based on the below formula:

$$n = \left[\frac{z \cdot \sigma}{E}\right]^2$$

Substitute the values:

$$n = \left[\frac{1.96 \cdot 1000}{100}\right]^2$$

$$\approx 384.16$$

where can round up this number as $n = 385$ individuals required for the study.

Activity:

The Ministry of Works wants to know whether a sample of towns in Bahrain has a bus stop. If they have 0, then $x = 0$ and if they have at least 1, then $x = 1$. They want to know how many towns they should include in their sample.

- The margin of error that the ministry will tolerate is $E = 0.1$.
- The confidence level should be at 90%.
- The population proportion is unknown (remember that this means $\pi = 0.5$).

Calculate the appropriate sample size.

Solution:

Since x is a discrete binary variable, we can use the below formula:

$$n = \pi(1 - \pi)\left(\frac{z}{E}\right)^2$$

Substituting the values:

$$n = 0.5(1 - 0.5)\left(\frac{1.64}{0.1}\right)^2$$

$$= 67.24$$

where we round this up to $n = 68$, which is the appropriate number of towns to conduct this study.

8.5 Ethics

- Convey results honestly. A point estimate alone omits meaningful uncertainty.
- Be transparent about what a confidence interval does and does not mean.
- Resist choosing a confidence level to produce a favourable-looking result.

In using AI:

- AI must be prompted to report the margin of error and confidence level, conveyed honestly.
- Do not ask AI to adjust a confidence interval to produce a more favourable-looking result.

Chapter 9

Hypothesis Testing

9.1 Introduction

In the previous chapter, we established that under suitable conditions, we can appeal to the **Central Limit Theorem (CLT)** to approximate the sampling distribution of the sample mean using a normal distribution:

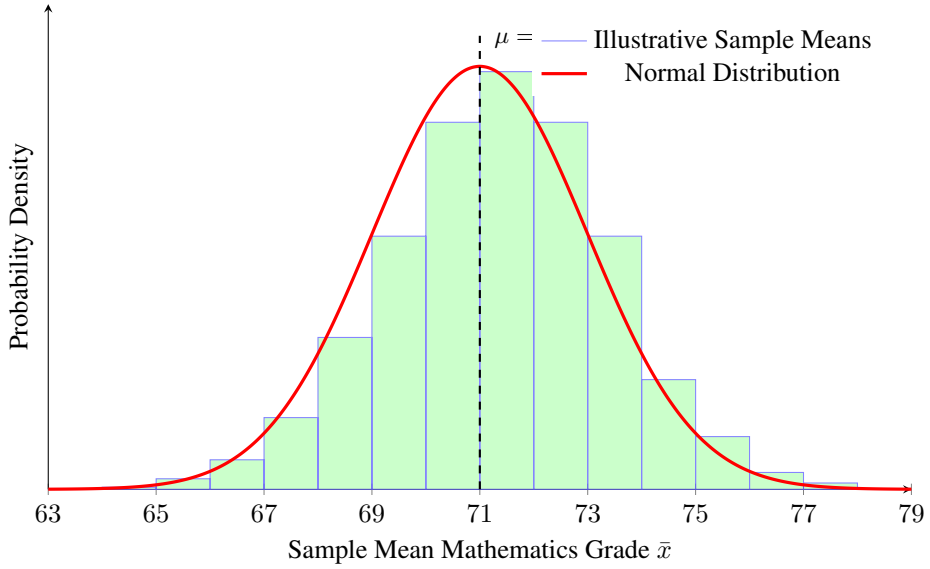
$$\bar{X} \overset{\text{approx}}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right)$$

where $\bar{X}$ represents the sample mean as a random variable, μ is the population mean, σ^2 is the population variance, and n is the sample size.

For example, recall our previous illustration involving mathematics grades across schools in Bahrain. Suppose the population mean is $\mu = 71$, the population standard deviation is $\sigma = 9.95$, and we repeatedly collect random samples of $n = 25$ schools.

We can illustrate the sampling distribution of the sample mean as follows:

Figure 9.1: Illustrative Sampling Distribution of the Sample Mean ($n = 25$)



Notice that although the sample means vary across different samples, their sampling distribution is centred around the true population mean $\mu = 71$.

We also established the concept of **confidence intervals**. These allow us to construct a range of plausible values for an unknown population parameter using information obtained from a sample.

Recall that when the population standard deviation σ is known, a confidence interval for the population mean takes the form:

$$\bar{x} \pm z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

Alternatively, when the population standard deviation is unknown, we can use the sample standard deviation s and the t-distribution:

$$\bar{x} \pm t_{\alpha/2, n-1} \left(\frac{s}{\sqrt{n}} \right)$$

The confidence level determines the critical value used to construct the interval. For example, a 95% confidence interval is constructed using a procedure that would contain the true population mean in approximately 95% of repeated random

samples, provided the relevant statistical assumptions hold.

9.2 Hypothesis Testing when σ is Known

We now turn to a related concept known as **hypothesis testing**. To properly appreciate this, you must remember what the principle issue is when we conduct statistical analysis. As a researcher, we **often do not know** what the value of μ is. This means that we have no choice by to rely on our sample estimate $\bar{x}$.

In the previous chapter, our objective was to **estimate the unknown population mean**. In hypothesis testing, however, we are interested in **evaluating a specific claim about the population mean**.

For example, suppose the Bahraini Ministry of Labour wants to investigate the average number of employees working at companies in Bahrain. A researcher collects a random sample of 25 companies and records the number of employees working at each company.

The sample produces the following information:

$$n = 25$$

$$\bar{x} = 12$$

This means that the average number of employees per company in our sample is 12. However, since this is only a sample, we cannot immediately conclude that the true population mean is also 12.

Now suppose that we wish to evaluate a claim that the true population mean is actually:

$$\mu = 10$$

Recall that different random samples can produce different sample means, even when they are drawn from the same population. Now what we are interested in asking is if the true population mean is $\mu = 10$, how unusual would it be to observe a sample mean of $\bar{x} = 12$?

Hypothesis testing provides a statistical framework for determining whether an observed sample result is sufficiently

unusual to reject a particular claim about the population.

9.2.1 The Null and Alternative Hypotheses

The first step in conducting a hypothesis test is to specify two competing hypotheses, which we call the null hypothesis H_0 and the alternative hypothesis H_1.

The null hypothesis represents the claim that we wish to evaluate. In our example:

$$H_0 : \mu = 10$$

When conducting a hypothesis test, we temporarily assume that the null hypothesis is true.

The alternative hypothesis represents the competing claim. For example, if we are interested in whether the population mean is greater than 10, we would specify:

$$H_0 : \mu = 10$$
$$H_1 : \mu > 10$$

The choice of alternative hypothesis depends on the research question being investigated and your selection will change the way to we conduct the test. We will discuss this shortly, but for now let us focus on the null hypothesis that $\mu = 10$ and examine how unusual our observed sample mean would be under this assumption.

9.2.2 When σ is Known

Suppose we are told that the population standard deviation of the number of employees per company is:

$$\sigma = 5$$

We therefore have the following information:

$$n = 25$$

$$\bar{x} = 12$$

$$\sigma = 5$$

$$\mu_0 = 10$$

where μ_0 represents the population mean specified under the null hypothesis.

Since we have a random sample of 25 companies, suppose that the population distribution is sufficiently well behaved for the Central Limit Theorem to provide a reasonable normal approximation.

Under the null hypothesis:

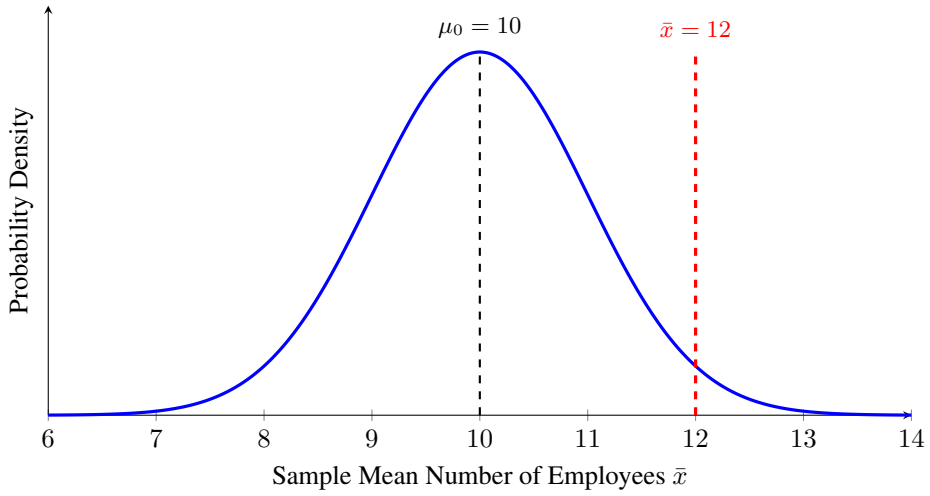
$$\bar{X} \overset{\text{approx}}{\sim} N\left(\mu_0, \frac{\sigma^2}{n}\right)$$

$$\bar{X} \overset{\text{approx}}{\sim} N\left(10, \frac{5^2}{25}\right)$$

$$\bar{X} \overset{\text{approx}}{\sim} N(10, 1)$$

We can illustrate this distribution as follows:

Figure 9.2: Sampling Distribution of the Sample Mean Under the Null Hypothesis



The blue curve represents the sampling distribution of the sample mean under the assumption that H_0 is true.

The black dashed line indicates the hypothesised population mean $\mu_0 = 10$, while the red dashed line indicates our observed sample mean $\bar{x} = 12$.

Notice that the observed sample mean is larger than the hypothesised population mean. However, the important question is not simply whether $\bar{x}$ differs from μ_0. Instead, we want to determine **how far the observed sample mean is from the hypothesised population mean**.

9.2.3 The Test Statistic

To measure how unusual our sample mean is, we need to start by standardising the sampling distribution.

Recall that the standardisation formula for an individual observation is:

$$Z = \frac{X - \mu}{\sigma}$$

Since the standard deviation of the sampling distribution is $\frac{\sigma}{\sqrt{n}}$, we instead use:

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

Substituting our values:

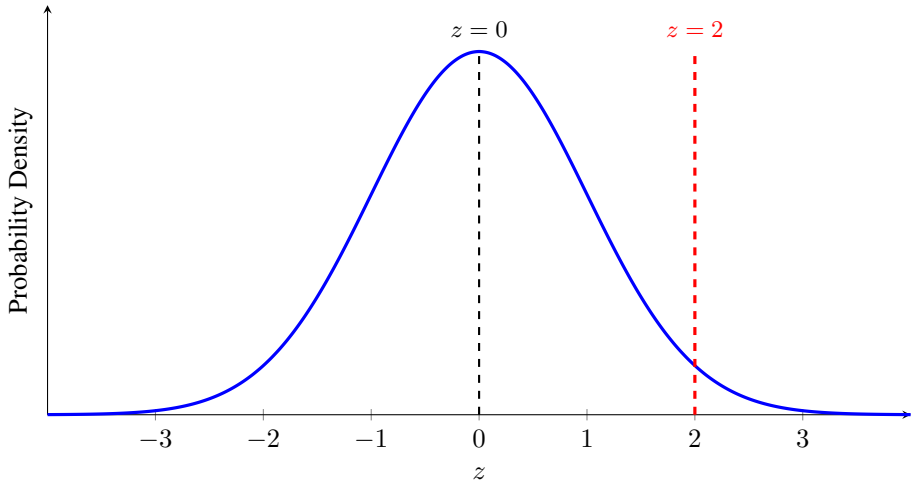
$$z = \frac{12 - 10}{5/\sqrt{25}}$$

$$= \frac{2}{1}$$

$$= 2$$

This tells us that the observed sample mean $\bar{x} = 12$ is **two standard errors away from** the population mean specified under the null hypothesis.

Figure 9.3: Standardised Sampling Distribution Under the Null Hypothesis



Now we are interested in the following question: is 2 standard deviations sufficiently far enough from $\mu = 10$ that we can reject the null hypothesis H_0?

The **decision rule** is to simply ensure that your z-value for $\bar{x}$ must be greater than α:

$$z > \alpha$$

In this case:

$$2 > \alpha$$

In order to determine what the value of α is, we will need to know what the **alternative hypothesis** H_1 is and what the **significance level** of our test will be. It will be one of the values that come from the below table:

Table 9.1: Critical Values of the t-Distribution for Hypothesis Testing

	Level of Significance for One-Tailed Test						
	20%	**15%**	**10%**	**5%**	**2.5%**	**1%**	**0.5%**
	Level of Significance for Two-Tailed Test						
df	**40%**	**30%**	**20%**	**20%**	**5%**	**2%**	**1%**
1	1.376	1.963	3.078	6.314	12.706	31.821	63.657
2	1.061	1.386	1.886	2.920	4.303	6.965	9.925
3	0.978	1.250	1.638	2.353	3.182	4.541	5.841
4	0.941	1.190	1.533	2.132	2.776	3.747	4.604
5	0.920	1.156	1.476	2.015	2.571	3.365	4.032
6	0.906	1.134	1.440	1.943	2.447	3.143	3.707
7	0.896	1.119	1.415	1.895	2.365	2.998	3.499
8	0.889	1.108	1.397	1.860	2.306	2.896	3.355
9	0.883	1.100	1.383	1.833	2.262	2.821	3.250
10	0.879	1.093	1.372	1.812	2.228	2.764	3.169
∞	0.842	1.036	1.282	1.645	1.960	2.326	2.576

9.2.4 One-Tailed Test

Suppose we test the below hypothesis:

$$H_0 : \mu = 10$$

$$H_1 : \mu > 10$$

Essentially, we are saying that if we reject the null hypothesis H_0, then there is sufficient evidence supporting the alternative must, which is that μ is some number that is greater than 10.

This is known as a **one-tailed test**. This is because if $\mu > 10$, then we are only testing half of the distribution (i.e, only the right tail):

Recall that our observed Z-statistic is:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

$$= \frac{12 - 10}{5/\sqrt{25}}$$

$$= 2$$

In order for us to reject the null $H_0 : \mu = 10$, we need the z-value of $\bar{x}$ to be greater than the critical value:

$$z > z_\alpha$$

In a one tailed test when the value of σ is known, we only need to observe the row for ∞ to get the value of α:

Table 9.2: Critical Values of the t-Distribution for Hypothesis Testing

	Level of Significance for One-Tailed Test						
	20%	**15%**	**10%**	**5%**	**2.5%**	**1%**	**0.5%**
∞	0.842	1.036	1.282	1.645	1.960	2.326	2.576

The percentages on each column represents the **significance level** α of the test. The smaller the percentage, **the more plausible** your results become. This is because the significance level represents the chance that you are wrong when rejecting the null hypothesis.

For example, a significance level of 10% means that, if the null hypothesis is true, our testing procedure has a 10% probability of incorrectly rejecting it.

In order to reject the null hypothesis, we need to check:

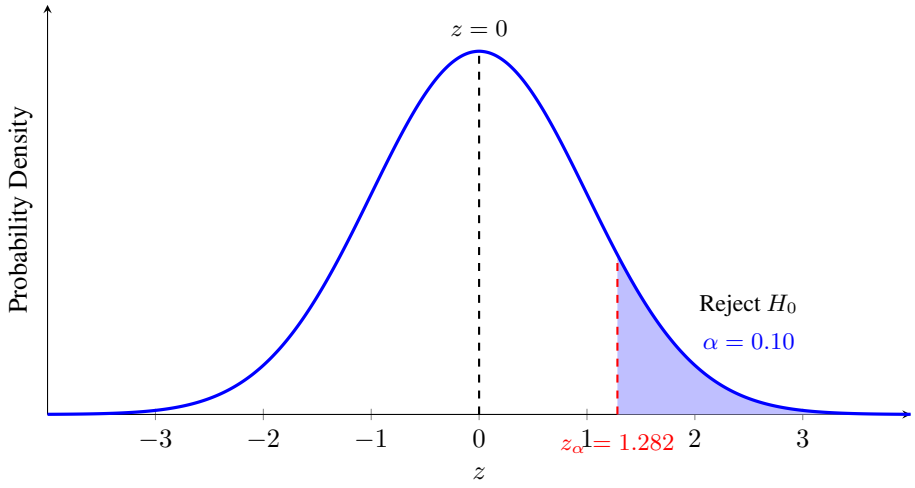
$$z > z_{10\%}$$

$$2 > 1.282$$

This shows that $\bar{x} = 12$ is indeed statistically different from $\mu = 10$. We reject the null hypothesis and accept the alternative H_1 that $\mu > 10$.

Graphically, this is shown as follows:

Figure 9.4: Right-Tailed Hypothesis Test



Intuitively, this means that $\bar{x} = 12$ is indeed an unusually large number compared to $\mu = 10$ based on how many standard deviations away it is from the centre. We would therefore conclude that μ is some number that is greater than 10.

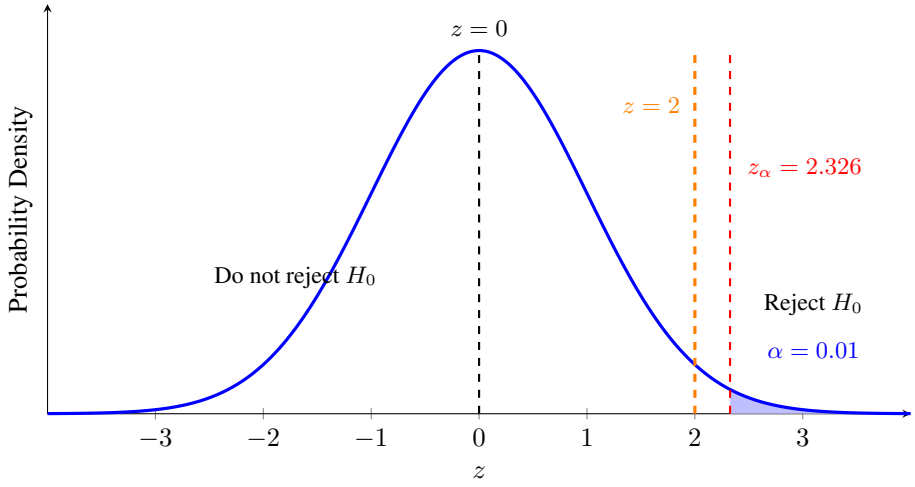
Now let us instead test our hypothesis with a 1% significance level. This means that to reject the null:

$$z > z_{1\%}$$

$$2 < 2.326$$

In this case, there is **insufficient evidence** to reject the null hypothesis. It is therefore plausible that $\mu = 10$:

Figure 9.5: Right-Tailed Hypothesis Test ($\alpha = 0.01$)



One-tailed tests can also be done on the negative tail of the distribution. Consider the below activity.

Activity:

A bank in Bahrain claims that the average waiting time for customers is 20 minutes.

Following improvements to its customer service system, the bank wants to investigate whether the average waiting time has decreased.

A researcher collects a random sample of 25 customers and obtains the following information:

$$n = 25$$

$$\bar{x} = 17$$

$$\sigma = 6$$

Assume that the population standard deviation is known and that the sampling distribution of the sample mean is normally distributed.

We wish to test the following hypotheses:

$$H_0 : \mu \geq 20$$

$$H_1 : \mu < 20$$

1. Calculate the Z-test statistic.
2. At the 5% significance level, identify the critical Z-value and use the decision rule to determine whether we should reject the null hypothesis.
3. Draw the standard normal distribution. Shade the rejection region and indicate the critical Z-value and the calculated Z-test statistic.

Solution

1. **Calculate the Z-test statistic.**

   Recall that when the population standard deviation is known, the Z-test statistic is:

   $$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

   Substituting our values:

   $$z = \frac{17 - 20}{6/\sqrt{25}}$$

   $$= \frac{-3}{6/5}$$

   $$= \frac{-3}{1.2}$$

   $$= -2.5$$

   Therefore, our observed sample mean is **2.5 standard errors below** the hypothesised population mean.

2. **Identify the critical value and analyse the decision rule.**

   We are conducting a left-tailed hypothesis test because our alternative hypothesis is:

$$H_1 : \mu < 20$$

   At the 5% significance level:

$$\alpha = 0.05$$

   From the standard normal distribution table, the critical value is:

$$z_{\text{critical}} = -1.645$$

   Notice that the critical value is **negative** because the rejection region is located in the left tail.

   The decision rule is:

$$z < -1.645 \quad \Rightarrow \quad \text{Reject } H_0$$

   We compare our calculated Z-statistic with the critical value:

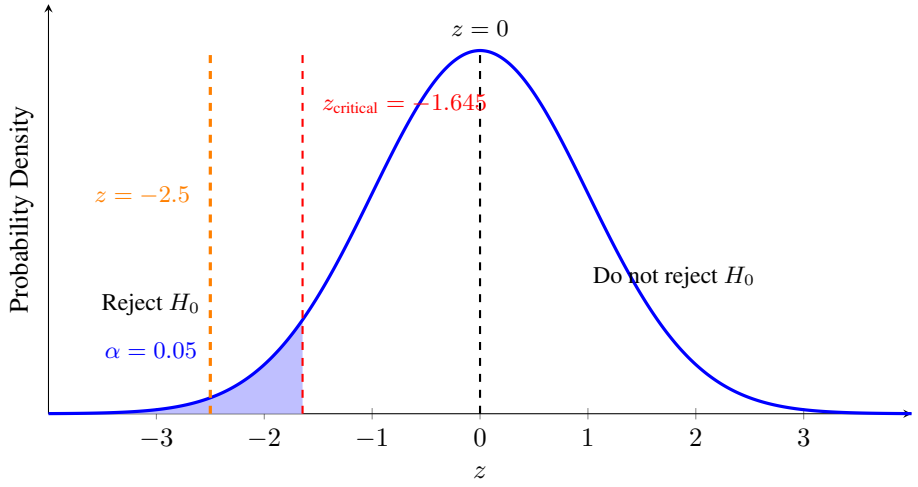
$$-2.5 < -1.645$$

   Since the calculated Z-statistic is smaller than the critical value, we **reject the null hypothesis**.

   Therefore, at the 5% significance level, there is sufficient statistical evidence to conclude that the population mean waiting time is less than 20 minutes.

3. **Illustrate the rejection region.**

Figure 9.6: Left-Tailed Hypothesis Test ($\alpha = 0.05$)



The shaded region represents the **5% rejection region**. Notice that our calculated test statistic $z = -2.5$ falls inside this region.

Therefore, we reject H_0.

9.2.5 Two-Tailed Test

Suppose now we test the below hypothesis:

$$H_0 : \mu = 10$$

$$H_1 : \mu \neq 10$$

Notice this time the alternative hypothesis does not make any assumptions about whether the value of $\bar{x}$ is over or under 10. It is simply saying that it is not equal to 10.

This is known as a two-tailed test. This is because we are essentially testing the hypothesis **under both tails of the distribution**.

The decision rule for rejection is still the same. We simply have to select the z_α which corresponds for a two-tailed test:

Table 9.3: Critical Values of the t-Distribution for Hypothesis Testing

	Level of Significance for One-Tailed Test						
	20%	**15%**	**10%**	**5%**	**2.5%**	**1%**	**0.5%**
	Level of Significance for Two-Tailed Test						
∞	0.842	1.036	1.282	1.645	1.960	2.326	2.576

Suppose we wanted to test the hypothesis under the 5% significance level. We would simply select $z_{5\%} = 1.960$:

$$z > z_{5\%}$$

$$2 > 1.96$$

We would therefore reject the null hypothesis that $H_0 : \mu = 10$.

9.3 Hypothesis Testing when σ is Unknown

Now suppose that the population variance σ was unknown to us. Just as before, we would have to instead use the sample estimator s instead.

However the consequence of doing this is that we are no longer able to say that $\bar{x}$ has a normal distribution. We instead say that it follows the t-distribution:

$$\bar{x} \sim t_{n-1}$$

Remember that the t-distribution still has the same symmetric shape that a normal distribution has. For a sufficiently large sample $n = \infty$, our t-distribution approximates a normal distribution. The procedure for hypothesis testing is slightly different in this circumstance.

Recall that when σ is known, we use the Z-test statistic:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

When σ is unknown, we replace it with the sample standard deviation s:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

This is known as the **t-test statistic**.

If the observations are independently sampled from a normally distributed population, then under the null hypothesis:

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}$$

where t_{n-1} represents the t-distribution with $n - 1$ degrees of freedom.

Suppose a researcher wants to investigate whether the average height of male university students in Bahrain exceeds 170 cm.

The researcher collects a random sample of 10 students and obtains the following information:

$$n = 10$$

$$\bar{x} = 174 \text{ cm}$$

$$s = 6 \text{ cm}$$

Let us start with our hypothesis:

$$H_0 : \mu \leq 170$$

$$H_1 : \mu > 170$$

To calculate the t-statistic:

The formula is:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$t = \frac{174 - 170}{6/\sqrt{10}}$$

$$= \frac{4}{1.897}$$

$$\approx 2.108$$

Therefore, our observed sample mean is approximately **2.108 estimated standard errors above** the hypothesised population mean.

Let us conduct our test at the 5% significance level. In this case, we would need to look at the table and select the row where $df = 10 - 1 = 9$:

Table 9.4: Critical Values of the t-Distribution for Hypothesis Testing

	Level of Significance for One-Tailed Test						
	20%	**15%**	**10%**	**5%**	**2.5%**	**1%**	**0.5%**
	Level of Significance for Two-Tailed Test						
df	**40%**	**30%**	**20%**	**20%**	**5%**	**2%**	**1%**
8	0.889	1.108	1.397	1.860	2.306	2.896	3.355
9	0.883	1.100	1.383	1.833	2.262	2.821	3.250
10	0.879	1.093	1.372	1.812	2.228	2.764	3.169
∞	0.842	1.036	1.282	1.645	1.960	2.326	2.576

The corresponding critical value is:

$$t_{0.05,9} = 1.833$$

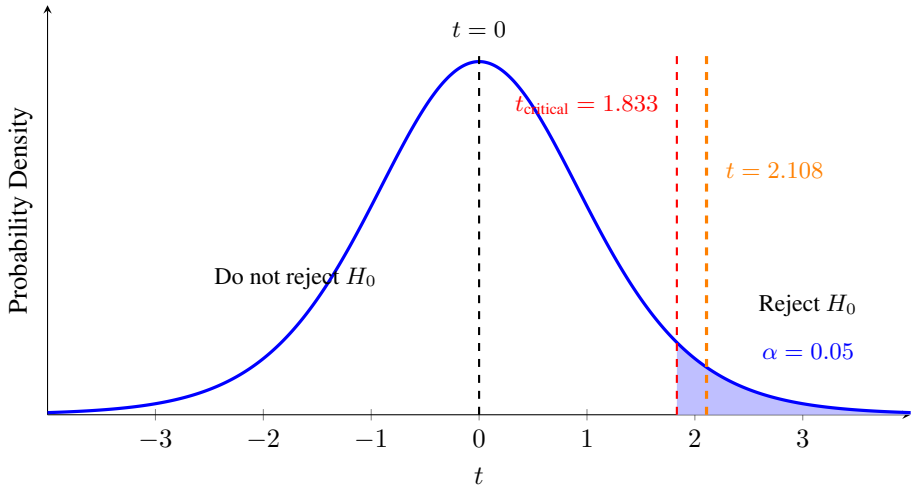
For a right-tailed t-test, the decision rule is:

$$t > t_{\alpha,df}$$

$$2.108 > 1.833$$

Since our calculated t-statistic exceeds the critical value, we **reject the null hypothesis**.

Figure 9.7: Right-Tailed t-Test ($df = 9$, $\alpha = 0.05$)



Therefore, at the 5% significance level, there is sufficient statistical evidence to conclude that the population mean height of male university students exceeds 170 cm.

Notice that the procedure is almost identical to the Z-test. The main differences are that we use the sample standard deviation s instead of the population standard deviation σ, and we obtain our critical value from the t-distribution using the appropriate degrees of freedom. The same principle follows for a two-tailed test.

9.4 Hypothesis Testing with Proportional Variables

What we have discussed so far relates to testing hypotheses when the random variable X is continuous. When analysing discrete variables, we would need to use the appropriate estimators.

Recall from the previous chapter that a proportion and its variance is defined as:

$$\pi = \frac{\sum x_i}{N} \qquad p = \frac{x}{n}$$

The variance and standard deviation is given by:

$$\sigma^2 = \frac{\pi(1-\pi)}{n} \qquad \sigma = \sqrt{\frac{\pi(1-\pi)}{n}}$$

In this module, we only examine the case when σ is known.

In order to conduct a hypothesis test for discrete variables, the value of z for our decision rule will have the below formula:

$$Z = \frac{|p-\pi|}{\sqrt{\frac{\pi(1-\pi)}{n}}}$$

> **Activity:**
>
> A large retail company in Bahrain aims to maintain a customer satisfaction rate of at least 80%. The company's management is concerned that customer satisfaction may have fallen below this target.
>
> To investigate this, the company conducts a random survey of 2,000 customers. The survey reveals that 77.5% of customers are satisfied with the company's services.
>
> Using a **5% significance level**, conduct a hypothesis test to determine whether there is sufficient statistical evidence that the population proportion of satisfied customers has fallen below 80%.

Solution

We are given the following information:

$$n = 2000$$

$$p = 0.775$$

$$\pi_0 = 0.80$$

We want to determine whether the population proportion of satisfied customers is less than 80%.

$$H_0 : \pi \geq 0.80$$

$$H_1 : \pi < 0.80$$

This is a **left-tailed hypothesis test** because the alternative hypothesis states that the population proportion is less than 0.80.

We are conducting our hypothesis test at the 5% significance level:

$$\alpha = 0.05$$

Since this is a left-tailed Z-test, the corresponding critical value is:

$$z_\alpha = -1.645$$

We need to calculate the z-value:

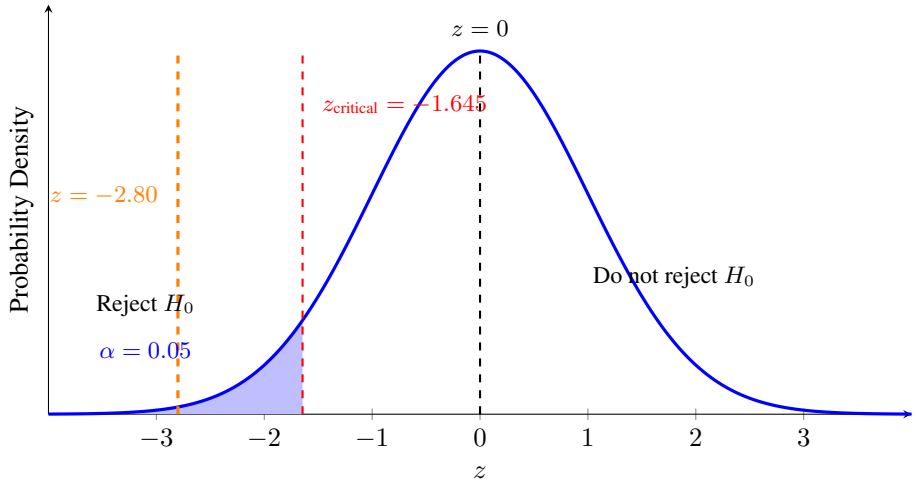
= 0.775-0.80$\frac{}{\sqrt{\frac{0.80(1-0.80)}{2000}}} = \frac{-0.025}{\sqrt{\frac{0.80(0.20)}{2000}}} = \frac{-0.025}{\sqrt{0.00008}} = \frac{-0.025}{0.008944} \approx -2.80$

For a left-tailed hypothesis test, we reject the null hypothesis if the calculated Z-statistic is smaller than the critical value:

Where we can see:

$$-2.80 < -1.645$$

Figure 9.8: Left-Tailed Z-Test for a Population Proportion ($\alpha = 0.05$)



Our observed Z-statistic of -2.80 falls inside the rejection region. Therefore, we reject H_0.

9.5 Hypothesis Testing for Paired Samples

Consider an example where a company in Bahrain introduces a new training programme to improve employee productivity. The company wants to investigate whether its employees become more productive after completing the programme.

One approach would be to measure the productivity of the **same employees** before and after receiving the training.

This is an example of **paired samples**. That is, when two observations are collected from the same individual. The

objective of a paired-sample hypothesis test is to determine whether there is a statistically significant difference between the two measurements.

Let us analyse the results of this training programme:

Table 9.5: Employee Productivity Before and After Training

Employee i	**Before Training**	**After Training**	**Difference** d_i
1	20	25	5
2	25	28	3
3	18	24	6
4	22	24	2
5	30	34	4

Notice that each employee has two observations.

For example, Employee 1 completed 20 tasks before training and 25 tasks after training. Therefore, their productivity increased by 5 tasks. Similarly, Employee 2 increased their productivity by 3 tasks.

The central idea behind paired-sample hypothesis testing is that we transform our two measurements into a single variable representing their difference.

Let:

$$X_{i,1} = \text{Productivity before training}$$

$$X_{i,2} = \text{Productivity after training}$$

We define the difference for employee i as:

$$D_i = X_{i,2} - X_{i,1}$$

For example, the difference for Employee 1 is:

$$d_1 = 25 - 20$$
$$= 5$$

The difference for Employee 2 is:

$$d_2 = 28 - 25$$
$$= 3$$

Therefore, instead of working with two separate columns of productivity measurements, we can work with a single variable containing the differences:

$$d = \{5, 3, 6, 2, 4\}$$

A **paired-sample hypothesis test** is essentially a **one-sample hypothesis test** applied to the differences between paired observations.

Once we calculate the differences, we can calculate their sample mean.

The formula is:

$$\bar{d} = \frac{\sum_{i=1}^{n} d_i}{n}$$

where:

- $\bar{d}$ is the sample mean difference.
- d_i is the difference for observation i.
- n is the number of paired observations.

Using our example:

$$\bar{d} = \frac{5+3+6+2+4}{5}$$

$$= \frac{20}{5}$$

$$= 4$$

Therefore, employee productivity increased by an average of 4 tasks following the training programme.

However, this does not necessarily mean that the improvement is statistically significant. As usual, the hypothesis test would measure how many standard deviations away we are from the the null.

We specify the following hypotheses:

$$H_0 : \mu_D \leq 0$$

$$H_1 : \mu_D > 0$$

The null hypothesis states that there is no positive average change in employee productivity following the training programme. The alternative hypothesis states that the population mean difference is positive.

Recall that when the population standard deviation is unknown, we use the t-test statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

For paired samples, we use almost the same formula:

$$t = \frac{\bar{d} - \mu_{D,0}}{s_D/\sqrt{n}}$$

where:

- $\bar{d}$ is the sample mean difference.
- $\mu_{D,0}$ is the hypothesised population mean difference, usually zero.
- s_D is the sample standard deviation of the differences.
- n is the number of paired observations.

We previously calculated:

$$\bar{d} = 4$$

Using a **5% significance level**, conduct a paired-sample hypothesis test to determine whether employee productivity has increased following the training programme.

Recall that the sample variance is:

$$s_D^2 = \frac{\sum_{i=1}^{n}(d_i - \bar{d})^2}{n-1}$$

Substituting our values:

$$s_D^2 = \frac{(5-4)^2 + (3-4)^2 + (6-4)^2 + (2-4)^2 + (4-4)^2}{5-1}$$

$$= \frac{1+1+4+4+0}{4}$$

$$= \frac{10}{4}$$

$$= 2.5$$

Therefore, the sample standard deviation of the differences is:

$$s_D = \sqrt{2.5}$$

$$= 1.581$$

The paired-sample t-statistic is:

$$t = \frac{\bar{d} - \mu_{D,0}}{s_D/\sqrt{n}}$$

Substituting our values:

$$t = \frac{4 - 0}{1.581/\sqrt{5}}$$

$$= \frac{4}{0.7071}$$

$$\approx 5.657$$

Therefore, the calculated test statistic is:

$$t = 5.657$$

We are conducting a right-tailed hypothesis test at the 5% significance level:

From the t-distribution table, the critical value for a one-tailed test with $df = 4$ and $\alpha = 0.05$ is:

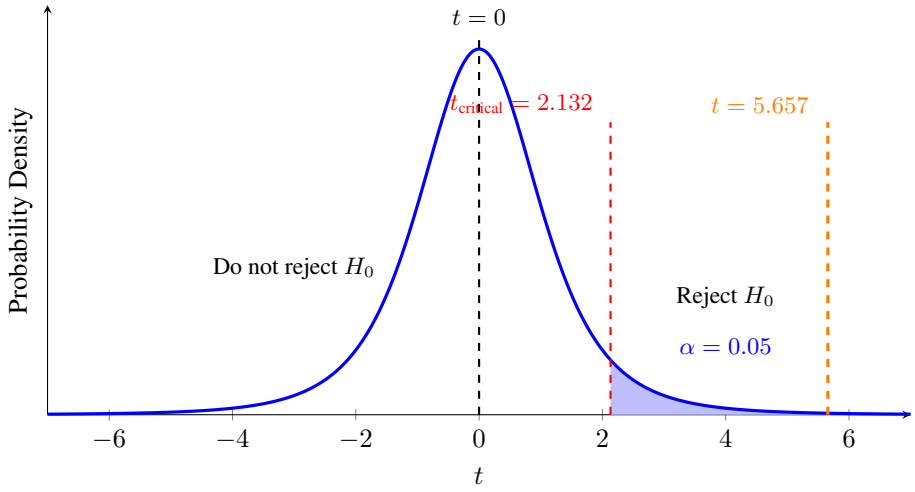
$$t_{0.05,4} = 2.132$$

For a right-tailed hypothesis test, we reject the null hypothesis if:

$$t > t_{\alpha,df}$$

$$5.657 > 2.132$$

Figure 9.9: Paired-Sample Right-Tailed t-Test ($df = 4$, $\alpha = 0.05$)



Since our calculated t-statistic exceeds the critical value, we **reject the null hypothesis**.

9.6 Ethics

9.6.1 Ethical Considerations in Hypothesis Testing

- Disclose when multiple comparisons/tests are run, and any adjustments made.
- Convey "fail to reject H_0" honestly. It is not proof H_0 is true.

- Be transparent about a priori vs. post hoc choices, including when α was set.

Ethical Use of AI

- AI can compute a test correctly, but data collection soundness must still be disclosed by the practitioner.

- Do not ask AI to help reframe a non-significant result as significant.

Chapter 10

Analysis of Variance